

(Front Cover)

Green Ideals: Exploring Sustainability in ASEAN and Beyond

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Green Ideals: Exploring Sustainability in ASEAN and Beyond

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“Imagine a verdant forest, a tapestry of life woven from countless trees. We stand beneath its canopy, gazing upwards. Light filters through the leaves, dappling the ground, as we crane our necks to envision the future stretching before us. Just as each tree plays a vital role in the health of the forest, so too does each individual and nation contribute to shaping a sustainable tomorrow. This design embodies this interconnectedness (Paticcasamuppāda - បដិច្ចសមុប្បាទៈ) reminding us that a collective effort is necessary to cultivate a ‘Green’ future.”

Edited by CHHEM Kieth Rethy, CHEM Phalla and KHIENG Sothy

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Foreword

Every country needs a strategy to address the great challenges of our age: rapid technological advance, the urgent need for advanced education and skills, the rapid transformation to green technologies, the need for regional solutions that transcend national borders, and the need for diplomacy in order to operate effectively in a multipolar world. In sum, every country needs a strategy for Sustainable Development.

This volume proves that Cambodia is undertaking an intensive transformation to an advanced knowledge economy that aims to utilize science, technology, and innovation (STI) to address the many dimensions of sustainable development. In this exceptionally strong compendium of essays, Cambodian scientists, scholars, policymakers, and thought leaders, as well as co-authors from other ASEAN nations, explore every dimension of Cambodia's sustainable development challenges, from considerations of the local environment to the opportunities for global diplomacy. This expansive perspective – ranging across issues, scientific disciplines, and scales of action – is just what is needed for Cambodia to meet its national objective of becoming a high-income and sustainable society by mid-century, if not sooner.

What is most impressive to me is Cambodia's recognition that all success will depend on the quality of education of Cambodia's young people, and their preparation for an interconnected, knowledge-intensive, and digitally driven world economy. Cambodian scholars and policy leaders recognize that every sector of the Cambodian economy will soon have to be a "high-skilled" sector, including even the so-called traditional sectors such as textiles, tourism, and agriculture. Each sector of the economy, and all successful enterprises, will be digitally empowered, linked to artificial intelligence and other machine learning, and in need of continual innovations in order for Cambodia to remain globally connected and competitive. In short, Cambodia must prepare to thrive in an age of rapid technological advancement.

If the digital age were all that is challenging Cambodia, that alone would be a major challenge. But Cambodia faces even greater challenges. As a remarkable tropical environment of unique biodiversity and ecosystems, Cambodia depends on its natural capital for its sustenance, culture, and global competitiveness. Yet Cambodia's vital natural capital is already under deep stress, both because of past poor practices in land use, waste management, and industrial pollution and because ongoing global environmental changes (most importantly, global warming) will create major new risks for Cambodia regarding water management, agricultural production, urban habitability, and protection of biodiversity.

As the essays here underscore, the Royal Government of Cambodia, together with Cambodia's leading universities and scientific centers, have turned their curricula, research priorities, and operating budgets to face these new challenges. The returns will be enormous. Cambodia's investments in

STI will result in enormous advances in Cambodian productivity, efficiency of energy and materials use, and sustainability. These advances in turn will enable Cambodia to achieve rapid economic growth in the decades ahead, also taking advantage of the overall rapid growth of East Asia.

The essays here touch on every dimension of the STI agenda: water resource management, agricultural productivity, circular economy, and achievement of a zero-carbon energy system. In each area, the urgency is clear. Cambodia is already facing the risks of intensified droughts and floods, heatwaves, and land degradation. Yet the opportunities are even greater. Through smart technologies, Cambodia will be able to raise agricultural productivity, reduce food wastes, build a regional power grid based on renewable energy, and enhance Cambodia's standing as a global destination for sustainable tourism.

One of the great innovations ahead lies in Cambodia's future Universities 4.0, meaning Cambodian universities that are technologically enabled and organizationally structured to train young people to become national and global leaders of sustainable development. A University for Sustainability will teach the interconnections among the Earth's physical systems, the engineered infrastructure systems, and the socio-politico-economic systems. It will train students and researchers in the policy methods needed to achieve sustainable development. It will guide students through the ethical underpinnings needed for a high-income sustainable society. And it will train future problem solvers in the diplomatic arts of regional and global collaboration.

The essays in this volume recognize at every juncture that Cambodia is not alone in facing these great challenges, and that indeed, Cambodia can meet its national objectives only through extensive diplomacy and global cooperation. International cooperation will be needed at many scales. The ASEAN countries working together will need to build an interconnected zero-carbon power grid extending across ASEAN and connecting ASEAN with neighbors including Australia, China, and others. ASEAN countries will need to coordinate on the protection of ASEAN's rainforests, fisheries, coastal areas, and river sheds. ASEAN nations will need to join together in fruitful diplomacy with other regions of the world, such as with the European Union. In this regard, ASEAN's Green Deal will be an important diplomat counterpart to the European Green Deal and will enable the EU and ASEAN to set mutually agreed rules for trade, finance, and technological change.

ASEAN will also need to play a major global leadership role as well, in the UN General Assembly, the UN Framework Convention on Climate Change, the UN Convention on Biological Diversity, the G20, the BRICS, and other global venues. As the chapter on Climate Finance makes clear, Cambodia's success will also depend on access to high-quality international financing – long-term loans, green bonds, foreign direct investments, and various kinds of grants from international institutions – in order for Cambodia to invest at the scale needed to achieve rapid growth combined with sustainable development. Cambodia's

leadership, both through the national government and as a leading member of ASEAN, will be very important in forging the needed reforms of the Global Financial Architecture.

In sum, the coming years will be a period of rapid global change. With leadership, vision, ethics, good governance, and strong national institutions working in collaboration with other nations, Cambodia is poised for very exciting advances. The volume's lead editors are Prof. Chhem Kieth Rethy, a scientist, physician, and former UN diplomat, Dr. Chem Phalla, an engineer specializing in water resources management, hydrology, and hydraulics and Dr. Khieng Sothy, an accomplished scholar, educator, and higher education administrator. Their dedicated partnership during the past decade has focused on development and foreign policy research and has emphasized the interconnected dynamics of environmental, social, and economic dimensions. The editors and the authors of this volume deserve high praise and careful attention to their many important insights and recommendations, which together provide an important framework for Cambodia's path to sustainable development.

Prof. Jeffrey Sachs

President of the UN Sustainable Development Solutions Network

Preface

Cambodian Scholars' Vision of the Green Deal

In an era defined by unprecedented environmental challenges and the urgent need for sustainable solutions, the world's attention has turned towards the ambitious concept known as the "Green Deal." This comprehensive framework has since captured the imaginations of policymakers, scholars, and citizens around the globe. The Green Deal represents a holistic approach to addressing pressing issues related to climate change, environmental sustainability, and socio-economic development.

The "Green Deal" concept encompasses practical policies, frameworks, and strategies aimed at mitigating the consequences of climate change and promoting sustainable development. Conversely, "Green Ideals" are the aspirational objectives, principles, and visions that serve as the impetus for these pragmatic initiatives. The Green Ideals are the visionary foundation that propels these endeavours, while the Green Deal concentrates on implementation. It is imperative for ASEAN to adopt Green Ideals as a Green Deal aspiration, as this ensures that practical actions are consistent with high standards and long-term environmental objectives, thereby promoting a more comprehensive and unified approach to sustainability. In 2022, former Samdech Prime Minister Hun Sen introduced the concept of the green deal. To strengthen early efforts from governments to further define the ASEAN Green Deal—currently a vision and early regional initiative—this book aims to clarify the concept. By offering diverse perspectives, it endorses the critical mission of the 2022 Green Deal and underscores the region's need to take it seriously, thereby transforming ideals into actionable ideas and impactful outcomes.

As we embark on this journey through the perspectives and insights of Cambodian scholars and experts, we explore a diverse array of topics, each intertwined with the overarching theme of the Green Deal. From the intricate geopolitics of this transformative initiative to the vital connections between Sustainable Development Goal 13 (SDG 13) and circular economy principles, this book offers a multifaceted exploration of the Green Deal's far-reaching impact. In our exploration, we uncover the intricate connections between international trade and the circular economy, demonstrating how economic policies can either reinforce or hinder our progress toward a greener future. We investigate the realms of climate and water diplomacy, shedding light on the crucial role of diplomacy in addressing shared environmental challenges. As technology continues to shape our world, we examine the concept of Green Digital Transformation and how it can revolutionize our approach to sustainability. The book looks into the complexities of energy diplomacy within the Green Deal framework and explores the nexus between food systems, security, and sustainable agriculture.

We also highlight the critical intersections of science, technology, and innovation, including the advent of Industry 4.0, in driving forward the Green Deal's objectives. Environmental scientists, as well as their invaluable

contributions, take center stage in this discourse, emphasizing the pivotal role they play in shaping a greener future.

The Green Deal is not just about the environment; it is about global health too. Climate change's impact on public health is a pressing concern, and we investigate how it intertwines with this grand endeavor. The concept of a University for Sustainability is presented as a potential game-changer in educating and equipping the next generation of leaders with the tools to navigate the challenges of our time.

Financing is at the heart of any transformative initiative, and we explore the various mechanisms and strategies for funding Green Deal programs. Sustainable tourism, as a key economic sector, is examined in-depth, considering its potential to drive sustainable development. Lastly, we explore the creation of a green-built environment, showcasing innovative architectural and design approaches that prioritize sustainability.

The chapters within this book offer a panoramic view of the Green Deal's multifaceted dimensions, weaving together the perspectives of Cambodian scholars and experts with global implications. Through these insightful contributions, we hope to inspire discussions, spark innovation, and foster a deeper understanding of the transformative potential of the Green Deal on both local and global scales. It is our collective hope that this book will serve as a valuable resource for policymakers, academics, students, and anyone passionate about building a sustainable and resilient future for our planet.

In practice this book was conceived as a compilation of chapters contributed by a diverse group of scholars, practitioners, and policy researchers. All of them are thoughtful and clear-thinking researchers. The intention was not to impose a strict formatting style on the individual chapters. Instead, the editors chose to provide authors with the freedom to express their research and ideas creatively. Our goal was to encourage each author to explore the urgent need for a Green Deal in their own unique way.

By doing so, we aimed to foster brainstorming and gather the most innovative, unrestrained ideas and solutions to address the pressing challenges posed by climate change and its far-reaching impacts on both humanity and the environment. In summary, the aim was to create a book that is both approachable and thorough, serving as the starting point for our intellectual exploration of the intricate and captivating aspects of the Green Deal. The wide array of ideas from diverse perspectives can serve as a foundation for conducting comprehensive interdisciplinary research on the Green Deal that aims to inform future policies that transcend national borders, emphasizing a more systematic and globally relevant approach.

CHHEM Kieth Rethy

CHEM Phalla

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Acknowledgments

We extend our heartfelt appreciation to the dedicated individuals who contributed to the meticulous review and refinement of each chapter in this book. Their invaluable efforts have significantly enhanced the quality and coherence of the content, ensuring a seamless reading experience for our audience.

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Their collective contributions have facilitated the work of the editors and have undoubtedly elevated the overall quality of this book. We are deeply grateful for their commitment and dedication to excellence.

Last but not least, we would like to extend our sincere gratitude to the Kirirom Institute of Technology (KIT) and the Kang Meng Huy (KMH) Foundation for their generous sponsorship. This support significantly facilitated the completion of the Green Ideals book. Their commitment to sustainability perfectly aligns with the book's message, and their input was invaluable.

With sincere appreciation,

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CHEA Vatana was a Vice President of Cambodia University of Technology and Science and holds a master's and a doctoral degree in demography. His research interest includes population projection, migration and development, human capital development, and the economics of household. Dr. Chea has authored and co-authored more than thirty scientific articles, including book chapters and journals in ISI and SCOPUS databases.

CHEM Phalla, an Adjunct Professor at Cambodia University of Technology and Science and an esteemed scholar holding a Ph.D. in Water Resources Management from the University of Sydney, delves into the transformative realm of water diplomacy in transboundary river basins and is also a co-editor of this book. Serving as the Vice-Chairman, with the rank of Minister, at the Economic, Social, and Cultural Council office of the Council of Ministers in the Kingdom of Cambodia, Dr. Phalla brings unparalleled expertise in climate change and water governance, water resources management and sustainability, and policy design and implementation in the water resources sector. His dedication to fostering connections among societies through water highlights the critical intersection of diplomacy, sustainability, and societal well-being in our shared water resources.

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SRUN Pagnarith is a prominent figure in Chapter 6, focusing on Energy Diplomacy and the ASEAN Green Deal. He is a beacon of expertise in his field, earning a Doctor of Engineering degree from the University of Tokyo. Currently serving in the Ministry of Industry, Science, Technology, and Innovation, Dr. SRUN specializes in policy research, path analysis, and environmental management. His dedication to sustainable practices and innovative solutions makes him a key player in driving positive change within the ASEAN region.

THONG Mengdavid is a distinguished expert featured in Chapter 6, focusing on Energy Diplomacy and the ASEAN Green Deal. With a master's degree in public administration from Tsinghua University and another master's degree in Political Science from Shanghai Jiao Tong University, both in the People's Republic of China, he brings a wealth of knowledge and experience. Currently serving as a Research Supervisor at the Asian Vision Institute and a Lecturer at the Institute for International Studies and Public Policy at the Royal University of Phnom Penh, THONG Mengdavid is deeply engaged in areas such as U.S.-China Relations, ASEAN Studies, Regional Peace, and Maritime Security, as well as International Relations and Small State Strategy. His dedication to fostering understanding and cooperation in these critical areas makes him pivotal in shaping regional dynamics and promoting sustainable development.

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INTRODUCTION

As global temperatures rise, the future existence of humans and other living species is called into question. Climate change and the environment have emerged as global concerns. The United Nations (UN), as a global governance actor, has played a critical role in developing and promoting policies to decrease and influence the norm in response to climate change. For example, the majority UN of member states have supported the United Nations Sustainable Development Goals, which are scheduled to be fulfilled by 2030. It is important to highlight that although development is vital and the state must seek to provide a better quality of life for its residents, it should be carried out or at the very least, considered a long-term option. In other words, if a country continues to use coal and fuel as a source of energy for houses, businesses, and daily life, it will have long-term consequences; thus, sustainable development is critical. However, although it is a great notion, implementing sustainable development is not as simple as expected. For one thing, the notion of sustainability may be greenwashed by many actors, notably a lack of good faith on the part of the superior, including governments or corporate actors, towards the weak. In addition, actors with less capability are more likely to face and be harmed by climate change. In light of the above, it is critical to investigate the green development funding model to learn how it functions and how much it contributes to mitigation efforts. Knowing the possible effects, risks, and repercussions in each area is also crucial for matching the budget allocation.

However, it should not be assumed that there is a generally applicable solution to this dilemma; rather, it is critical to situate the issue in its appropriate local context. With this immediate need for action in mind, the ASEAN Green Ideals Book was created to analyze the region's sustainable or green development dilemma while also providing a chance for additional intellectual research into the Green Deal's numerous and interesting features. To provide a thorough understanding and introduce a wide range of ideas from diverse perspectives from various authors with multidisciplinary backgrounds, it is important to note that the ASEAN Green Ideals book was designed to conduct comprehensive interdisciplinary research on the Green Deal with the goal of informing future policies that transcend national borders, emphasizing a more systematic and globally relevant approach. To ensure a comprehensive yet holistic approach to sustainable development, the book strives to incorporate the green ideal suggestion into each chapter, either as a practical application or as a framework for the green deals. This will be particularly relevant in the development of the ASEAN Green Deals as a structured and implementable document.

In chapter one, Mr. Chhort Chhoravuth and Dr. Chheang Vannarith introduce the concept of the ASEAN Green Deal, which has been derived from the United Nations Green Economy Framework. The authors also introduced the driving factors, such as pollution, market, and social factors. The authors

also drew upon the framework of the EU Green Deal as a source of motivation and model for ASEAN, which is nothing more than a concept as of now.

In chapter two, the authors, Mr. Gupta Aditya and Dr. Keo Piseth, expand from the first author to specifically cover Cambodia's decarbonization. In this chapter, the author will focus on specific relevant issues such as climate change, carbon neutrality, forest conservation, renewable energy, environment, social and governance (ESG), and green investments, all of which have been taken into consideration and put into action by Cambodia. This chapter can be a source of inspiration for Cambodia's action and policy response to achieving decarbonization with the support of green investment and financing.

In the next chapter, chapter three, the author, Dr. Hing Vutha, focuses on the issue of the European Green Deal and its implications for Cambodia, which highlights that the Green Deal is a new growth strategy that will seek to push forward Europe into a resource-efficient and competitive economy. The author highlighted that with the implementation of the European Green Deal, Cambodia, to some extent, will be affected because the standard of commercial activities will be altered and Cambodia's industry may need to adapt to the EU's standard of green development. With that being said, it is interesting to compare this chapter with the previous chapter to have a holistic view of Cambodia's green efforts and their potential external consequences.

Along the lines of diplomacy and external issues, Dr. Chem Phalla's chapter four focuses on a specific issue: water and how it will affect transnational borders and interactions, as well as societal well-being in the Mekong region. The author's goal is to explain sustainable development by depicting Cambodia's approach, as the water sector is one of the critical prioritized areas contributing to poverty reduction by improving access to safe and adequate water for domestic consumption, irrigation, and industry, addressing and minimizing all forms of threat, loss of life, and livelihoods caused by water-related hazards, and sustainably managing water resources.

Furthermore, Dr. Prum Sophea and Mr. Chhem Siriwat wrote chapter five to provide a Cambodian viewpoint on digital transformation, which is a topical and current development for environmental sustainability. The chapter aims to describe the influence and repercussions of technology on the environment and the manner of production in the economy. The growth of technology itself may be both a gift and a burden, so a careful and measured approach to cultivating technology for long-term development in Cambodia and elsewhere is critical.

Chapter six, by three writers (Mr. Norn Samnangseymony, Dr. Srun Pagnarith, and Mr. Thong Mengdavid), expands on the preceding chapter by concentrating on energy and its ties to the Green Deal. Energy has become a critical concern around the globe, especially during the COVID-19 pandemic and the Ukraine-Russia conflict. The authors emphasized that, if implemented, the ASEAN

Green Deal is a significant initiative that might assist ASEAN in adapting to challenges and threats, fostering inclusion and resilience, and supporting the regional transition to sustainable development and recovery.

Dr. Hul Seingheng wrote chapter seven, which discussed science, technology, and innovation. Linking to earlier chapters such as technology and energy, Science, Technology & Innovation play an important role in providing solutions for improving and enhancing environmental quality. In this chapter, the author illustrates Cambodia's national innovative system and Cambodia's target, as well as the importance of green technology for manufacturing industries because it is associated with efficient resource use, and green technology transformation is a recommended new trend for resource conservation.

Dr. Kouk Fidero and Professor Promentilla Michael Angelo B. have offered a perspective through chapter eight on the importance of scientists and engineers in the green deal movement, particularly highlighting their role in promoting sustainable development, reducing carbon emissions, and addressing climate change, as well as communicating scientific facts and discoveries to the public and engaging them in discussion. The writers have emphasized the significance of research and development, as well as solid policy and monitoring, while also reflecting Cambodia's attempts to disseminate scientific facts and discoveries to the public and engage them in conversations to reach a more sustainable future.

Furthermore, chapter nine by Dr. Chhem Sirika and Dr. Sudipta Adonis focuses on the problem of health as it relates to climate change. The authors have said that, in addition to the other sectoral challenges highlighted in the preceding chapter, Cambodia, like the rest of the globe, is facing an urgent global danger that impacts human health via several exposure routes. These include direct exposure to severe weather events like heatwaves and storms, as well as indirect exposure due to air pollution, poor food and water quality, and changes in vector distribution and ecology. The authors argued that climate change is a global problem, not simply one state's, and that diplomacy is critical to implementing preventative and adaptation measures that reach beyond the scope of the health sector.

As Dr. Chea Vatana, Mr. Lim Seakleng, and Dr. Khieng Sothy authored chapter 10, grassroots or other actors other than the government, such as universities, influence the mitigation of climate change and the implementation of sustainable development measures. With the demands for greater skills from first-time job seekers, the need for higher wages, and the changing features of the Cambodian economy, the Cambodian educational system must change to meet these expectations. This chapter will study Cambodian universities' transition from a conventional teaching-only purpose to a more contemporary and up-to-date mission that includes social entrepreneurship education, research and innovation, and contributions to inclusive and sustainable development.

Along with attempting to determine the best path ahead for Cambodia's sustainable development, chapter 11 by Dr. Seng Touch and Mr. Ros Sayumphu examines one component of sustainable development: tourism. The authors argue that by properly leveraging technology to enhance the tourist business, notably culturally and socially, sustainable tourism may increase social and cultural involvement, particularly between visitors and locals. Tourists also assist citizens and the nation in promoting their image, reputation, and culture.

Designing and building a green infrastructure is just as important as the other factors discussed in chapter 12 by Mr. Nath Kananda. According to the chapter, many environmental issues have a significant impact on how we design and practice the built environment, and the two are inextricably linked. The author argues that a new design, a green architecture, is required in view of the multiple harmful consequences of climate change and other environmental issues.

In chapter 13, Ms. Surn Marnick, Mr. Soth Sereyboth, and Professor Chhem Kieth Rethy have written about the farm-to-fork approach, which is important in the EU green agreement and addresses the problem of having a sustainable food system. The chapter will discuss how implementing the Farm-to-Fork Strategy aligns with ASEAN's and how Cambodia may benefit from this strategy in order to achieve sustainable development goals and food security objectives. It is a crucial instrument for reaching Cambodia's objective of achieving high-middle-income status by 2030 and becoming a high-income nation by 2050.

While it is true that climate change will impact several sectors, and the chapters above have thoroughly explained mitigation plans, it is also necessary to stress the key implementor, which is the young. The last part, chapter 14, authored by Ms. Chea Angela, Mr. Say Vithya Arthur, and Mr. Ros Sayumphu, focuses on youth and their involvement in ASEAN and global sustainable development efforts. The chapter said that the present state of play and the possible consequences of environmental crises would outlast future generations if not addressed urgently and seriously today. The sheer magnitude and passion of ASEAN's and the world's young generation places them as a force capable of bringing about substantial change.

It is crucial to recognise that "Green Ideals" and "Green Deals" are distinct yet complementary concepts that contribute practical measures and visionary goals to the implementation of sustainable development. The goal of this book is to motivate and direct readers by combining aspirational environmental sustainability objectives with practical frameworks and strategies. The book employs a multidisciplinary approach to provide a comprehensive understanding of the transformation of green ideals into effective green deals. This ensures that the region not only aspires to high environmental stewardship standards but also takes concrete steps to achieve them, with a particular emphasis on concrete measures.

Chapter 1: The Political Economy of the ASEAN Green Deal

CHHORT Chhorravuth and CHHEANG Vannarith*

1. Introduction

The standard economic system of “business as usual (BAU)” has been unsuccessful in achieving comprehensive growth or social progress and has frequently resulted in damage to the environment and shortages (United Nations Economic Commission for Africa [UNECA], 2020). To address these issues, it is necessary to create a “green economy” that can promote the structural transformation of the economy, increase productivity and value-added, and address distributional impacts (UNECA, 2020). In 2011, the United Nations Environment Programme (UNEP) defined a “green economy” as an economic system that leads to better human welfare and fairness while significantly minimizing environmental hazards and ecological scarcity (UNECA, 2020). Similarly, the Organisation for Economic Co-operation and Development (OECD) defined a “green economy” as an economic development that ensures that natural assets continue to provide the resources and environmental services on which development relies (OECD, 2011).

In 2022, as the rotating Chair of ASEAN, Cambodia proposed the ASEAN Green Deal to harness the green economy and combat climate change in the region. The deal is based on the European Green Deal, which aims to make the European Union carbon-neutral by 2050 (Chheang, 2023). The ASEAN Green Deal is a framework that aims to promote sustainable, inclusive, and resilient development in Southeast Asia. This chapter aims to shed light on the green economy, the objectives and driving forces of the ASEAN Green Deal, and potential implications on political and economic transformations.

2. Green Economy: A New Game in Town

The concept of a green economy has gained momentum over the past decade due to the increasing global concern over climate emergency and environmental degradation caused by rapid, unsustainable economic growth. The green economy aims to promote sustainable development by minimizing the adverse impacts of economic activities on the ecosystem while maximizing the positive impact on social and economic well-being. There are several key indicators to measure the green economy, including the following:

- Environmental and resource productivity
- Economic and environmental assets
- Environmental quality of life
- Economic opportunities and policy responses (OECD, 2011).

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A green economy should embed the five following principles (Green Economy Coalition, 2020):

First, the Wellbeing Principle aims to provide all individuals with the ability to generate and enjoy prosperity.

Second, the Justice Principle strives for fairness and equity both for current and future generations through an impartial system that ensures that decisions, benefits, and costs are distributed equally.

Third, the Planetary Boundaries Principle emphasizes the significance of safeguarding, repairing, and investing in nature.

Fourth, the Efficiency and Sufficiency Principle promotes sustainable consumption and production through various means.

Fifth, the Good Governance Principle is based on the idea of having institutions that are inclusive, transparent, and accountable.

3. Driving Factors

There are three main driving factors. First, political factors play a significant role in the emergence and development of the green economy. Governments are adopting policies to promote it. For a green economy transformation to take place effectively and successfully, it is necessary to establish a conducive political framework for negotiating institutional and legal reforms. Without such a framework, there is a risk that influential stakeholders may impede the process or push it in conflicting directions (UNECA, 2020). There are instances where green growth policies may compel countries to revise their national strategies in manners that can increase the financial burden and production costs for local economic players (Resnick et al., 2012).

Second, a market-based approach is crucial to the green economy. The transition to a green economy involves restructuring production and consumption patterns, which can have significant economic impacts. Market-based mechanisms can encourage flexibility in discovering solutions to enhance resource utilization, environmental quality and justice, as well as provide incentives to minimize the expenses of achieving green objectives. Moreover, they foster innovation that reduces the expense of implementing environmental sustainability criteria (Asian Development Bank, 2021).

Third, social factors also play a role in the green economy. Public attitudes and consumer behavior toward sustainability can influence political and economic decisions (Gugushvili, 2021). The political economy of the green economy is complex and shaped by a range of political, economic, and social forces. While the emergence of sustainable policies and practices has gained momentum in recent years, there are still challenges that need to be addressed to fully realize the potential of the green economy.

4. The EU Green Deal: A Source of Inspiration for ASEAN

The European Green Deal is a comprehensive plan introduced by the European Union (EU) to tackle climate change and create a more sustainable future for Europe. The plan was announced by the European Commission in December 2019 and aims to make Europe carbon-neutral by 2050. The EU Green Deal covers a wide range of topics, including energy, transport, agriculture, biodiversity, and waste management, and sets out a roadmap for achieving a more sustainable and environmentally friendly Europe (Fetting, 2020).

One of the primary goals of the EU Green Deal is to reduce greenhouse gas emissions. The EU aims to cut emissions by at least 55 percent by 2030, compared to 1990 levels, and reach net-zero emissions by 2050 (European Commission [EC], 2021). Another key element of the EU Green Deal is the protection of biodiversity. The European Union's biodiversity strategy for 2030 is an extensive, ambitious, and long-term blueprint for safeguarding the environment and reversing the deterioration of ecosystems. The objective of the plan is to lead Europe's biodiversity towards recovery by 2030 and includes specific measures and commitments (EC, 2020). Furthermore, the Commission recommends reducing the use and hazards of chemical pesticides by 50 percent by 2030 (EC, 2022).

The EU Green Deal also includes plans to transition to a circular economy. This means reducing waste and increasing the use of recycled materials. By 2030, the European Union intends to cut residual municipal waste by 50 percent, and the recycling goals for municipal waste are inherently related (European Environment Agency, 2023).

The EU Taxonomy Legislation came into effect on July 12, 2020, with the aim of promoting sustainable investment, which will contribute to the realization of the European Green Deal. Moreover, it is a significant tool for encouraging and facilitating changes and transitions towards sustainability. The Taxonomy Regulation outlines six fundamental environmental objectives, including (a) "mitigating climate change"; (b) "adapting to climate change"; (c) "sustainably utilizing and protecting water and marine resources"; (d) "transitioning towards a circular economy"; (e) "preventing and controlling pollution"; and (f) "protecting and restoring biodiversity and ecosystems" (Keo & Chheang, 2022, p. 126).

5. Green Development in Southeast Asia

Over the past few decades, Southeast Asia's economy has grown significantly, but there are a myriad of economic, social, and environmental issues and challenges that need to be tackled. The current growth model relies heavily on exploiting natural resources, particularly in the least developed economies. To tackle these challenges, it is important to strengthen public and private institutions, structurally

reform the economic systems, integrate green growth into national policy design and implementation, and steer sustainable growth to ensure the well-being and prosperity of the people (OECD, 2014). In other words, a people-centric approach needs to be adhered to.

Notably, green development is becoming increasingly important as the region faces mounting environmental challenges such as climate change. One of the main benefits of green development in Southeast Asia is the potential to reduce environmental degradation and promote sustainability. Southeast Asia is home to some of the world's most biodiverse ecosystems, but these ecosystems are under threat due to unsustainable practices, such as deforestation and mining. Green development can help to protect these ecosystems by promoting sustainable land use, such as agroforestry and sustainable forestry practices. Another potential benefit of green development in Southeast Asia is the creation of green jobs and economic growth. Green industries, such as renewable energy and sustainable agriculture, have the potential to create new jobs and stimulate economic growth. The growth of green industries can also help to reduce poverty and improve social equity.

However, there are also challenges to implementing green development in Southeast Asia. One of the main challenges is the lack of political will and institutional capacity. Many governments in the region have not yet prioritized environmental sustainability, and there is often a lack of political will to implement green policies. In addition, many countries in the region lack the institutional capacity to enforce environmental regulations and promote sustainable development.

Another challenge is the lack of funding for green development projects. Many green projects require significant investments, and there is often a lack of funding from both public and private sources. In addition, the high costs of green technologies, such as renewable energy, can make it difficult for Southeast Asian countries to transition to a more sustainable economy.

To conclude here, green development is an important concept for Southeast Asia as the region faces significant environmental challenges. Green development has the potential to reduce environmental degradation, promote sustainability, and create new jobs and economic growth. A green recovery from the COVID-19 pandemic could transform Southeast Asia into an "economic powerhouse" (Argyriou, 2020). However, there are also challenges to implementing green development in the region, including the lack of political will and institutional capacity and the lack of funding for green projects. To promote green development in Southeast Asia, governments and international organizations can take several steps to support sustainable practices and invest in green technologies and industries.

6. The ASEAN Green Deal

The ASEAN Green Deal is a framework that aims to promote resilient, inclusive and sustainable development in Southeast Asia. Overall, the ASEAN Green Deal should (a) promote knowledge sharing and capacity building on green economy and green innovation; (b) generate new opportunities from the adoption of green technologies and improved production capacity; (c) generate new opportunities for investment in green infrastructure through public-private partnerships and innovation in new sustainable technologies, as well as new programs for ecological maintenance; (d) promote sustainable production and consumption; and (e) develop integrated and innovative solutions to green economy and green transformation.

Some of the key elements of the ASEAN Green Deal might include (a) green energy, infrastructure, and logistics to support the global transition to a zero-carbon, affordable, and resilient energy system; (b) green production and green consumption by harnessing new technologies and applications that reshape value chains to improve efficiency, lower costs and reduce environmental impact; and (c) green finance platform to mobilize public and private investment for green transformation in energy and infrastructure¹.

The ASEAN Green Deal will contribute to the ASEAN region's transition to a green future, sustainability, resource efficiency, resilience, and economic competitiveness. It covers a wide range of areas, from infrastructure, energy, and manufacturing to consumption and agriculture. The detailed concept and action plan of the ASEAN Green Deal have yet to be developed, but generally it seeks to reduce greenhouse gas emissions, enhance energy security and efficiency, and promote sustainable agriculture and circular economy practices.

Although ASEAN and its member states have shown strong commitment to climate action, the region's emissions are among the fastest growing in the world. Southeast Asia has experienced rapid population growth, urbanization, industrialization, rising income levels, and development gains over the past few decades, resulting in a significant increase in emissions. While region-wide action, especially around initiatives related to renewable energy deployment and energy efficiency, has been increasing, it is essential to transition to clean energy sources to mitigate the impacts of these trends. Without such efforts, energy-related emissions in ASEAN are predicted to nearly double by 2040 (ASEAN Green Future, 2021).

Various ASEAN Member States (AMS) have set targets for reducing greenhouse gas emissions. For instance, Cambodia has committed to reducing 42 percent of emissions below "business as usual (BAU)" levels by 2030. Indonesia

1 Policy inputs from the ASEAN Economic Club submitted to the Cambodian Government on the ASEAN Green Deal (January 2023).

has committed to a 29 percent reduction by 2030, with the possibility of reaching 41 percent depending on international support. Lao PDR has committed to a 60 percent reduction by 2030, while Malaysia has set a target of a 45 percent reduction in carbon intensity by 2030. Thailand has committed to a 40 percent reduction by 2030, conditional on capacity building, finance, and technological support.

The ASEAN Green Deal has the potential to have a significant impact on the region. By promoting sustainable economic growth and reducing greenhouse gas emissions, the deal could contribute to the global effort to mitigate climate change. The promotion of renewable energy and sustainable agriculture could also enhance energy and food security in the region, while the promotion of circular economy practices could reduce waste and promote more sustainable consumption and production patterns.

However, there are challenges to implementing the ASEAN Green Deal. The deal relies on the cooperation of all ASEAN Member States, and some countries may be reluctant to adopt sustainable policies that could impact their economic growth. Additionally, the costs of transitioning to a more sustainable economy could be high, particularly for the less developed economies. At the national level, governments play a crucial role in the implementation of the Green Deal. National governments will need to develop and implement policies to achieve the objectives of the Green Deal.

Local governments also play a crucial role in the implementation of the Green Deal. Local governments can implement policies to reduce emissions from transport, such as promoting the use of bicycles and public transportation and promoting energy-efficient buildings. Local governments can also implement policies to protect biodiversity and promote sustainable land use.

One of the main benefits of domestic politics in promoting green development is the ability to create policies and regulations that incentivize sustainable practices and penalize unsustainable practices. Governments can use a range of policy tools, such as taxes, subsidies, and regulations, to promote sustainable practices. For example, governments can provide subsidies for renewable energy, tax incentives for energy-efficient buildings, and regulations to reduce emissions from industries.

Another benefit of domestic politics in promoting green development is the potential to create green jobs and stimulate economic growth. Green industries, such as renewable energy and sustainable agriculture, have the potential to create new jobs and stimulate economic growth. Governments can create policies and programs to support the growth of green industries, such as providing funding for research and development or creating training programs for workers.

However, there are also challenges to promoting green development through domestic politics. One of the main challenges is the lack of political will and institutional capacity. Governments may not prioritize environmental

sustainability, and there may be resistance from industries and interest groups. In addition, governments may lack the institutional capacity to enforce environmental regulations and promote sustainable development. Another challenge is the potential for policy to be too rigid or inflexible. Policies and regulations can be effective in promoting sustainable practices, but they may also be too rigid and not able to adapt to changing circumstances. In addition, policies and regulations may have unintended consequences, such as creating negative impacts on small businesses or low-income communities.

To address these challenges, governments should prioritize environmental sustainability and invest in green technologies and industries. They can also create policies and regulations that incentivize sustainable practices and penalize unsustainable practices. In addition, governments can engage with businesses, civil society, and communities to ensure that policies and regulations are effective, fair, and equitable. United Nations Economic and Social Commission for Asia and the Pacific (UNESCAP)'s Asia Pacific Green Deal for Business can be integrated in the policy framework of the ASEAN Green Deal.

The following are key areas to focus on for achieving a zero-carbon, affordable, and resilient energy system, in line with the Paris Agreement and the 2030 Agenda for Sustainable Development:

- Advancing the decarbonization of the energy system to provide reliable and affordable access to zero-carbon energy for all.
- Developing smart, low-carbon, water-secure, and climate-resilient cities, infrastructure, and mobility to reduce greenhouse gas emissions and enhance water security.
- Developing blended finance for sustainable investments in energy, infrastructure, logistics, food production, and water supply.
- Accelerating and scaling up innovations in sustainable development by implementing new business models and solutions to reduce environmental impact.
- Transitioning towards a more circular economy to maximize resource efficiency and minimize environmental impact, which will create new business and employment opportunities (UNESCAP, 2022).

To achieve the policy objectives mentioned earlier, several strategic interventions need to be implemented. These include developing green and clean energy system; building green infrastructure and logistics for smart, green, and climate-resilient cities and buildings; promoting green finance to mobilize public and private investment for innovation in the energy sector, infrastructure, logistics, and other sectors of a green and circular economy; promoting green innovation and scaling up new technologies and applications to reduce environmental impact; and promoting a circular economy with new business and employment opportunities (UNESCAP, 2022).

The ASEAN Framework on Circular Economy adopted in 2021 aims to achieve three strategic goals which are to promote resilient economy, resource efficiency, and sustainable growth. There are five strategic priorities, including (a) “standard harmonization and mutual recognition”, (b) “trade openness and trade facilitation”, (c) “innovation, digitalization and emerging technologies”, (d) “sustainable finance and innovative investments”, and (e) “efficient use of energy and other resources”. Additionally, there are six guiding principles, namely (a) encouraging the integration of ASEAN and the development of value chains within the region; (b) taking into account the overall impact on the economy and society; (c) acknowledging the distinctive circumstances of each ASEAN Member State; (d) fostering collaboration across ASEAN for the sharing of knowledge and technology; (e) assessing the financial and institutional viability and sustainability before implementing initiatives; and (f) operating within the context of international production connection (ASEAN Secretariat, 2021). The Circular Economy Framework presents a forward-thinking, far-reaching goal for the circular economy, leveraging the advantages of current ASEAN programs and projects. It also pinpoints crucial areas of emphasis and factors that will facilitate the advancement of a circular economy in ASEAN, with the aim of hastening its realization.

The green transition transformation cannot be realized without significant international support. ASEAN and its member states need financial and technical support from the dialogue partners to develop and implement the ASEAN Green Deal. For instance, ASEAN and the EU have closely cooperated on green economy and climate change. At the national level, it is important to establish a national-level cross-ministerial governance body in order to build political buy-in, build political consensus, formulate a holistic transition strategy, and strengthen inter-agency coordination to implement the ASEAN Green Deal.

The ASEAN region has the potential to expedite the transition to a low-carbon economy and gain significant economic advantages. Leveraging its history of working together to address shared issues, the region can benefit from a coordinated approach to tackle one of the most pressing challenges of our era. By collaborating to recognize and tackle common obstacles, ASEAN can speed up its transition to a low-carbon future and facilitate the establishment of priority projects relating to green and circular economy across the region. ASEAN Member States have identified the establishment of a regional climate fund, carbon market, and power grid as the top three collective opportunities to accelerate the shift to a low-carbon economy (ASEAN Green Future, 2021).

In 2021, the ASEAN Taxonomy on Sustainable Development was introduced as a comprehensive guide for all ASEAN Member States to complement their respective national sustainability initiatives, and to serve as a common language for sustainable finance in the ASEAN region. It will cover capital markets, banking, and insurance sectors and will be multi-tiered and inclusive,

facilitating an orderly and effective transition towards a sustainable ASEAN. The development of the ASEAN Sustainable Finance Taxonomy will benefit all ASEAN Member States, providing an overarching guide that complements their national sustainability initiatives (Keo & Chheang, 2022).

7. Conclusion

The ASEAN Green Deal is an important framework for promoting resilient, inclusive, and sustainable development in Southeast Asia. By promoting clean and renewable energy, circular economy, green production, and green innovation, the deal has the potential to contribute to the global effort to mitigate climate change, while promoting regional transition into a green economy. However, there are challenges to implementing the deal, and it will require the cooperation of all ASEAN Member States and the private sector to achieve its objectives. International support is essential to assist ASEAN and its members to develop and carry out the ASEAN Green Deal.

Five strategic policy interventions are recommended to make the ASEAN Green Deal a reality. First, it is critical to invest in a carbon-neutral, dependable, affordable, and resilient energy system. Second, more emphasis should be placed on developing smart, green, and climate-resilient infrastructure. Third, public and private funds for green investment and transformation should be raised. Fourth, stakeholders should continue to accelerate and scale up green innovation. To implement UNESCAP's Asia Pacific Green Deal and the Framework for Circular Economy for the ASEAN Economic Community, all parties should promote multi-stakeholder consultation and partnership building, as well as develop a green economic zone. In addition to green economy development, we require a holistic approach to wellbeing that integrates economic, social, and environmental solutions.

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Chapter 2: Cambodia's Decarbonization: Climate Change, Forest Conservation, and Renewable Energy

GUPTA Aditya and KEO Piseth*

1. Introduction

“..... At this rate, the planet will soon be pushed past the 1.5-degree limit.

Countries' ambitious new national climate plans – which are due next year – are essential to avert this calamity.....”

The United Nations Secretary-General's, António Guterres, remarks at the UNDP Climate Promise Launch (United Nations [UN], 2024).

The message is not only loud but very clear. Only sincere, concrete and diligent global efforts can limit temperature increase to 1.5°C. The Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report, largely a confirmation of the previous five reports, reiterates the scientific consensus on global warming, greenhouse gases, and the climate change impact. The very recent Conference of Parties (COP) 28's global stocktake on the situation, however, revealed a slow progress across different areas including greenhouse gas emissions, resilience to climate change, and lack of financial & technological support to vulnerable nations (United Nations Climate Change [UNCC], n.d.). As Kennedy said, “First is danger of futility” (Kennedy, 1966, para 32), the global initiatives don't seem to bearing fruit (UNCC, 2022), given the notion that one individual's actions may be insufficient to address the vast variety of global problems, it seems that climate change worldwide activities are not yielding positive results.

The European Green Deal's aims to reach net-zero greenhouse emissions within the European Union (EU) and deliver a pollution-free environment by 2050 (EU-ASEAN, 2022b). Following commitments by other nations such as India, the United States, Japan, and China keep the hopes alive. Cambodia, a nation known for its remarkable resilience, not only in South East Asia but all over the world, has resolved to reduce greenhouse gas emissions by about 42 per cent by 2030 and by 2050, decarbonize its rapidly growing economy.

This chapter discusses the ongoing global debate around climate change in light of decarbonization strategies, forest conservation, and renewable energy. It provides a snapshot of how Cambodia is shaping the discourse through its carbon friendly stance on environment. Since 1996, Cambodia has been a party to the United Nations Framework Convention on Climate Change (UNFCCC) and around 2.3% of its GDP has been allocated in public resources for climate actions and to increase the insistence on renewable energy (Say, 2022).

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The chapter comprises six sections including 1) introduction, 2) climate change, 3) carbon neutrality, 4) forest conservation, 5) renewable energy, 6) environment, social, and governance (ESG) and green investments, and 7) conclusion and policy recommendations.

2. Climate Change

Climate change is real. The overall increase in earth's temperature (global warming) can be attributed to the significant increase in the greenhouse gases¹ due to human activities. These are produced by the burning of fossil fuels², agricultural activities, wastes from production and consumption, refrigeration and air conditioning, and other activities³.

The Climate Change 2023 ('Synthesis Report') brings home the fact that humans have greatly contributed to the global heating for more than 195 years leading to a current temperature increase of 1.1°C above pre-industrial levels. This in-turn has led to hazardous and frequent weather events that are extremely detrimental to the people and the planet (United Nations Environment Programme, 2023). The seven years from 2014-20 rank as the warmest years since 1880 (United Nations Environment Programme, 2023, p. 108). Lakes have shrunk, glaciers have melted, and the sea level has risen. Around 3.3 billion people live in areas vulnerable to climate change (Chatterley et al., 2023). According to the World Health Organization (2023), climate change could cause 250,000 additional deaths per year from malaria, malnutrition, heat stress, and diarrhoea, between 2030-50. Fires, floods, and droughts in various parts are some other consequences of change being felt in different parts of the world. Hans Joachim Schellnhuber, one of the leading climate researchers whilst underscoring the gravity of the problem has warned about a civilisational collapse (Moses, 2020).

The climate change mitigation and adaptation goals of the Association of Southeast Asian Nations (ASEAN) countries are aligned closely with their developmental goals (World Health Organization, 2023). With GDP growth per capita nearing 6% and urbanization in the region growing by about 3.2%, Cambodia's climate change vulnerability as one of high disaster-prone countries globally makes it a priority contender for sustained actions (Climate Change Knowledge Portal, 2021). Between 2011 and 2018, droughts have affected nearly over seven and a half million hectares of land, and floods and storms have

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- 1 These greenhouse gasses include carbon dioxide (CO₂), methane, nitrous oxide, sulphur hexafluoride, hydrofluorocarbons, and perfluorocarbons.
 - 2 Over four-fifths of the world's energy is derived from fossil fuels: natural gas, oil and coal. When exhumed, they release waste gases along with energy that pollute the environment (ADB, 2021, 25).
 - 3 Decomposing plants and animals make Fossil Fuels. Found in Earth's crust, these fuels contain carbon and hydrogen which can be burned to produce energy. Natural gas, coal and oil are examples of fossil fuels. (<https://education.nationalgeographic.org/resource/fossil-fuels/>)

affected nearly a million households (The NGO Forum on Cambodia, 2021). The country is classified in the high-risk category when it comes to riverine flooding⁴, urban floods, coastal floods⁵, landslide, cyclones, extreme heat, and wildfires (Think Hazard, 2020). Moreover, climate change could reduce Cambodia's absolute GDP by 2.5% in 2030 for which it is already working on a National Adaption Plan (National Council for Sustainable Development [NCS], 2020a; 2020b). The identified vulnerable sectors (NCS, 2020a) include i) Agriculture and Water Resources—flooding and droughts contributing about 62% and 36%, respectively, to loss of productivity; ii) Forests⁶—agriculture and fisheries contribute to a quarter of the nation's GDP and employ almost half of the country's labor force (Asian Development Bank, 2021); iii) Coastal Zones—around 30,700 people on average might be affected due to flooding with a rise in sea levels between 2070 and 2100 and about 69,800 additional people might be at risk due to riverine floods; iv) Human Health—increase in the annual temperature (estimated to be about 4.2°C from 1900 to 2100) is likely to affect the vulnerable including children and the elderly population. Other effects of variable precipitation and higher temperatures could be microbial growth, transmission and virulence leading to vector borne and water borne diseases.

There is an immediate need to move from climate procrastination to climate activation⁷. The paranoia surrounding climate change has no doubt prompted countries around the world to come up with sui generis measures. The Committee on Science and Technology, and Plan of Action on Science, Technology and Innovation (PASTI) in the ASEAN region have identified climate change as a key priority area. The consensus is unequivocally in favor of emission reduction and decarbonization. The ASEAN Working Group on Climate Change (AWGCC) is working towards developing a climate change plan promoting co-operation in the region (Asian Development Bank, 2021). Globally, the average carbon footprint⁸ is close to four tons which should ideally drop to under two tons by 2050 (The Nature Conservancy, 2023). Cambodian statistics seem optimistic with per capita emissions much lower at one ton (2019) (World Bank, 2020)⁹. Whilst in

4 Southwest monsoons bring in about three quarters or more of annual rainfall for Cambodia from May to October flooding Mekong River and Tonle Sap Lake region. Approximately 80% of the country's population lives along the Mekong.

5 Sea level rises could affect the country's areas that are low-lying, including settlements, sea ports, beach resorts, coastal, mangrove forests and coastal fisheries.

6 Will be discussed in the next section of the Chapter.

7 Inger Andersen, Under-Secretary-General, United Nations and Executive Director, UN Environment Program.

8 Greenhouse gases including methane and carbon dioxide generated in totality due to human actions.

9 Whilst in the other ASEAN states, energy use due to electricity and heat production (Power Sector) have contributed to a linear increase in CO₂ production, in Cambodia, the emissions are mainly attributable to the Transport sector (ASEAN, 2020).

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3. Decarbonization or Carbon Neutrality

Carbon neutrality is expressed as the balance sought to be achieved between emitting carbon and absorbing carbon emissions by carbon sinks such as forests, soils and oceans. It is a term used interchangeably with 'net zero' which envisages a situation where no carbon is emitted at the very outset. The Paris Agreement was a breakthrough for achieving the common consensus for carbon neutrality to limit global warming to below 2°C (preferably 1.5°C). Signatories to the Accord were required to submit Nationally Determined Contributions (NDC's).

Speaking of Cambodia's resolve towards decarbonization, it has already ratified the United Nations Framework Convention on Climate Change (UNFCCC), acceded to the Kyoto Protocol and subsequently has adopted the Paris Climate Agreement in 2016. It presented a long-term strategy in 2021 to reflect its firm commitment towards the Paris Agreement and to achieve the goal of carbon neutrality in the next 30 years¹⁰. The strategy presents the institutional arrangements for development of a monitoring and evaluation framework, long-term planning, and analysis of socioeconomic impacts and financial options of selected actions that need to be prioritized (NCSD, 2021). Based on rules for reporting and review under the Enhanced Transparency Framework of the Paris Agreement, Cambodia also put in place the NDC tracking tool, which traces the progress towards achieving and implementing its NDC¹¹.

Since the early 2000s, the country has pro-actively been involved in, and has implemented voluntarily, projects for mitigation of Green House Gasses (GHG's) including Angkor Bio Cogen Rice Husk Power Project, a methane fired power generation plant in Samrong Thom, Animal Husbandry Biogas, TTY Cambodia Biogas Project and Kampot Cement Waste Heat Power Generation Project (KCC-WHG)¹². In 2013, the Climate Change Strategic Plan (CCCSP) (2014-2023) was adopted to steer the country towards achieving a green, climate resilient, low-carbon, sustainable, equitable, and knowledge-oriented society. According to the First Biennial Update report to the UNFCCC (2020), Cambodia emitted an average of 163.59 megatons of CO₂-equivalent in 2016, 285 per cent more than that in the year 1994 (NCSD, 2020b). Forest and Other

10 The strategy to be reviewed every 5 years to confirm to the requirement of the Paris Accord as a part of the nation's responsibility to update its NDC.

11 As of 2020, approximately USD 211.57 millions of out of 1,863.27 million (only 11.3%.) have been invested for the implementation of the proposed climate actions, Department of Climate Change 2022.

12 Torii 2010, Ministry of Environment (MoE) 2002).

Land Use (FOLU), representing 80.08 per cent of the total GHG emissions¹³. Deforestation for agricultural plantation and water-logging are the major drivers of GHG emission from FOLU.

Further, the Rectangular Strategy for Growth, Employment, Equity, and Efficiency (2019-2023), and the National Strategic Development Plan (2019-2023) are the guiding policy documents for sustainable and inclusive national development. Strengthening adaptation capacity and resilience to climate change, whilst contributing to the global goal of GHGs reduction have been main goals. Apart from many other policies, the “Long-Term Strategy for Carbon Neutrality (LTS4CN)” endeavours to transform Cambodia into a carbon-neutral economy by 2050.

Cambodia has been a beacon of hope for achieving the global goal of carbon neutrality and has assumed a catalytic role through its increasing emphasis on forest conservation and renewable energy.

4. Forest Conservation

Forests are fundamental components for the planet's recovery (Fothergill et al., 2020). They are nature's carbon sinks and their importance in maintaining the ecological balance is acknowledged from time immemorial. Forests are not only essential for wildlife but people as well. Over 1.6 billion people need forests for food, timber, fuel, shelter and jobs (United Nations Department of Economic and Social Affairs, 2020), most importantly, for oxygen. Forests play an integral role in maintaining the Earth's water cycle. Plant and tree roots keep soil from washing away in heavy rainfall and clogging rivers (National Geographic Kids, 2016, p. 288). World's forests could provide ecosystem services worth 16.2 trillion USD annually (National Geographic Kids, 2016, p. 26). Afforestation could absorb 66 per cent of the carbon emissions in the atmosphere (Fothergill et al., 2020).

A crucial factor responsible for high vulnerability in the ASEAN region in relation to climate change is massive deforestation that leads to negative implications for environmental feedback effects and most importantly local resilience. The ASEAN Joint Statement on Climate Change to COP 28 reaffirms the commitment of the ASEAN Member States to promote sustainable management of forests through Reducing Emissions from Deforestation and Forest Degradation and the role of conservation, sustainable management of forests and enhancement of forest carbon stocks in developing countries (REDD+) (ASEAN, 2023). Due to persistent droughts and climatic disturbances, around 4 million hectares of lowland forests, especially in the southwest and northeast, are likely to be affected which may result in serious repercussions. High deforestation

13 Followed in that order by agriculture (11.25 per cent), energy (5.87 per cent), Waste (1.69 per cent), and Industry (1.11 per cent).

rates, alongside high biodiversity and habitat losses of up to 40 per cent by 2100 (Estoque et al., 2019), may leave the countries and communities in the region bereft of benefits that forests provide, making them vulnerable to variability and natural phenomena (ASEAN, 2020).

Cambodia has lost an average of 132,733 hectares and 579,280 hectares of forest cover every year from 1994-2009 and 2010-2016 respectively (NCSD, 2020b). Relying on sustainable forest management to maintain at least 60 per cent forest cover could massively reduce GHGs from FOLU. Approximately, 41 per cent of Cambodian land was covered by forests in 2016, 97 or more of which were semi-evergreen, deciduous and evergreen forests whilst flooded, mangrove and planted forests covered only a small percentage (Chou & Chinh, 2023; Ministry of Environment, 2020¹⁴).

During 1965-2014, Cambodia's forest cover declined from a mighty 73.04 per cent to 49.48 per cent (Open Development Cambodia, 2022a, p. 14; 2022b, p. 39). This figure remains the highest in terms of forest coverage. Around 9 million hectares of forests are under the jurisdiction of the Ministry of Environment (MoE) and the Ministry of Agriculture, Forestry and Fisheries (MAFF). The Royal Government of Cambodia has also taken a decision to integrate Cambodia's national REDD+ institutional arrangements with the MoE and the NCSD (Royal Government of Cambodia [RGC], 2017).

Large scale reforestation and afforestation efforts are essential to reverse deforestation that contributes to increased CO₂ emissions. Cambodia's REDD+ program is in conformity with the decisions adopted the United Nations Framework Convention on Climate Change (UNFCCC) (RGC, 2017). Both local governments and community participation is vital for the success of REDD+ Program. Once full financed, the program is likely to reduce the deforestation by 50 per cent in 2030 in comparison to the REDD+ baseline set in the updated NDC (reducing 50 per cent of historical emission) (NCSD, 2021).

Cambodia has long since assumed the role of a leader in the ASEAN region when it comes to initiatives concerning community involvement. Community-Based Natural Resource Management (CBNRM) was introduced in Cambodia as early as 1991 as a new system of collectivization aimed to empower local people to manage their own common resources. The co-author has elucidated the successes of the CBNRM and the Sahak Khom in his doctoral thesis and other distinguished works (Piseth, 2017).

From the early 2000s, the government has welcomed and considered the local community as one of the key actors for forest conservation and protection. Currently, there are more than 500 community-based institutions being formed and recognized by the government for forest protection and protected areas

14 Ministry of Environment (MoE) 2020.

management (Open Development Cambodia, 2022a; 2022b). Local participation has proven to be more cost effective, better protects forests, and addresses local issues better than distant government bureaucrats.

While the REDD+ initiative is bearing fruit and the recovery of the forest cover is under way, Cambodia could well be an important participant in the carbon credit scheme in negotiating a fair market rate in the world markets, especially the EU. The Government of Cambodia has selected Everland to market the carbon credits from its REDD+ projects with large global corporations as buyers. The offering yielded a total bid of 15 million tons of emission reductions (Jennifer, 2022).

5. Renewable Energy

The changing geopolitical dynamics following the Ukraine war have tilted the balance in favor of clean energy and the European nations greatly acknowledge its importance. Germany's energy strategy (Energiewende) aims to achieve 80 per cent renewable energy target by 2050. State of the art fossil-free technologies such as green hydrogen, sustainable aviation fuels, fossil-free steel coupled with carbon capture and storage capabilities are set to change the ways in which the markets function and operate.

ASEAN too has identified its key areas- renewable energy diffusion and energy efficiency improvement. Ensuring energy efficiency would entail proper management of advanced coal and gas power plants whilst offering a greater impetus to renewable energy portfolio diversification (i.e., solar, wind, large scale hydroelectricity, geothermal etc.). The ASEAN Power Grid project aims to prioritize rural electrification and ensure energy security, transborder energy interconnection and renewable power penetration in the future - so much so that an aspirational target has been set to increase the share of renewable energy in the ASEAN energy mix to 23% and the total installed power capacity to 35% by 2025 (ASEAN Centre for Energy, 2020).

As Cambodia embarks upon its growth journey, the demand for energy has increased at an unprecedented rate. The per capita electricity consumption has risen around 18 per cent year-on-year during 2010-2019, whilst electricity demand has increased more than four-fold over the same period (Open Development Mekong, 2022). Investments in clean energy projects are extremely important to meet the increased energy requirements to ensure sustained, balanced, and efficient growth. The National Energy Efficiency Policy provides the long-term vision to transform the consumption of energy in Cambodia to create a strong vibrant economy whilst fostering sustainable development. It also envisages the policy, regulatory and sector specific intervention programs and a mechanism to facilitate coordination amongst various governmental agencies.

Extraordinary times call for extraordinary measures. Specific measurable targets to achieve energy efficient goals are the key. The national target for

reducing the total energy consumption (electric and thermal) by at least 19 per cent by 2030, seems ambitious but achievable. In the last decade, between 2040 and 2050, of the Long-Term Strategy for Carbon Neutrality, the power sector will increase its decarbonization efforts by increasing the share of renewable energy in the generation mix, to 35 per cent (12 per cent to come from solar) in 2050 (NCSD, 2021). The Cambodian energy mix in 2030 is projected to have hydropower (55 per cent), coal/gas (35 per cent) and renewable energy (biomass, solar, and wind) (10 per cent)¹⁵. Statistics indicate that 6.7 per cent of the electricity generated in Cambodia comes from solar energy, and around 53 per cent from hydropower¹⁶. As a developing economy with increasing energy requirements, Cambodia imports about 59 per cent of its total energy requirements. Increased capacity for solar, hydropower and other renewable energy projects would augment the existing supply and help in making the nation self-reliant whilst catering to the carbon neutrality goal.

Keeping in mind, Cambodia's rivers and highlands in its geographical makeup, the potential is enormous potential for the better development of hydropower capabilities alongside advanced research and development in renewable technologies.

6. ESG, Green Financing, and Sustainable Financial Inclusion

The focus of investments should be the emerging and developing nations. It is extremely important that ESG compliant companies (or seemingly so) walk the talk rather than just talking the talk. Transformative greenification of businesses is seen with scepticism by some corporations and investors. Profit is certainly not a dirty word. However responsible, businesses are expected to be profitable ones. Green washing may thus be preferred to risk taking. The Securities and Exchange Commission in the United States has adopted rules to standardise and enhance climate related disclosure for investors (US Securities and Exchange Commission, 2024). The European Union's Corporate Sustainability Reporting Directive (CSRD) entered into force on 5 January 2023 aims to bring around 50,000 companies under its ambit (European Parliament, 2022).

Financing forest action is needed to address climate crisis. Green Climate Fund (GCF) has been set up to scale up financing and catalyse investment for sustainable forests and enabling environments needed for countries to take forest action. GCF has been proactive in supporting countries in the implementation of the REDD+ to mainstream the role of forests in climate resilient development processes whilst ensuring benefits for indigenous people. Mobilization of private

15 (Economic Research Institute for ASEAN and East Asia (ERIA) 2019- https://www.eria.org/uploads/media/CAMBODIA_BEP_Fullreport_1.pdf Refer page 18 of the PDF

16 Electricity Authority of Cambodia, 2023- Page 1. https://eac.gov.kh/uploads/salient_feature/english/salient_feature_2022_en.pdf

sector financing is also essential to augment such initiatives. It also ensures that it leverages public funds to derisk equity investments, such as through the Arboro fund. Initiatives such as the Inclusive Green Financing Initiative could also help in strengthening national financial institutions to green wall investments. The GCF portfolio as of 31 December 2023¹⁷ revealed 243 projects were approved with GCF funding of USD 13.5 Billion (Green Climate Fund, 2024). The project titled Public Social Private Partnerships for Ecologically Sound Agriculture and Resilient Livelihood in Northern Tonle Sap Basin (PEARL) accredited by the Food and Agriculture Organization has been approved for Cambodia with total GCF financing of 36.2 million USD (Green Climate Fund, 2023).

The Enhanced Transparency Framework (ETF) under the Paris Accord aims at ensuring the provision of technology transfer, financial and capacity building support by developed nations to the developing ones as a part of their climate adaptation and mitigation initiatives. The EU Green Deal is a shot in the arm for the ASEAN region in supporting a greener sustainable tomorrow. The EU under the ASEAN Green Finance Facility would commit 50 million EUR to encourage private investments in green infrastructure projects in ASEAN Member States (EU-ASEAN, 2022a). Strategic actions for green growth in Cambodia would involve policy level reforms to facilitate public private partnerships and capacity building for resilient, sustainable and environment friendly solutions (RGC, 2013).

Climate mitigation and adaptation requires enough funds to be available for technology development, human capacity building and climate change intervention (RGC, 2013). Green bonds might act as important pillars for clean growth and instruments of sustained financial inclusion in the Cambodian financial markets. They could be backed by the government for public private ESG projects and offer fixed returns to investors with issuers making a non-binding voluntary commitment to utilize the funds for green projects. Around 269.5 billion USD worth of green bonds were issued worldwide in 2020.

Recently, the European Union (EU) has also agreed upon the Carbon Border Adjustment Mechanism levy to equalize the price of carbon paid for EU products and non-EU products. A similar arrangement could be envisaged by considering the introduction of an ASEAN-wide carbon tax or working out an inter-regional mechanism for pricing carbon (Nurdianto & Resosudarmo, 2016). Cambodia has vast reserves of carbon credits estimated to be around 15 million and is an important trading partner of the US and the EU. Initiatives such as establishment of global marketplaces, such as Climate Impact X in Singapore, could go a long way in the monetization of carbon credits. The price that the carbon credits would fetch would be much greater in the developed countries' markets.

The essential elements of legislation such as the Inflation Reduction Act in the US merit consideration for adoption in the Cambodian legislative framework.

17 38th Board Meeting held at Kigali, Rwanda from 4th-7th March, 2024.

The Ministry of Environment in Cambodia is the apex institution actively involved in ensuring the feasibility of these projects by entering a contractual arrangement with the investors (RGC, 1999). It is reflective of the good governance framework ensuring greater engagement between government and the private sector. Empowering businesses in clean capital deployment is a step in the right direction for a nation on the cusp of great economic reforms.

7. Conclusion and Policy Recommendations

In light of the above discussion, it is essential to ponder upon whether we owe the youth a better planet - what David Attenborough calls “a rich, diverse and sustainable future in the world.” (Fothergill et al., 2020).

- We recommend for community-based natural resource management/ gender inclusive climate change responses to ensure that farmlands can turn into wilderness again to restore the ecological balance - an initiative in which Cambodia already has the first mover advantage.
- Embarking upon environmentally sustainable projects for promoting the use of renewable energy using the Public Private Partnership (PPP) model.
- Effective legislative measures could be introduced by the Ministry of Environment and Ministry of Commerce in consultation with the Ministry of Industry, Science, Technology & Innovation to ensure seamless technology transfer with stringent anti-dumping measures in place. With the fast-paced developments waiting to see the light of the day including carbon capture and storage technologies and fusion technology, Cambodia should not remain bereft of these developments.
- Strengthening the existing ESG framework by ensuring accountability mechanisms and penalties for companies indulging in green washing or window dressing the ESG expenditure¹⁸.
- We also recommend that the financial system and the markets should be more resilient and conducive for green projects that aim to greenify every sector. In this regard, the Government could consider issuing green bonds with attractive interest rates to bring citizens in the mainstream¹⁹.

18 The authors have been working towards designing a comprehensive Corporate Governance Code for Cambodia in line with the best practices from the developed jurisdictions.

19 The authors are also in the process of developing an ESG manual which could serve as a ready reckoner in this regard.

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Chapter 3: The European Union Green Deal and Its Implications for Cambodia

*HING Vutha**

1. Introduction

In the past, trade and development policies have prioritized promoting economic growth and eradicating poverty over the long-term health of the environment. However, this strategy has resulted in gradual environmental degradation and resource depletion, both of which pose a threat to the economic growth and poverty reduction that these policies aim to achieve. As the effects of climate change worsen, there is a growing recognition of the need for sustainable development policies that strike a balance between economic growth and environmental protection; consequently, climate change and environmental sustainability appear to be the primary focus of the development strategies of the majority of countries. The emergence of green industries and technologies, which foster not only economic growth but also lessen environmental impact, is a result of the shift towards sustainable development policies.

Furthermore, international programs like the Paris Agreement have pushed nations to cooperate for a more sustainable future. As of March 2023, 127 nations, including the United States, China, and the European Union (EU) — the three largest emitters — have committed to achieving net-zero emissions by 2050 (United Nations, n.d.-b). A concerted effort from all facets of society, including governments, corporations, and individuals, is necessary to achieve this goal of net-zero emissions, which is a significant step in the direction of reducing the effects of climate change. To ensure sustainable development for future generations, it is essential that we keep sustainable development policies as a top priority and continue to invest in green businesses and technologies.

The EU, the third-largest greenhouse gas emitter and responsible for approximately 8% of the world's greenhouse gas emissions (European Parliament, 2018), has made significant efforts to reduce greenhouse gas emissions, particularly through its climate and energy policies. The EU Green Deal (hereinafter referred to as EGD) is one of the most important and integral components of the EU's strategy to implement the 2030 Agenda and the sustainable development goals of the United Nations. The EGD was adopted in December 2019 and became a comprehensive plan to transform the EU's economy and society towards a more sustainable and resilient future while also addressing the urgent need to combat climate change and biodiversity loss. It also aims to create new job opportunities and ensure a just transition for workers in sectors that will be impacted by the transition to a greener economy (Fetting, 2020). The EGD includes a wide range

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of initiatives and policies including increasing energy efficiency and the use of renewable energy, promoting sustainable agriculture, and reducing waste.

This ambitious plan has far-reaching implications not only for EU member states but also for other nations worldwide, particularly the least developed countries (LDCs) like Cambodia. The EGD is anticipated to have a significant impact on Cambodia in areas such as trade, investment, and development cooperation. As the EU works to transition to a low-carbon economy, it is likely to place more of an emphasis on environmentally friendly trade, renewable energy, sustainable production and consumption, and green technologies. This could open up new opportunities for Cambodia to engage in environmentally friendly trade, investment, and development assistance. At the same time, the EGD could also present challenges for Cambodia, particularly in the areas of trade and competitiveness. As the EU makes its rules and standards about the environment stricter, it may be harder for Cambodia to get into the EU market or compete with EU producers. This could significantly impede Cambodia's trade and development. This study aims to explore the implications of the EGD for Cambodia. In particular, it will identify potential opportunities and challenges and provide some suggestions on how Cambodia can take advantage of these opportunities.

2. EU Green Deal—A Comprehensive Plan to Combat Climate Change

In 2019, the European Commission launched the EGD in an effort to create a sustainable and climate-neutral economy in the continent that provides its citizens with a high quality of life while preserving the natural environment. The EGD specifically aims to accomplish a number of important objectives, including raising the EU's emissions reduction target from 40% to at least 55% by 2030 relative to 1990 levels and achieving net-zero emissions by 2050; separating economic growth from resource exploitation; increasing resource efficiency by switching to a clean, circular economy; and halting climate change, reversing biodiversity loss, and lowering pollution.

The European Commission has labelled the EGD as a 'new growth strategy' that aims to transform the EU into a fair and prosperous society, with a modern, resource-efficient and competitive economy (Fetting, 2020). In order to achieve these goals on schedule, the European Commission developed a comprehensive action plan and policy framework that covers several areas, including:

- **Climate action:** The EGD aims to accelerate the EU's efforts to become climate neutral by 2050 by reducing greenhouse gas emissions in sectors such as energy, transportation, buildings, and industry. This includes the use of more renewable energy sources, the promotion of energy conservation, and the adoption of low-emission transportation. By 2030, the EU has set a target to increase the proportion of renewable energy in its energy mix to at least 55% and increase energy efficiency to at least 32%

(European Commission [EC], n.d.-b). The latter objective will necessitate the modernization of buildings, the enhancement of public transportation, and the promotion of more efficient industrial processes. In March of 2020, the European Commission proposed the climate law to ensure that all EU policies contribute to achieving climate neutrality goals. The law establishes a framework for achieving this objective, including the establishment of legally binding reduction targets for greenhouse gas emissions. In addition, the law establishes a European Climate Change Council to provide impartial scientific advice and monitor progress towards the targets. The climate law also mandates that member states develop and implement national plans for climate and energy policies, ensuring a coordinated effort towards achieving the EU's climate objectives. This law is an important step towards a more sustainable future for Europe and the entire world.

- **Clean, affordable and secure energy:** The EU has set a target of producing 32% of its energy from renewable sources by 2030 (EC, n.d.-b) and is investing in clean energy technologies such as wind and solar power. Furthermore, the EU prioritizes energy efficiency while at the same time, ensures that the energy supply needs to be secure and affordable for consumers and businesses. The clean energy transition will require smart infrastructure. In this direction, the EU will foster the deployment of innovative technologies and infrastructure such as smart grids, hydrogen networks or carbon capture, storage and utilization and energy storage.
- **Circular economy:** The circular economy is an essential part of the EGD that aims to reduce waste, increase the use of recycled materials, and promote sustainable production and consumption practices. The circular economy model aims to keep resources in use for as long as possible, minimizing waste and pollution. It also encourages innovation and the development of new technologies that can help achieve more sustainable production and consumption. By promoting a circular economy, the EGD can create new business opportunities and jobs in sustainable industries. The Commission has issued a new 'circular economy action plan' to guide the transition of all sectors on various issues including the sustainable and circular design of products, reducing and reusing materials, measures on the release of plastics, and the packaging (EC, 2020a). The circular economy has the potential to reduce waste significantly and contribute to greenhouse gas emission reduction. Furthermore, the circular economy action plan will help the EU modernize its economy and stimulate the development of lead markets for climate-neutral and circular products in the EU and beyond.
- **Sustainable mobility:** As transport accounts for a quarter of the region's greenhouse gas emissions, the EGD aims to promote sustainable and clean transport by increasing the use of low-emission vehicles, improving public transport, and supporting cycling and walking. This will not only

reduce greenhouse gas emissions but also improve air quality and reduce noise pollution in urban areas. In terms of targets, the EU aims to reduce greenhouse gas emissions from transport by at least 55% by 2030 and by 90% by 2040, compared to 1990 levels (EC, 2021). This will require investments in charging infrastructure for electric vehicles, promoting the use of sustainable fuels, and improving public transport services. Additionally, the EU plans to invest in the research and development of new technologies to further promote sustainable mobility.

- **Sustainable food systems:** The EGD aims to promote sustainable food systems by reducing food waste, promoting sustainable agriculture and fisheries, and encouraging healthier and more sustainable diets. This will require significant changes to the way food is produced, distributed, and consumed. To achieve these goals, the EU plans to provide financial support to farmers who adopt sustainable practices and invest in research to develop new technologies that reduce the environmental impact of agriculture. Furthermore, the EU will work with food industry stakeholders to promote sustainable production and consumption patterns. The Farm to Fork Strategy is at the heart of EU plans that aim to boost sustainable food systems. The strategy sets out both regulatory and non-regulatory measures for sustainable food production, sustainable food processing and distribution, sustainable food consumption and food loss and waste prevention (EC, 2020b). The strategy also includes measures to improve food labelling and increase consumer awareness of the environmental impact of their food choices. Additionally, the strategy aims to promote the adoption of sustainable farming practices and reduce food waste throughout the supply chain. It is anticipated that these actions will not only lead to a more sustainable and resilient food system in the region and benefit the environment but will also support the economic viability of the European agriculture sector.
- **Biodiversity:** The EGD aims to protect and restore ecosystems and biodiversity by preserving natural habitats, reducing pollution, and promoting sustainable agriculture and forestry practices. It also aims to reduce the use of pesticides and fertilizers, as well as to promote organic farming practices. These efforts will help to ensure the long-term health and sustainability of Europe's ecosystems and food systems. The Biodiversity Strategy for 2030 is a key pillar of the EGD that sets out new commitments, measures, targets, and governance mechanisms to achieve the goals. The strategy aims to protect and restore biodiversity, promote the use of sustainable agricultural practices, and reduce pollution to ensure a healthier environment for future generations (United Nations, n.d.-a) It also emphasizes the importance of engaging stakeholders and promoting international cooperation to achieve these goals.

- **Pollution:** The EGD aims to reduce pollution in the EU by implementing stricter regulations on emissions and pollutants, promoting clean technologies, and investing in sustainable infrastructure. This includes initiatives to reduce air and water pollution, promote sustainable waste management, and improve the environmental performance of the industry. The Commission has recently adopted a zero-pollution action plan for air, water and soil that aims to tackle pollution in all forms and to ensure a sustainable and healthy future for EU citizens. The Commission will also propose a chemical strategy for sustainability in order to ensure a toxic-free environment. The strategy includes a range of measures including strengthening chemical legislation, reducing hazardous substances, promoting safe and sustainable chemicals, and supporting research and innovation in the chemical sector.

Figure 3.1 Core Components and Objectives of the EGD



Source: Fetting (2020)

3. Implications of the EU Green Deal for Cambodia

While the EGD is primarily focused on the EU's internal social, economic, and environmental affairs, its impact will be felt beyond European borders. Cambodia, in particular, is likely to be affected in several ways. One of the major implications is that the EGD will present Cambodia with a number of potential opportunities to transition to a more sustainable economy while promoting environmental conservation and social development. More precisely, Cambodia can leverage the EU's commitment to promoting sustainable production and consumption practices such as organic farming and agroforestry by adopting sustainable agricultural practices, which could help the country both reduce

its carbon emissions and attract environmentally conscious consumers and investors. Furthermore, it is expected that the EGD will increase the demand for agricultural goods especially sustainably produced foods. This presents a significant opportunity for Cambodia to tap into the EU market and increase its agricultural exports.

Access to funding is yet another opportunity presented by the EGD. The EU has committed to mobilizing at least €1 trillion in investment over the next decade to support the transition to a more sustainable economy (EC, n.d.-a). This funding will be directed towards projects that aim to reduce greenhouse gas emissions and increase energy efficiency, creating new job opportunities and boosting economic growth in the process. As promoting global cooperation and encouraging other countries to adopt similar measures to combat climate change is part of the EGD's strategic action (Fetting, 2020), the EU is likely to expand the development aids, and further strengthen diplomatic efforts and partnerships involving funding for projects that promote economic growth, job creation, and environmental sustainability. This could potentially lead to increased cooperation between the EU and Cambodia on environmental and sustainable issues. In fact, the EU has already started cooperating with Cambodia on environmental issues, including extending support for sustainable forest management and the promotion of renewable energy. However, the EGD will open up additional avenues for financial and technical assistance, including waste management, renewable energy sources, sustainable agricultural practices, and a circular economy. These efforts will not only assist Cambodia in lowering its carbon footprint and reducing the effects of climate change, but they may also generate new employment opportunities and stimulate economic growth. Further collaborations and partnerships on the issues essential to the EGD could strengthen the relationship between Cambodia and the EU, leading to an increase in trade and investment opportunities in sustainable industries.

Although the EGD offers Cambodia opportunities to further develop its trade, investment, and development partnerships, it also presents Cambodia with many challenges, especially with respect to adhering to EU rules. As part of the EGD, the new Circular Economy Action Plan (CEAP) introduces new regulations, non-regulatory measures, and sustainability standards for imports to accelerate the transition to a circular economy. A few examples of more stringent requirements include sustainable product design with an emphasis on circularity, durability, and repairability, and a production process requiring the use of sustainable materials (EC, 2020a). As these changes may require significant investments in new technologies, as well as changes in business models and supply chain management practices, it is likely to increase costs and responsibilities for businesses exporting into the EU. Apart from that, adhering to these rules might be challenging for Cambodia, and as a result, it could affect its ability to export goods to the EU.

CEAP also imposes a comprehensive set of measures for the textiles industry, the fourth most-pressurized sector for the use of primary raw materials and water and the fifth for greenhouse gas emissions (EC, 2020a). It specifically calls for the application of a sustainable product framework by the producers, along with environmental standards for the collection of textile waste, to guarantee that textile products are suitable for circularity. The implementation of CEAP in Cambodia's textile industry may result in higher production costs, which could reduce their ability to compete in the EU market. However, compliance with these regulations could allow Cambodia to enhance its sustainability practices and attract environmentally conscious consumers.

4. Conclusion

In summary, the EGD is a comprehensive and critical policy initiative that seeks to combat climate change and sustainable development globally. As the EGD underscores the importance of global cooperation and leadership in promoting sustainable development, it represents a significant shift in global trade and development cooperation landscapes. Given that the EU is one of Cambodia's largest trading partners and providers of development assistance, a shift in trade and development financing under the Green Deal framework could have a substantial impact on Cambodia's economy. Cambodia can benefit from the EU's growing demands and support for environmentally friendly products in promoting sustainable agricultural practices and exporting sustainable products to the EU. Cambodia could also benefit from the technical and financial assistance that the EU can provide to promote sustainable development and address the challenges of climate change.

To maximize the benefits of the EGD, Cambodia may need to invest more in green initiatives, such as sustainable production methods, renewable energy, waste management, and recycling systems. In addition, Cambodia may need to collaborate closely with the EU to increase policy coherence. This requires the nation to develop a green growth strategy and align its policies and programs with the EGD. To ensure compliance with the new regulations, Cambodia may need to invest in strengthening its institutional capacity, regulatory framework, and research and development capabilities. At the same time, Cambodian businesses must immediately begin preparing for and adapting to these changes in order to remain competitive in the EU market.

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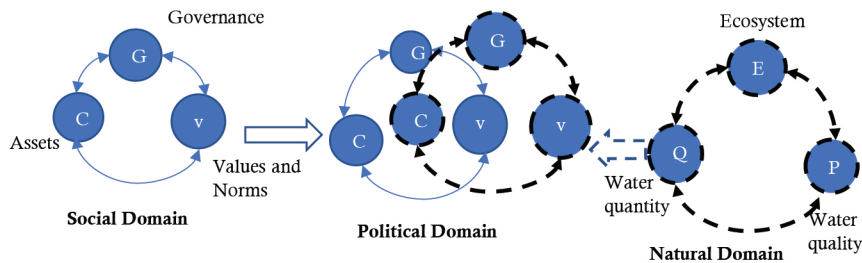
Chapter 4: Connecting Societies Through Water: The Potential of Water Diplomacy in Transboundary River Basins

CHEM Phalla*

1. Introduction

This chapter explores the water diplomacy framework considering the three-dimensional interaction between (1) the natural domain or river nodes, (2) the social domain, and (3) the political domain that creates water problems linking water-user communities from upstream to downstream through hydrological connectivity (Islam & Susskind, 2013).

Figure 4.1 Interactions Between Natural and Social Processes Within a Political Domain



Source: Islam and Susskind (2013)

This paper hypothesizes that to manage the complex water problems involved in transboundary-river basin management, such as the Mekong River basin, riparian states require a water diplomacy framework that would help solve problems through robust cooperation. Dealing with such management involves interactions that are usually unknowable and unpredictable due to many variables, including economic, political, and environmental factors, and requires much feedback. The engineering approach, which water managers have relied on for decades, needs to work better because hydrological processes and social and political boundaries significantly shape water problems.

According to Islam and Susskind (2013), *water networks are interconnected nodes representing natural (ecological), societal, and political variables*. Therefore, information and data sharing between nodes is vital to enable managers to regulate and coordinate the use of shared water resources (Islam & Susskind, 2013). The most challenging aspect of water management as a common pool resource is how best to share the information among riparian stakeholders to formulate a policy that meets the desired outcomes.

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The water diplomacy framework is suitable because water flow connects society upstream and downstream into a complex interaction. Therefore, water problems such as conflicts occur when the three dimensions—natural river nodes, societal, and political systems—interact (Islam & Susskind, 2013). Therefore, water diplomacy can connect scientists and policymakers to address such problems.

In the following sections, the chapter introduces water diplomacy, its importance in water management cooperation, and other aspects such as green economy, sustainable development, and water sector development in more detail.

2. What is Water Diplomacy?

Water diplomacy employs diplomatic tools to resolve or alleviate disputes and conflicts arising from shared water resources to foster cooperation, regional stability, and peace (Schmeier & Shubber, 2018). Diplomatic tools encompass treaties, conventions, agreements, and protocols. A treaty is a formal agreement between cooperating countries, while a convention sets rules for member countries in multilateral cooperation. Water diplomacy tools may involve negotiation, dispute-resolution methods, and diplomatic discussion platforms.

It is crucial to distinguish between scientific tools and water diplomacy instruments. For example, organizing collaborative research trips, which are considered a technical tool like creating comprehensive management plans or shared monitoring systems, does not fall under the realm of water diplomacy.

Water diplomacy is a strategic approach that focuses on resolving disagreements and conflicts over scarce water resources. These conflicts often arise when different user groups have competing interests, potentially destabilizing communities, countries, or regions. Disputes frequently center around differing views on whether water development should prioritize economic gains or mutual benefits, especially in shared water basins like the Mekong River Basin. It is crucial for parties involved to address these disagreements early on to prevent them from escalating into conflicts, highlighting the importance of diplomacy in managing water resources. Water diplomacy is a resourceful tool that promotes peace and stability by fostering diplomatic engagement and cooperation toward common goals.

Water diplomacy has been previously introduced in the Mekong River Basin, with organizations like the Mekong River Commission (MRC) using it to develop strategies. However, there is still disagreement among riparian nations on the term. A common term needs to be integrated into policies and strategies. Managing the Mekong River, a transboundary issue, is complex due to physical, social, and political factors. Water diplomacy has been crucial in promoting cooperation and peacebuilding for sustainable river basin development and management.

The MRC has played an influential role in policy dialogue and cooperation between member states and has engaged with regional partners and stakeholders on transboundary water management. For example, it has intensely engaged

with China and Myanmar, organized a basin-wide ASEAN-Mekong forum, and incorporated academic and public opinions in its planning. This existing mechanism and practice paved the way for a more structural water diplomacy framework (MRC, 2021). The role of the regional coordination of the MRC significantly impacts the future water diplomacy framework in promoting individual riparian members to participate in the regional decision-making process.

3. Why Water Diplomacy?

The Mekong River provides an enriching aquatic biodiversity. It is vital to the social, cultural, environmental, and economic wealth of more than 65 million people in the Basin (Soukhaphon et al., 2021). The riparian states have developed more infrastructure along the Mekong basin in their territories to gain significant benefits. However, such infrastructure development causes adverse effects on local livelihoods and the environment. In addition, water management requires scientifically supported data and politicians' commitments in their evidence-based policymaking to support regional governance decision-making in water sharing and allocation. Governments and regional coordination bodies, such as MRC, can use water diplomacy to talk about how to fix problems that people are worried about because of how climate change and development might affect the Mekong basin's physical, environmental, and ecological processes (Hoanh et al., 2010). However, such impacts lead to a potential disagreement, leading to a regional conflict between upstream and downstream communities, in which stakeholders should use diplomatic tools to mitigate the risks (Kittikhoun & Staubli, 2018).

4. What is a Green Economy?

This chapter argues that development leads to economic benefits but alters river flows and affects the environment. One approach to dealing with environmental degradation is to adopt a green economy—a path of economic growth that promotes a sustainable environment and social inclusion. Nevertheless, the question is how to translate “green development” into real natural resource management. In the case of transboundary river flow, incorporating water diplomacy principles into the green economy and development is critical for achieving a balance between economic growth and environmental sustainability. Water diplomacy also provides a framework for cooperation among riparian states, ensuring that development activities do not disturb the ecological balance of shared water resources.

With limited experience, attempting to achieve a sustainable environment while simultaneously considering economic growth and social harmonization is challenging. From an environmental point of view, “greener” economic development means supporting resource efficiency, greener tourism and markets, and “cleaner” agricultural products (Vuola et al., 2020). Moreover,

the Cambodian government promotes cooperation to achieve sustainable development goals. Cambodia could use this to capitalise on water diplomacy as a tool for facilitating dialogue and negotiations among stakeholders in order to achieve inclusive and sustainable green economy goals throughout the Mekong River Basin.

5. Cambodia's Sustainable Development Goals

Sustainable economic growth must ensure environmental sustainability and social harmonization. Therefore, it requires a multidisciplinary approach and a high level of multistakeholder support for goals, especially Sustainable Development Goal 6 on access to clean water and sanitation and Goal 9 on industry, innovation, and infrastructure.

Preserving good environmental health supports sustainable economic growth. However, a single state cannot solve environmental problems because of the three-dimensional interactions that shape such problems. Therefore, a basin-wide multistakeholder cooperation approach is required (Charoula, 2015). This approach is reinforced by water diplomacy, which facilitates collaboration among riparian nations to address water-related challenges, ensuring that economic development does not undermine the region's environmental and social integrity. In such regional multistakeholder-approach cooperation, states exercise sovereign development policy for sustainable development. Cooperation among states occurs through international treaties and ratification procedures. International cooperation also takes the form of economic and technological cooperation and governance that brings together economic and environmental resolution.

Thus, water diplomacy plays a crucial role in realising Cambodia's SDGs and highlighting Cambodia's commitment to SDGs by integrating sustainable water management into broader development policies and ensuring that economic growth aligns with environmental and social sustainability objectives. The Cambodian Government has a policy on Development Cooperation and Partnership Strategy (DCPS 2014-2018 and now 2019-2023). Through this policy, the government considers cooperation financing for economic development through promoting technological development in every sector of economic and social development. In addition, institutional capacity building in science is part of practical cooperation through promoting bilateral, regional, and multilateral diplomacy. Using the framework of DCPS (2019-23), Cambodia will continue to engage with national and international economic, technological, and scientific institutions at regional and global levels. Such an approach will establish monitoring, evaluation, and reporting systems that enhance results-based development cooperation. Development cooperation in the Cambodian context has progressively taken a result-based approach. The Rectangular Strategy is a comprehensive framework that supports achieving development results. Cambodia has integrated its Sustainable Development Goals framework

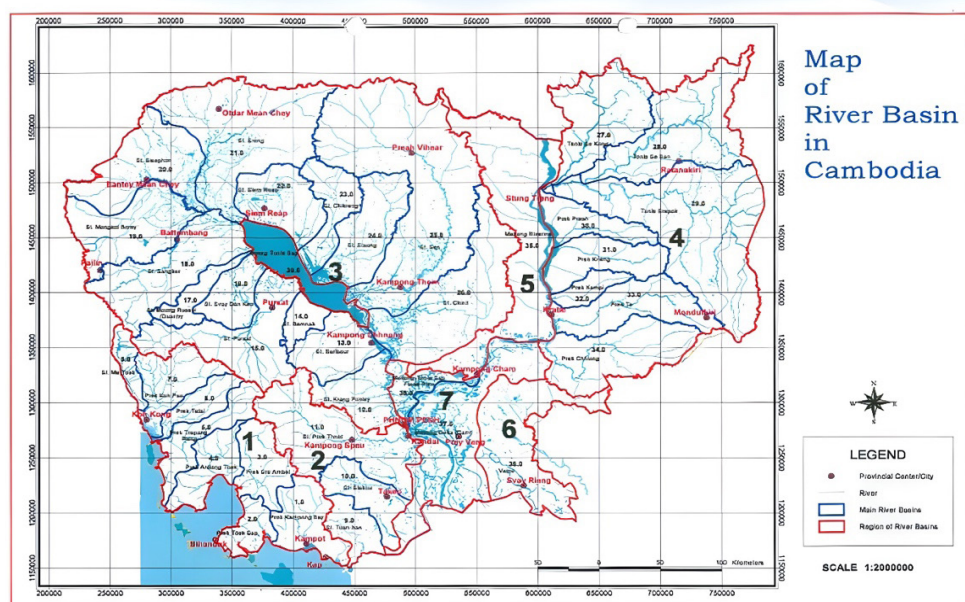
into its National Strategic Development Plan. At the same time, Cambodia also integrates its development cooperation into national budget planning (Cambodian Rehabilitation and Development Board, 2020).

6. Water Sector Development

The source of the Mekong River is the Lancang River on the Tibetan Plateau in China. It flows through several countries, including Myanmar, Thailand, Lao PDR, and Cambodia, before reaching the South China Sea in Vietnam. This river sustains diverse ecosystems, with about 86 percent of Cambodia's inland water draining into the Mekong system, while only 14 percent flows into coastal areas. The annual surface water availability is around 75,000 million m³ per annum, excluding the portion that drains into the coast. Additionally, there is an estimated 17,600 million m³ of groundwater (Ministry of Water Resources and Meteorology, 2019). Cambodia's varied geography supports rich biodiversity and allows for growing different crops. The country's agricultural practices are divided into upland, lowland, and floodplain areas, with the central plain having high population densities, rich ecological diversity, and significant potential for agriculture and fisheries development.

The Tonle Sap Great Lake (known as the 'Great Lake') connects to the Mekong River through the Tonle Sap River, which meets the Mekong River at the Chatomouk confluence in Phnom Penh.

Figure 4.2 General Situation of Cambodia's Rivers, Lakes, and Provinces



Source: Ministry of Water Resources and Meteorology (2011)

Cambodia utilizes its water for domestic water supply, irrigation, drainage, flood protection, recreation, and energy generation (MRC, 2021). The government's vision is to graduate from a low-middle-income to a high-middle-income country by 2030 and a high-income country by 2050. Such a vision requires the government to gain high economic growth. In addition, such economic growth must be environmentally sustainable and socially harmonious. Therefore, the government supported the Sustainable Development Goals (SDGs) at the UN General Assembly in 2015. Since then, Cambodia has adopted the Cambodian SDGs. First, the government provides a strategy and policy intervention to improve Cambodia's competitiveness by promoting SMEs and attracting Foreign Direct Investment (FDI). Secondly, the government fosters Public-Private Partnerships to increase investment projects, including BOT (Build Operate Transfer) in economic infrastructure development. Thirdly, the government continues to improve connectivity to all corners of the country (Ministry of Planning, 2019).

When there is enough water, there are not many social problems. On the contrary, when there is not enough water, competition occurs between upstream and downstream areas of the basin, which is often intense and can lead to conflict. In such a situation, there is often a question: who should get water first, how much, and when? With such a scarce resource, while the country needs economic growth, we must maintain a good environment and social harmonization, nationally and regionally. In such a situation, Islam and Susskind (2013) suggest that a negotiation involving diplomats using a collaborative-adaptive technique would be a water diplomatic tool, ideal for building informed agreements among conflicting interests.

7. A Call to Action in Water Resources Management

The government strives to reduce poverty by diversifying economic activities, increasing social expenditure, promoting public administrative reform, and improving national food security. The water sector is one of the critical prioritized areas contributing to poverty reduction by improving access to safe and adequate water for domestic consumption, irrigation, and industry, tackling and minimizing all forms of threat, loss of life, and livelihoods according to water-related hazards, and sustainably managing water resources. Hence:

- The Government fosters effective and sustainable water management to improve the national economy and social welfare (Ministry of Water Resources and Meteorology, 2007). However, investment in irrigation development remains an urgent priority mode with no basin-wide approach to irrigation management in a catchment management context (Chem & Craig, 2008). Therefore, the Government will consider a scientific approach to water management.

- Remarkably, since 1994, the Government has considered transferring irrigation management responsibility for small-scale irrigation operation and maintenance (O&M) to the Farmer Water User Community (FWUC). However, the FWUC lacks support from the technical departments leaving the FWUC to struggle in performing its O&M roles. FWUC plays a crucial role in sustainable irrigation development by performing its small-scale O&M. Therefore, the Government supports them financially and technically in performing their roles and responding effectively.
- FWUC in downstream areas cannot manage its irrigation schemes in isolation from the upstream areas because the hydrological processes have connected them spatially and temporarily. The Government will manage irrigation schemes following the principle of Integrated Water Resources Management (IWRM¹).
- The Government will facilitate institutional arrangements supporting irrigation management, urban water supply, urban sanitation, rural water supply and sanitation, and ecological conservation. Strategies, including electricity, fisheries, agriculture, forestry, and the environment, are also in place in the form of strategic frameworks and action plans.
- The Government will define policies and develop water resources management and development strategies. Other plans should include Research and Development (R&D) in water conservation, such as flood control and drought warning, gathering and managing data and information about surface and groundwater, and meteorology. Another key role is coordinating international collaboration in the Mekong River Basin.
- The Cambodia National Mekong Committee (CNMC) advises the Cambodian representative to the MRC Council on all matters relating to the Mekong River Basin activities that could affect Cambodian interests. CNMC also examines suggestions that agencies of the Royal Government of Cambodia produced considering the Mekong-1995 Agreement and collaborates between MRC and the appropriate ministries.

8. Conclusion

In the national context, the central plains around Tonle Sap Lake and the Mekong lowland are ideal for irrigation development in Cambodia's geographical condition. However, the above geographical locations pose limitations for suitable, cost-effective large-scale reservoir storage sites, with many suitable only

1 Principles of IWRM include: (1) water supports life, but it is vulnerable to change and pollution; (2) it is, therefore, water management must be effective; (3) women play crucial roles in water management; and (4) water has economic goods.

for shallow reservoirs. Moreover, without significant storage capacity, we cannot keep the abundant water during the wet season for use in the dry season. As a result, irrigation in the dry season in Cambodia is vulnerable to scarcity and shortage.

Addressing water scarcity has become a complex issue due to the complexity of natural, social, and political systems. Therefore, a new approach to water scarcity management must accept the increased interconnectivity and the consequences of the political decision and reject the unquestioned authority of the hierarchical governance structure. Most importantly, water managers must ask themselves, as water is a common pool resource, how to ensure adequate water to support (1) sustainable economic growth that can feed the growing population, (2) social harmonization, and (3) sustainable ecology.

Additionally, there is a connection between the Mekong Basin and the Tonle Sap Lake. Many natural-rainfall phenomena and the Mekong water level contribute to water scarcity, affecting the flow of the Tonle Sap Lake. The flow from upstream directly impacts the Mekong River's water levels and flow, which in turn affects the water supply of Tonle Sap Lake. Therefore, when responding to the question above, we need to consider water diplomacy to solve water issues at every level, especially regional river-basin development. Maintaining the water level in the Mekong River would increase the flow from the Mekong into Tonle Sap Lake. Having enough water level in the Mekong River would provide Cambodia with great potential for agricultural production and fishing industries while maintaining the unique ecosystems of the Tonle Sap Lake.

Globally, there is widespread adoption of the new approach to water management, known as Integrated Water Resources Management (IWRM). However, water professionals generally lack awareness and knowledge of it. Therefore, water professionals and managers must change management techniques by accepting the water diplomacy framework according to the river's hydrological-social-political connectivity. Nevertheless, the approach is standard; for example, at the national level, there is a need to establish a catchment management committee or river basin management committee to take responsibility for coordinated multistakeholder catchment management. However, Cambodia has not yet established such a catchment management committee. Instead, it considers implementing catchment management under the water law and sub-decree on river basin management.

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Chapter 5: Digital Transformation for Environmental Sustainability: Impact and Responsibility

PRUM Sophea and CHHEM Siriwat*

1. Introduction

The advent of Industry 4.0 (IR4.0) has brought about a significant shift in the ways we live, work and interact with each other, largely due to the transformative impact of digital technology. This revolution has led to increased efficiency and productivity across various sectors, resulting in reduced carbon emissions in areas such as agriculture, transport, planning, building, waste management, and public services. It is imperative to acknowledge that our usage of digital technology has a significant impact on the environment, primarily in terms of energy consumption. In this context, it becomes vital to realize the intrinsic relationship between environmental health and human wellbeing. According to the World Health Organization (Sphera's Editorial Team, 2020; World Health Organization, n.d.), around 24 percent of all worldwide deaths can be attributed to environmental factors. Therefore, ensuring access to clean air, fresh water, and safe living conditions free from pollutants and hazards assumes critical importance for promoting human health and quality of life. While digital transformation has the potential to enhance socio-economic progress, it can also impose costs on the environment. Therefore, it is crucial to establish clear roles and responsibilities for various stakeholders to maintain a balance between socio-economic development and environmental sustainability. Moreover, it is important to devise digital transformation diplomacy to ensure this equilibrium.

2. The Impact of Digital Transformation on the Environment

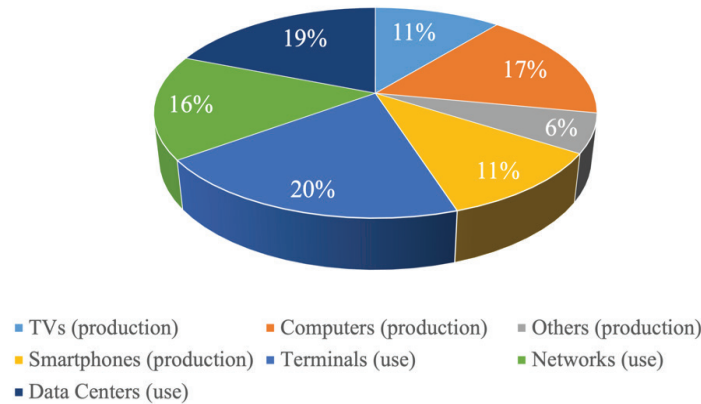
According to recent estimates, the number of active internet users worldwide has reached around 4.66 billion, which accounts for approximately 60 percent of the global population (ESCP Business School, 2021). While digitalization has positively impacted on various aspects of our lives, it is important to recognize that it also comes with certain costs, particularly in terms of energy consumption and greenhouse gas emissions. Recent reports indicate that digital technologies are responsible for emitting approximately 1.6 billion annual tons of greenhouse gasses, accounting for roughly 4 percent of all GHG emissions (The Shift Project, 2019; Efoui-Hess, 2019). Furthermore, this energy consumption is increasing at a rate of 9 percent per year, and it is expected that by 2025, the digital industry will account for 20 percent of the world's total electricity consumption (Ferreboeuf et al., 2021).

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2.1. Impact of Digital Transformation Caused by Data Usage

In a study that was published by Ferrebouef et al. (2018), it was found that 55 percent of energy consumption is attributed to the use of data, which includes activities such as operating data centers, networks, and terminals. On the other hand, the remaining 45 percent of energy consumption is associated with the production of digital devices such as TVs, computers, smartphones, and other devices (see Figure 5.1.).

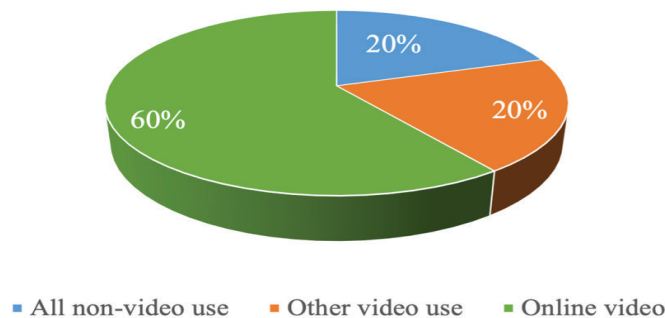
Figure 5.1 Distribution of Energy Consumption Per Digital Workstation for Production and Use in 2017



Source: Ferrebouef et al. (2018)

It is important to emphasize that global data traffic is expanding at a rate of over 25 percent annually, according to Cisco Public (2019). Of this content, video makes up 80 percent of world traffic, with online video accounting for 60 percent and other video comprising the remaining 20 percent (see Figure 5.2.). In 2018, online video viewing alone generated over 300 MtCO₂e, which is roughly equivalent to the carbon footprint of Spain's annual emissions.

Figure 5.2 Distribution of Online Data Flows Between Different Uses of Digital Technologies and of Online Video in 2018 in the World



Source: Cisco Public (2019)

2.2. Impact of Digital Transformation Caused by Manufacturing

The widespread use of digital devices like desktops, laptops, tablets, and smartphones has led to a significant increase in energy consumption. According to a report by Luna (2023) and Cisco (2020), the number of networked devices is expected to reach 29.3 billion by 2023, up from 18.4 billion in 2018, which translates to an average of 3.6 networked devices per person. Furthermore, with the advent of the Internet of Things (IoT), it is predicted that there will be 55.7 billion connected IoT devices by 2025, as per International Data Corporation's forecast.

According to the findings presented in Statista (2023), it has stated that around 1.5 billion smartphones, nearly 126 million desktop computers, and approximately 659 million laptops were sold in 2021. These figures are expected to experience an exponential surge in the next five to ten years due to the emergence of new advanced technologies that are more intelligent and interactive. The extensive production of digital devices has sparked environmental concerns due to their significant carbon footprint, water consumption, and depletion of resources throughout their entire life cycle, from mining to manufacturing processes.

According to Luna (2023), it has been found that the majority of smartphones can accommodate 17 rare-earth elements. However, the process of mining these essential materials is highly energy-intensive and results in a significant environmental impact, affecting forests, mountains, and water sources. Moreover, refining, separating, and purifying these elements is an arduous and resource-consuming task. This procedure necessitates the use of several chemical treatments and purification methods, which frequently cause the degradation of land and water resources.

The process of manufacturing digital devices, particularly semiconductors, consumes a significant amount of energy. Additionally, the production of these devices has a considerable impact on water resources. This is because the creation of complex devices such as smartphones and laptops involve numerous intricate steps, each requiring a certain amount of water. According to a report in Cisco (2020), it has been estimated that the production of a single smartphone, including mining operations, requires a whopping 12,760 liters of water.

3. Designing Responsible Digital Transformation

The environment bears the brunt of digital transformation through two major factors: excessive data usage and electronic device production. Video streaming is the most data-intensive activity, contributing significantly to this issue especially the online video uses. On the other hand, the production of electronic devices consumes a substantial amount of resources, accounting for almost half of the total electric consumption in the process of digital transformation.

3.1. Responsible Data Consumption and Usage

There are two ways to optimize the use of online videos: regulation by technologies and infrastructures, and regulation by content. Regulation by technologies and infrastructures ensures that the network infrastructures and their usage comply with legal requirements. For instance, a subscription model with unlimited data access would not limit the amount of data transferred, whereas offering lower-priced subscriptions with limited access to data transfer volume could reduce data consumption and its impact on the environment. On the other hand, regulation by content focuses on regulating how information is made available to users. One example is implementing regulations for addictive design techniques to minimize the time spent by users on platforms, leading to higher data consumption and transfer.

3.2. Responsible Manufacturing

A study conducted by Statista (2023) suggests that the manufacturing process of electronic devices has a significant impact on the environment compared to their usage. Hence, it is more environmentally friendly to use a device for an extended period rather than producing more devices. Therefore, the act of reusing electronic devices, such as laptops, desktops, or mobile phones, can significantly reduce our environmental footprint. The process of reusing these devices typically involves collecting them through a second-hand agent or a remanufacturer and then selling, renting, or redistributing them to another user. Companies or organizations may carry out larger-scale reuse operations with this aim. In fact, each component in each electronic device has a different lifespan. To maximize the benefits of reusing electronic devices, it is important to consider the reuse potential of each component. Eventually, after a few years, even with component replacements, the device or some of its components will become unusable. At this stage, it is appropriate to recycle the device in order to extract valuable raw materials. Recycling involves breaking down the device and retrieving useful materials from it.

4. Green Artificial Intelligence (AI)

In addition to the primary factors of digital transformation's impact, it is important to consider secondary factors such as distribution, the utilization of complex computational algorithms for fields like AI, and the recycling of electronic devices.

While AI could be viewed as a potential remedy to address the ongoing environmental crisis, it is important to acknowledge that AI and other energy-intensive algorithms consume significant amounts of energy themselves. Hence, it is crucial to factor in the environmental impact when implementing any applications of AI.

In today's digital era, it is important for individuals, industries, and governments to play their part in reducing the environmental impact of digital

transformation while maximizing its benefits. To achieve this, there are various examples of best practices for green digital transformation that can be adopted:

- **Green Digital Transformation Awareness:** Raising awareness about green digital transformation is essential because the digital world may seem abstract to most users, and they may not realize the environmental impact and carbon footprint it creates.
- **Managing Data:** As stated previously, data usage accounts for 55 percent of energy consumption in ICT, making it crucial to manage data storage and transactions efficiently. Organizations and companies should take responsibility for undertaking digital data clean-up activities to eliminate unused and unnecessary digital data. Furthermore, each country should establish policies to regulate data usage, covering technology and infrastructure regulations as well as content regulations.
- **Choose Sustainable Device Providers:** Every electronic device, whether in use or during manufacturing, contributes to the carbon footprint in the environment. Hence, it is crucial to select providers who prioritize promoting a green environment as a means of reducing the environmental impact.
- **Reuse and Repair before Replace:** The environmental impact of manufacturing electronic devices is much greater than their usage. Thus, promoting the reuse and repairability of these devices can significantly extend their average lifetime, leading to reduced emissions and environmental damage by millions of tons.
- **Support Green AI:** It is imperative to not only provide researchers with greater support to develop AI algorithms that consume less energy but also to focus on developing AI algorithms that can solve environmental problems.

5. Conclusion

The fourth industrial revolution heavily relies on technology to advance various sectors. Despite its benefits, technology adversely affects the environment and contributes to carbon footprint. By 2025, the digital industry will account for up to 20 percent of global energy consumption due to data usage and device manufacturing. To reduce this impact, it is essential to adopt best practices such as promoting green digital transformation awareness, efficient data usage and transactions management, sustainable manufacturing processes, extending the average lifetime of devices, and supporting research in green AI.

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Chapter 6: Energy Diplomacy and ASEAN Green Deal

NORN Samnangseymony, SRUN Pagnarith, and THONG Mengdavid*

1. Introduction

Throughout human history, energy, starting from the discovery of fire to electricity and the exploration of green and clean energy, has been strongly associated with human civilization and global modernization. Energy is not only a tool for economic development, but also a strategic and political resource for survival as its development directly impacts national security and wellbeing. As a matter of fact, since the beginning of the Industrial Revolution, people have thought about how to use energy to power their homes, factories, and other devices. Energy is a crucial component of sustainable development and a resource for all living things. Animal and human power have been employed for millennia, and under the impact of the Industrial Revolution, fossil fuels, including coal, oil, and natural gas, started to be exploited. The provision of essential human necessities, including food, water, and illumination, depends heavily on energy resources. The contemporary world depends on resources for everything from agriculture to transportation. The government must be aware of the political factors that cause numerous destructions of energy sources and reserves through wars in order for them to be able to contribute to ensuring energy security. Local administration and community leaders must be given the responsibility of raising public awareness of energy security. Small nations are aware of the important contribution multilateral diplomacy makes to raising their level of involvement and amplifying their voices, thereby leveling the playing field. Smaller states may have less negotiating power with larger governments due to their lesser resources, although this initial disadvantage may be overcome. This can necessitate a dependence on group cohesion and the rule of law, rigorous adherence to constrained goals, and the adoption of innovative solutions.

A number of factors, including geopolitical unrest and war, natural disasters, cybersecurity, the collapse of the energy infrastructure, and fluctuating energy prices, threaten energy security. The energy crisis poses serious problems directly to economic development and energy security failure is a tremendously complicated issue that involves a wide range of various elements and players, yet there are no people to blame for it. The government must seek external partners to advance international cooperation, foster increased conversation and negotiation between nations, and build energy commerce and agreements in order to reduce the dangers associated with energy security. In the face of hardships and global energy market disruption, the European Commission (n.d.a) is setting up the

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REPowerEU Plan, an EU plan to hasten the green transition and lessen reliance on fossil fuels, to help save energy, produce clean energy and diversify its energy supply. The initiatives have safeguarded EU citizens and businesses from energy shortages, supported Ukraine by weakening Russia's war chest, and accelerated the transition to clean energy (European Commission, n.d.b). Samdech Techo Hun Sen, then the prime minister of Cambodia, suggested an "ASEAN Green Deal" during the 55th ASEAN Foreign Ministers' Meeting in 2022, when Cambodia was the Chair of ASEAN. This project would change ASEAN's ability to adapt to challenges and dangers, foster inclusivity and resilience, and support regional transition towards sustainable growth and recovery. The agreement would also support efforts to increase energy investment and development, reduce emissions, and support investments in clean energy (Niseiy, 2022).

With the world's rapid increase in demand for energy and the urgent need to address climate change, energy diplomacy has emerged as an effective means for fostering cooperation, ensuring energy security, and promoting sustainable development. "Energy Diplomacy" is responsible for assisting governments in finding foreign investors for energy infrastructure in developing nations and for promoting the development of innovative energy technology. In its Basic Energy Plan, Cambodia has established numerous goals for each energy-related topic addressed by the plan. The plan intends to meet its energy policy targets and plan for the country's growth in its energy resources, so the goals should be reachable.

2. Definition and Scope of Energy Diplomacy

Energy diplomacy refers to any diplomatic movement intended to upgrade access to energy assets. Energy diplomacy supports a global, clean, and just energy transition to ensure sustainable, secure, and affordable energy, aiming to strengthen energy security, reduce dependency and accelerate an inclusive green energy transition globally. EU energy diplomacy aims to accelerate a global energy transition that is just, inclusive and leaves no one behind, promoting energy efficiency, renewable technologies and well-functioning global markets, amongst other things (Strategic Communications, 2021).

Protecting energy security, which is strongly linked to overall security and prosperity, is the goal of energy diplomacy. The security of the energy supply is a complicated topic with numerous facets. Clean energy from renewable sources is the main emphasis of energy diplomacy because of environmental issues and global warming. The complex energy transition will significantly alter the geopolitics of energy in a variety of ways, including shifting power relations between states and between energy producers and consumers. The globe is connected by globalization and has relied more heavily on fossil fuels for decades. Geographical factors can occasionally influence political decisions. As a result, the term "geopolitics" refers to the field where political, ideological, and economic choices are made based on a country's geography. The population,

climate, physical traits, and natural resources of a nation are all considered in geopolitics. These factors are taken into account when determining national policies and strategies. The diplomatic use of energy has two facets. One is the availability of energy resources and the nation's position in relation to international allies and energy security. On the other side, it may be utilized as a diplomatic tool by influencing governments and politicians to forward a foreign agenda. Clean energy from renewable sources is the main emphasis of energy diplomacy because of environmental issues and global warming.

However, the very intricate institutions and procedures involved in global diplomacy place a burden on their resources. For instance, they can lack the resources needed to garner support for their claims. Despite the restricted resources, this early deficit is frequently avoidable, manageable, and can be converted into a tactical advantage. The fundamental element in ensuring the unhindered and unrestricted development of the nation's energy supply might also be the development of progressive international bilateral ties.

3. Energy Landscape of Cambodia

The energy landscape in Cambodia has been undergoing tremendous changes in recent decades, as the country has been attempting to address its growing energy demand challenges on the one hand and transitioning towards more sustainable and diversified energy sources on the other. The demand for energy has constantly increased as a result of rapid economic expansion, population growth, industrialization, and urbanization, prompting the government to explore various means to ensure a reliable and affordable energy supply. Cambodia has historically been heavily dependent on imported electricity, primarily from neighboring countries such as Thailand and Vietnam. However, the reliance on imported energy has posed many problems, such as price stability and energy security. Understanding these challenges, the government has been working hard to improve domestic energy generation capacity and minimize dependency on imports (Economic Research Institute for ASEAN and East Asia, 2019).

According to the Cambodia Energy Technology Roadmap, published by the Ministry of Industry, Science, Technology & Innovation in 2023, Cambodia's primary energy sources are diesel power plants, biomass (agricultural and firewood), thermal power plants (coal), hydropower plants, and various renewable energy sources. Cambodia's energy landscape experienced a drastic shift from heavily dependent on diesel to diversified sources consisting of about only 10% of diesel, while hydropower, coal, and biomass are consistently leveraged as its primary energy sources (Mika et al., 2021). Even though these trends improve the energy status of Cambodia, the Asian Development Bank (Asian Development Bank, 2018) reports that around 5 million citizens were not able to access grid electricity and still used old methods to generate electricity, including batteries, firewood, and other fuels. Cambodia generated around 73.45% of its energy

domestically and imported around 26.55% from neighboring countries (Thailand, Laos, and Vietnam) including non-renewable energy (48.83%) and renewable energy (51.17%) from local generation. In short, non-renewable energy comprises coal (41.20%) and fuel oil (7.63%); and renewable energy sources ranged from hydropower (44.17%), solar power (6.36%) and biomass power (0.64%) in 2021 (Electricity Authority of Cambodia, 2021).

Cambodia utilizes renewable energy sources due to its favorable conditions. Hydropower, agriculture biomass, wind energy and solar energy are significant sources of renewable energy in Cambodia. Located in the Great Mekong sub-region, hydropower has a great potential for energy production (Poch, 2013). Cambodia is also beginning to look into the potential of its offshore wind energy. The government has initiated studies for offshore wind farms located in coastal areas to explore the potential of this untapped resource. Offshore wind could tremendously increase the domestic energy production capacity and reduce its dependency on fossil fuels as Cambodia's wind has the potential to generate up to 65 gigawatts in the coastal provinces, southwestern mountainous area and southern part of Tonle Sap Lake (Asian Development Bank, 2018).

4. Energy Security and Geopolitics

Geographical factors can occasionally influence political decisions. Geopolitics was initially utilized in relation to military conceptions. Then, the field of international relations also started to undertake geopolitical studies, which helped to better understand the interactions in the region and produce better future policies. Geopolitics in the framework of international relations defines the state economy and energy security. Resource scarcity is a major factor in how different countries approach their energy policies. For instance, between 1949 and 1991, the liberal capitalists, led by the United States, and the social communists, led by the Union of Soviet Socialist Republics (USSR), engaged in a global arms race to forge a bi-polar system of power that would rule the new world order and tip the scales in their favor. As the two blocs increasingly established their own spheres of influence and divided the world into two antagonistic blocs, forcing other nations and regions to pick a side between these two camps, their competition had a significant impact on the global political scene.

Southeast Asia has always been a region of great power struggle, according to Percival's (2007) argument, because of its wealth of natural resources and advantageous position. In order to better promote its interests during the Cold War, the United States thought it could control supplies from the Communist government that were more resilient in terms of their military and economy if it had political authority over this region. It was seen that superpowers like the United States (USA) and the Union of Soviet Socialist Republics (USSR) pushed towards western Asian states that were rich in energy resources during the Cold War, forming their strategies on this matter and using these nations to suit their demands.

From a realist perspective, it is possible to describe how superpowers will act in this situation. As a result, the game of bartering over the allocation of scarce interstate resources is played. According to Joseph Nye (2002), who had a similar viewpoint, “resource bargaining” was crucial in determining the regional energy supply. In other words, during the Cold War, the realism-based idea of military power started to be challenged, and under the influence of neorealism, the idea of military power, as well as rivalry, alliance relationships, and collective security, came to the fore. Energy security has unquestionably become significant in this situation (Mohapatra, 2016). The recent invasion of Ukraine by Russia in February 2022 had a significant impact on the world energy markets (Thomson, 2022). The International Energy Agency’s World Energy Outlook 2022 report explains the key changes to the energy sector following the invasion, including higher energy prices, altered trade flows and supply shortages, changes in energy policy, economic impacts of the energy crisis, impacts of emissions, and more pushes towards renewable energy sources.

5. Climate Change and Renewable Energy

Climate change poses significant challenges to Cambodia, as the country is considered to be among the most vulnerable countries in the world due to its geographic location and socio-economic factors. In recent years, Cambodia has recognized the importance of addressing climate change and has actively pursued renewable energy as a key component of its mitigation and adaptation strategies. For example, the National Strategic Development Plan 2019-2023 encourages the expansion of, and investment in, its capacity to generate electricity especially from renewable and clean energy sources (Royal Government of Cambodia [RGC], 2019). Cambodia’s Sustainable Development Goal aims to achieve the set goals under the UN’s 2030 Agenda (RGC, 2018). Cambodia is experiencing the effects of climate change, including rising temperatures, changing rainfall patterns, and increased frequency and severity of extreme weather events such as floods and droughts, which seriously impact agriculture, water resources, and food security, all of which are very critical for the Cambodian population, especially those in rural areas.

Cambodia has strived to be a low-carbon economy by boosting renewable energy sources in order to reduce greenhouse gas emissions which contribute to climate change. The increase in hydropower is a significant development in Cambodia’s energy environment. The country is endowed with numerous rivers and streams which have potential for hydropower, which offers great opportunities for renewable energy generation (Poch, 2013). The government has made use of this potential by building many large-scale hydropower plants. These projects have contributed to increasing domestic energy production, producing clean energy, reducing reliance on imported energy, and providing electricity to previously underserved rural areas. In addition to hydropower, Cambodia is also

exploring other renewable energy sources. Solar energy also gains popularity in Cambodia. The country has abundant sunlight, making solar power an attractive renewable energy option in addition to the previous energy option, if not entirely replacing oil. In recent years, solar farms and rooftop installations have been implemented, particularly in rural areas with limited access to the national grid. These solar initiatives provide clean electricity for households, schools, and healthcare centers, improving energy access and reducing dependence on traditional energy. Furthermore, Cambodia is exploring the potential of other renewable energy sources, including wind and biomass.

Realizing that we live in an era of unprecedented scientific and technological advancement, where innovative solutions have the potential to tackle energy challenges and reshape the energy landscape, the Royal Government of Cambodia has rolled out the National Research Agenda (NRA) 2025 to utilize research and development to pursue national economic interests and overcome the challenges the nation has faced through building a vibrant research ecosystem to strengthen science and technology capability and, in turn, leverage national innovation performance. NRA has prioritized eight research missions that Cambodia has to achieve by 2030, three of which are related to energy, which are “Reliable Energy Supply: 90 percent of energy consumption is generated locally”; “Electricity and potable water: All Cambodians have access to reliable electricity and safe potable water”; and “Carbon neutrality: Cambodia becomes a carbon neutral country” (RGC, 2023b, p. VI). To achieve these eight missions, four policy priorities have been inaugurated including investing in research to support the eight research missions, strengthening the role and capacities of research institutions, supporting research careers, and incentivizing research activities and collaboration. Moreover, the Energy Technology Roadmap has been rolled out as a comprehensive strategic actionable plan navigating the efforts toward the dynamic landscape of energy technology in the government’s quest to achieve energy security and sustainable development (RGC, 2023a).

In addition to mitigating greenhouse gas emissions, Cambodia is also focusing on building resilience and adapting to the impacts of climate change. To support the development of renewable energy, the Cambodian government has implemented various policies and incentives. International cooperation and partnerships have also been established to facilitate knowledge sharing, capacity building, and technology transfer in the renewable energy sector.

6. Current Landscape of Regional Frameworks on Energy

In the globalization era, regional cooperation is more important than ever to ensuring national security and promoting sustainable development to drive positive change on a regional and global scale. Hence, ASEAN has set ambitious targets for renewable energy in the Total Primary Energy Supply (TPES) and installed power capacity, aiming for a 23% share of renewable energy in TPES

and a 35% share in installed power capacity (Safrina, 2023). However, each ASEAN member state has its own clean energy targets, which vary based on their capacity, capability, and performance. Considering the diverse energy landscapes, economic conditions, and institutional capacities within ASEAN, it is natural for each member state to establish their own clean energy targets that align with their specific circumstances, resources, and development priorities.

Within ASEAN, Cambodia stands as an active participant in cross-border energy trade with neighboring countries, such as Laos, Thailand, and Vietnam. Cambodia has set its own ambitious renewable energy target, aiming to achieve a 25% share of renewable energy in its energy mix by 2030. To accelerate the transition, Cambodia has set a specific objective to install a minimum of 150 MW of solar capacity by 2023. Furthermore, Cambodia is actively pursuing waste-to-energy projects, focusing on the treatment of municipal solid waste (MSW). In terms of energy efficiency, Cambodia has identified several measures, including targets for reducing electricity consumption, strategies to mitigate the urban heat island effect (UHIE), the adoption of Minimum Energy Performance Standards (MEPS), and the implementation of energy labeling schemes (ASEAN Centre for Energy, 2023).

Apropos of the ASEAN Plan of Action for Energy Cooperation (APAEC) 2016-2025, Phase II (2021-2025)¹, which was adopted by the 38th ASEAN Senior Officials Meeting on Energy (SOME) and endorsed by the 38th ASEAN Ministers on Energy Meeting (AMEM), there are specific program areas of energy cooperation at the regional level, including (ASEAN Centre for Energy, 2020):

- ASEAN Power Grid: This program focuses on enhancing regional power grid interconnections to facilitate electricity sharing and trading among ASEAN member states, promoting energy security and supporting the integration of renewable energy sources into the grid.
- Trans-ASEAN Gas Pipeline: This program aims to develop a transnational natural gas pipeline network within ASEAN to ensure the reliable supply and efficient transportation of natural gas across the region.

1 In accord with ASEAN Centre for Energy (2020), the ASEAN Plan of Action for Energy Cooperation (APAEC) 2016-2025 is a dynamic and evolving document that provides guidance for implementing strategies and action plans aimed at accelerating the energy transition and enhancing energy resilience in ASEAN. It seeks to achieve energy security, accessibility, affordability, and sustainability for all within the framework of the ASEAN Economic Community (AEC). Serving as a continuation of APAEC Phase I: 2016-2020, APAEC Phase II: 2021-2025 incorporates comprehensive strategies to address the challenges of ASEAN's energy transition and resilience, taking into account the evolving global energy trends. The development of the APAEC Phase I for the years 2026-2035 is scheduled to commence in 2023, with careful consideration given to the implementation and advancements made during the APAEC Phase II: 2021-2025. This process will involve a comprehensive assessment of the various aspects of the APAEC Phase II to ensure the seamless continuation and progress of ASEAN's energy cooperation efforts.

- **Coal and Clean Coal Technology:** This program focuses on promoting the adoption of clean coal technologies, including high-efficiency and low-emission coal power generation, as well as carbon capture, utilization, and storage (CCUS) technologies, to reduce the environmental impact of coal-based energy production.
- **Energy Efficiency and Conservation:** This program aims to promote energy efficiency measures and conservation practices across various sectors to reduce energy consumption and improve energy efficiency in ASEAN member states.
- **Renewable Energy:** This program focuses on promoting the development and utilization of renewable energy sources such as solar, wind, hydro, and biomass to increase the share of renewable energy in the overall energy mix of ASEAN.
- **Regional Energy Policy and Planning:** This program aims to enhance regional energy policy coordination and cooperation, as well as to develop common approaches and frameworks for energy planning and management within ASEAN.
- **Civilian Nuclear Energy:** This program focuses on exploring the potential use of civilian nuclear energy in ASEAN member states, including research and development, capacity building, and the establishment of necessary regulatory frameworks and safeguards.

At the sub-regional level, individual ASEAN member states engage in cooperative efforts with both regional and non-regional countries, fostering partnerships and collaboration, namely: Laos-Thailand-Malaysia-Singapore Power Integration Project (LTMS-PIP); Brunei Darussalam-Indonesia-Malaysia-Philippines East ASEAN Growth Area (BIMP-EAGA); Greater Mekong Subregion (GMS); Indonesia-Malaysia-Thailand Growth Triangle (IMT-GT); Bangladesh, China, India, and Myanmar (BCIM); and Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation (BIMSTEC), to name a few.

7. Cambodia and ASEAN Green Deal

The Green Deal concept has emerged as a tool, adopted by the EU Commission, for climate, energy, transport and taxation policies to reduce the net emissions of climate change. The introduction of the 'ASEAN Green Deal' concept took place during the Opening Ceremony of the 55th ASEAN Foreign Ministers' Meeting and Related Meetings on August 3, 2022, during which Prime Minister Hun Sen presented two initiatives for ASEAN leaders to endorse. In addition to suggesting the creation of the Regional Comprehensive Economic Partnership (RCEP) Secretariat, he also proposed the establishment of a comprehensive

framework known as the ‘ASEAN Green Deal’. This newly-coined initiative, according to the Prime Minister, aims to facilitate a gradual transition towards a sustainable, resource-efficient, resilient, and competitive green economy in the region, which encompasses various sectors, including infrastructure, energy, manufacturing, consumption, agriculture, transport, environment, and finance. It would rely on technology, innovation, and circularity as enabling factors (Hun, 2022; Kunthear & Sochan, 2022).

One of the key objectives of the ASEAN Green Deal is to build new sources of competitiveness within the ASEAN Economic Community by 2030. This includes strengthening economic resilience and promoting inclusive economic community building. Additionally, the establishment of a highly innovative digital ASEAN Economic Community is envisioned, which will empower citizens, entrepreneurs, and businesses, and enhance ASEAN’s competitive edge in terms of skilled labor, professionals, enterprises, and industries. This will be achieved through continuous improvement of digital infrastructure, connectivity, and bridging the digital divide, as well as creating a secure and trustworthy digital space and services. The adoption of digital and innovative technologies, along with the promotion of circularity in business models, will support ASEAN’s drive towards a more sustainable, greener, carbon-neutral, and circular regional economy.

The ASEAN Green Deal also aims to strengthen the resilience of ASEAN’s economy by building adaptive capacity to manage systemic threats and emerging challenges, particularly those related to climate change, biodiversity loss, and health emergencies. This will involve mainstreaming resilience across various elements and ecosystems of the ASEAN Economic Community, such as food security, energy security, sustainable housing, clean air, macroeconomic stability, and economic security. Effective governance, both through regulatory practice and transparency, will be promoted, along with public and private investment to stimulate employment, income growth, and the creation of green jobs. Furthermore, a cohesive and enabling ecosystem and regulatory environment will provide the necessary financial, investment, infrastructure, and innovation support for the deployment of environmentally sustainable projects across sectors. The adoption of, and investment in, innovative and climate-friendly technologies will also be encouraged to drive sustainable green growth, integrate circular economy concepts, and improve the quality and efficiency of ASEAN’s urban and rural areas.

To achieve its objectives, the ASEAN Green Deal emphasizes green development and investment across various economic sectors, leveraging technology, innovation, and circularity. Drawing from the European Green Deal’s success, the ASEAN Green Deal seeks to reduce greenhouse gas emissions by at least 55% and increase the share of renewable energy by over 32% by 2030. Additionally, improving energy efficiency by at least 32.5% is a key target.

Similar to the European Green Deal, the ASEAN Green Deal proposes new legislation and funding mechanisms to support the transition to a green economy (Chheang, 2023).

Prime Minister Hun Sen reiterated his commitment to the ASEAN Green Deal during subsequent events. At the Opening Ceremony of the 2022 ASEAN Leadership and Partnership Forum on September 12, 2022, he expressed his intention to host the inaugural Green Economic Zone (GEZ) with the enactment of a new investment law as part of the ASEAN Green Deal framework. The Prime Minister further reiterated the importance of the ASEAN Green Deal at the opening of the 54th ASEAN Economic Ministers' Meeting and Related Meetings on September 15, 2022 and the ASEAN Business and Investment Summit (ABIS) 2022 on November 13, 2022, demonstrating his consistent emphasis on promoting the green agenda and its integration into ASEAN's economic strategies. At the latest, this initiative was raised at the Opening Ceremony of the 43rd ASEAN Inter-Parliamentary Assembly (AIPA) on November 21, 2022.

Albeit his consistent advocacy in various high-level meetings and forums, the ASEAN Green Deal remained in the early stages of development during Cambodia's ASEAN chairmanship. Regrettably, no tangible progress or concrete actions were taken to advance the initiative.

8. Challenges and Opportunities

During Cambodia's chairmanship of ASEAN, there appeared to be a lack of enthusiasm among member states for supporting the proposal of the ASEAN Green Deal. Despite the efforts made by Cambodia to promote the initiative, the response from other countries within the ASEAN community was relatively hesitant and reserved. The lack of tangible progress can be attributed to a combination of factors, primarily centered around issues of ownership and financial responsibility.

One major hurdle that hindered the advancement of the ASEAN Green Deal was the absence of clear ownership amongst ASEAN member states. Without a consensus on which country or countries would take the lead in developing and implementing the initiative, progress was stymied. Additionally, the question of financial resources emerged as a critical obstacle. The successful realization of any ambitious initiative requires substantial funding. The lack of commitment from member states to allocate the necessary resources further impeded the advancement of the ASEAN Green Deal.

It is important to note that the challenges faced by the ASEAN Green Deal were not driven by geopolitical considerations but rather by practical concerns related to ownership and financial burden-sharing. Unlike some initiatives, such as Singapore's proposal of the ASEAN Smart Cities Network (ASCN) during

their chairmanship of ASEAN, which have seen significant support from the proposed government through financial investments and the hosting of events and meetings, the ASEAN Green Deal's case lacks similar commitments. Leadership in multilateralism is often determined by a country's willingness to invest financially and create platforms for other nations to join and support the initiative. The absence of a strong financial commitment from member states limited the potential for leadership in driving the initiative forward. While the idea of the ASEAN Green Deal held promise, its progress was hindered by practical challenges that needed to be addressed for there to be any future efforts in this direction.

Despite the challenges and lack of progress in the development of the ASEAN Green Deal, there are still potential opportunities for advancing the initiative. These opportunities can be explored to overcome the existing barriers and pave the way for a more sustainable and resilient ASEAN community. Some potential opportunities include:

- **Enhanced collaboration and coordination:** Member states can seize the opportunity to strengthen cooperation and coordination within ASEAN to advance the goals of the ASEAN Green Deal. By sharing best practices, exchanging knowledge, and pooling resources, member states can collectively address the challenges and work towards a sustainable and green future.
- **Leveraging existing regional frameworks:** ASEAN can leverage existing regional frameworks and initiatives related to energy, sustainability, and climate change to support the ASEAN Green Deal. By aligning the goals and actions of the Green Deal with other regional initiatives, such as the ASEAN Plan of Action for Energy Cooperation (APAEC) or the ASEAN Framework on Sustainable Development, synergies can be created to maximize impact and efficiency.
- **Engaging with international partners:** ASEAN can explore partnerships and collaborations with international organizations, including the European Union (EU), development agencies, and other countries that have experience and expertise in green initiatives. For example, during the 3rd ASEAN-EU Dialogue on Sustainable Development in November 2021, the Team Europe Green Initiative collaborated with ASEAN to establish a work program aimed at fostering a greener future. To support this initiative, the EU has allocated an initial grant of \$32.2 million, which will be utilized alongside the European Green Deal and the ASEAN Community Vision 2025. Additionally, the EU has pledged \$53.68 million to the ASEAN Catalytic Green Finance Facility, encouraging private investments in green infrastructure projects across ASEAN member states (Chheang, 2023).

- **Mobilizing financial resources:** Member states can explore innovative financing mechanisms, such as green bonds, public-private partnerships, or dedicated funds, to mobilize the necessary financial resources for implementing the ASEAN Green Deal. Also, harmonizing national targets and implementing supportive policies, regulatory frameworks, and financial assistance mechanisms, such as Build-Operate-Transfer (BOT), Public-Private Partnership (PPP) and Feed-in Tariff (FiT), are crucial for attracting private investments in clean energy within ASEAN (Safrina, 2023). The active pursuit of funding and investments from international governments and organizations is vital, acknowledging the significant role of international financing in achieving national and regional energy targets.
- **Promoting capacity building and technology transfer:** Capacity building and technology transfer play crucial roles in enabling the implementation of sustainable practices and technologies. ASEAN can facilitate knowledge sharing, training programs, and technology transfer initiatives among member states to build the necessary skills and capabilities for implementing the ASEAN Green Deal.
- **Public awareness and stakeholder engagement:** Member states can actively engage with various stakeholders, including businesses, civil society organizations, and the general public, to raise awareness and garner support for the ASEAN Green Deal. By fostering a sense of ownership and promoting public participation, member states can create a conducive environment for the successful implementation of the initiative.

9. Policy Recommendations

As the government strives to achieve energy security and transition to green and sustainable energy sources, bold and transformative action to promote sustainable development and regional cooperation are indispensable. Hence, here are some recommendations to obtain the goals:

- **Develop a Sustainable Energy Transition Plan:** Cambodia should formulate a thorough, long-term energy transition strategy with precise objectives for transitioning to a sustainable energy landscape. Strategies for eliminating high-carbon sources, increasing the utilization of renewable energy sources, and incorporating energy storage technologies should all be part of this plan.
- **Enhance Energy Governance and Regulatory Frameworks:** Energy governance and regulatory frameworks are critical to promoting transparency, accountability, and efficiency in the energy sector. It can create a stable and conducive environment for investments, encourage competition, ensure consumer protection, and enable the transition to a cleaner and more sustainable energy ecosystem.

- **Develop a Comprehensive Energy Diplomacy Strategy:** Developing a comprehensive energy diplomacy strategy requires understanding the holistic view of a country's energy agenda, geopolitical factors, and global energy landscapes and trends. This plan shall cover a range of spectrums including energy security, economic interests, environmental sustainability, and diplomatic engagements. A comprehensive energy diplomacy plan should be flexible and responsive to shifting geopolitical conditions, developing technical landscapes, and changing energy demands.
- **Strengthen Bilateral and Multilateral on Green Deal and Energy Partnerships:** Cambodia should actively engage in bilateral and regional energy cooperation initiatives, such as the Association of Southeast Asian Nations (ASEAN) and the Greater Mekong Subregion (GMS) frameworks. Collaborative efforts can lead to the development of cross-border energy infrastructure, interconnections, technology transfer, joint energy projects, trading, collaborative research, fostering regional stability and economic integration.
- **Promote Green Deal and Energy Diplomacy and Dialogue:** Cambodia should actively participate in regional and international energy dialogues, forums, and negotiations. Engaging in energy diplomacy can enhance the country's diplomatic relations, foster knowledge sharing, and enable Cambodia to contribute to global energy governance.
- **Invest in Renewable Energy Research and Development:** Investing in research and development (R&D) for clean energy technologies, energy storage systems, and innovative solutions shall be prioritized. Collaboration with research institutions, academia, and the private sector to drive scientific and technological advancements and enhance innovation performance in the energy sector. Promotion of the commercialization of indigenous technologies and supporting startups and entrepreneurship in the energy sector shall also be considered.
- **Enhance Cross-Border Green Deals and Energy Trade:** Facilitating cross-border energy trade with neighboring countries and in the region can provide Cambodia with access to diverse energy resources and foster energy security. The country can explore opportunities for importing electricity or natural gas from neighboring countries with surplus production capacity.
- **Promote Capacity Building and Knowledge Exchange:** Cambodia should invest in building institutional capacity and expertise in energy planning, regulation, and policymaking. This can be achieved through training programs, knowledge exchange platforms, and partnerships with international organizations and developed countries in the energy sector. These policy recommendations aim to guide Cambodia towards a sustainable, diversified and dynamic energy landscape and energy security

while fostering regional cooperation and energy diplomacy. Implementing these strategies can help Cambodia address its energy challenges and leverage its energy resources for economic growth and prosperity.

10. Conclusion

Energy diplomacy and the green deal play a crucial role in addressing energy security challenges and achieving sustainable development. The global community is confronted with urgent threats of climate change and environmental degradation, necessitating a shift toward a modern, efficient, and competitive economy. Energy diplomacy serves as a vital tool in fostering cooperation, ensuring energy security, and driving sustainable development through international partnerships and agreements. Energy security faces risks from geopolitical unrest, natural disasters, cybersecurity threats, and volatile energy prices. To mitigate these risks, governments must engage in energy diplomacy to establish cooperation, facilitate dialogue, and create energy agreements. Collaborative efforts among nations can reduce vulnerabilities and establish a stable and resilient energy landscape.

The ASEAN Green Deal, as proposed during the ASEAN Foreign Ministers' Meeting, offers a comprehensive approach to address energy and environmental challenges. It aims to promote inclusivity, resilience, and sustainable growth in the ASEAN region by encouraging investments in clean energy and emission reductions. The Green Deal not only addresses the urgent need for energy transition but also supports renewable energy development, energy efficiency, and sustainable practices.

Energy diplomacy and the green deal are also essential for achieving the United Nations' Sustainable Development Goals (SDGs). By ensuring access to clean and affordable energy for all, countries can foster economic development, improve living conditions, and protect the environment. Collaborative initiatives like the UN-Energy partnership and the ASEAN Plan of Action for Energy Cooperation (APAEC) provide frameworks for cooperation and progress towards these goals.

Looking ahead, investments in technological advancements, energy infrastructure, and energy-efficient practices are critical. Governments, local administrations, and community leaders should raise awareness about energy security and engage in multilateral diplomacy to amplify their voices and level the playing field. By embracing innovative solutions and upholding the rule of law, smaller states can effectively contribute to international energy cooperation and sustainable development. In the context of the ASEAN region, the involvement of member states in the ASEAN Plan of Action for Energy Cooperation (APAEC) and ASEAN Green Deal represents a form of energy diplomacy. Collaborative efforts and cooperation with various stakeholders can help achieve the energy goals set at the regional level. While challenges, such

as unclear ownership and financial responsibility, need to be addressed, there are opportunities to be capitalized upon. Enhanced collaboration, leveraging existing regional frameworks, engaging with international partners, mobilizing financial resources, promoting capacity building and technology transfer, and fostering public awareness and stakeholder engagement can drive the progress of the ASEAN Green Deal.

The ASEAN Green Deal holds significant promise for shaping the future of Southeast Asia. It aligns with the global push for sustainability, drawing inspiration from successful initiatives like the European Green Deal. By fostering regional cooperation and commitment to environmental goals, the ASEAN Green Deal has the potential to drive meaningful change, promoting a greener, more resilient, and economically competitive Southeast Asia.

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Chapter 7: Science, Technology, and Innovation: Sustainable Industrial Development

*HUL Seingheng**

1. Sustainability of Industrial Development

The United Nations has adopted in principle the concept of sustainable development, which relies on three pillars, namely environment, society, and economy. The concept has faced challenges in achieving the ideal universal goal of sustainability. It has been observed that balancing the three pillars for common goods has not been met due to the fact that most nations value or materialize the concept in different forms based on national and/or aligned interests or preferences. As a result, realizing the goals is seen as challenging due to the fact that the environment and society are commonly left behind. It is important to note that human society and the natural environment are common environments in which people perceive no self-belonging in taking care. This falls under the concept of the tragedy of the commons, in which overpopulation is one of the factors that marginalizes the concerns of sustainability in the current context (Candelone et al., 1995; Hardin, 1968; Jonsson, 2012).

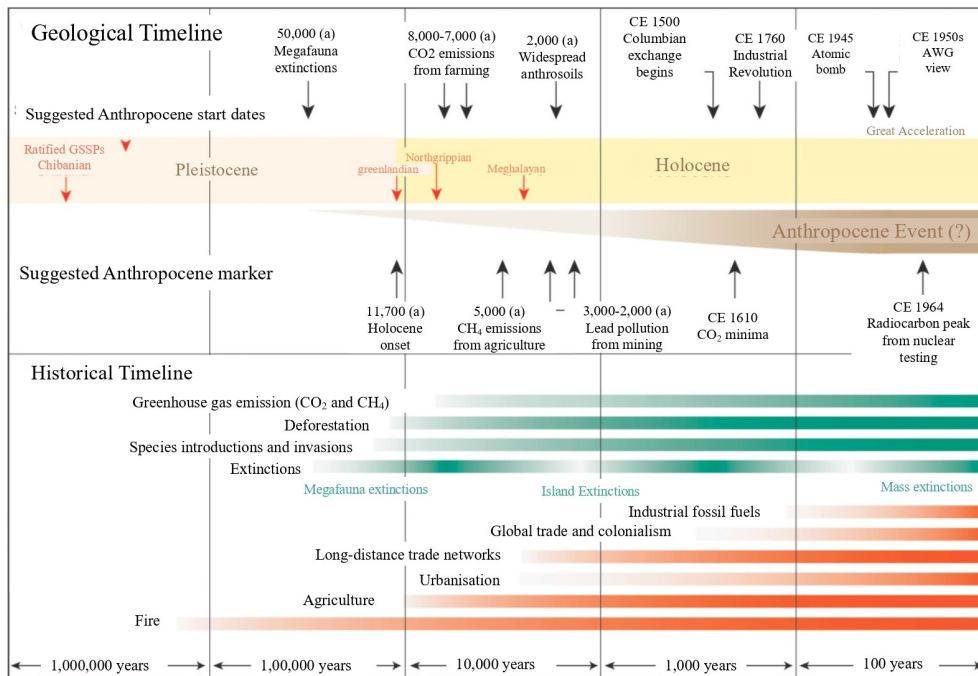
The natural changes show a strong correlation with human civilization's development. The records from agriculture to the industrial revolution appear to be evidence that scientists have agreed on positive development in one dimension and negative facts in another. The severity of the environmental damage has been quite obvious in the last 100 years (as shown in the Figure 7.1.) (Gibbard et al., 2022). For these reasons, the United Nations has called for more efforts to reduce degradation while ensuring inclusivity and growth in socio-economic development. Thus, many theoretical frameworks in regulation and/or science and technology have been articulated to maintain the development of the industrial sector. The industrial revolution in Anthropocene informs all players of a new paradigm shift in balancing development and environmental conservation (Jonsson, 2012). The accumulation of heavy metal deposition in human inhabitant such as Greenland and ice was attributed to tons of contaminants due to the industrial revolution (Candelone et al., 1995). This land pollution could be more severe for the atmospheric and water environments.

However, new studies have found some promising trends suggesting that Industry 4.0 could help manage environmental degradation in both the biophysical and social spaces. Additionally, green technologies are a priority for the top manufacturing industries. Studies suggest that environmental technologies and environmental industries are key factors in addressing sustainability issues

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in next-general society. For instance, the internet of things, green nano-tech manufacturing, and 3D printing are advanced technologies that could provide solutions to the improvement and enhancement of environmental quality (Choi et al., 2019). Global trends seem to be actively promoting dealing with environmental issues for the next generation of industrial growth models.

Figure 7.1 Environmental Change as Implication of Anthropocene



Source: Gibbard et al., (2022)

Even though the industrialization model is different from one nation to another based on the context, industrialization has posed a major change in human civilization. Obviously, the global population has remarkably increased, while the environment has been affected at the same time. Undoubtedly, the growth of industry requires the exploitation of natural resources, both renewable and non-renewable. The start of the industrial revolution has emerged unintentionally in various important considerations on environmental issues, particularly the viability of resources on the planet. The livability of industry has been adapted to market changes due to environmental regulations, norms, and standards, to name a few. In addition, some studies have revealed that technological advancement in industrialization is an influential factor in deteriorating environmental quality (Jovane et al., 2008; Cheng et al., 2021). Thus, competitive sustainable manufacturing is heavily affected as far as the concept of sustainable development is concerned.

2. National Innovation System

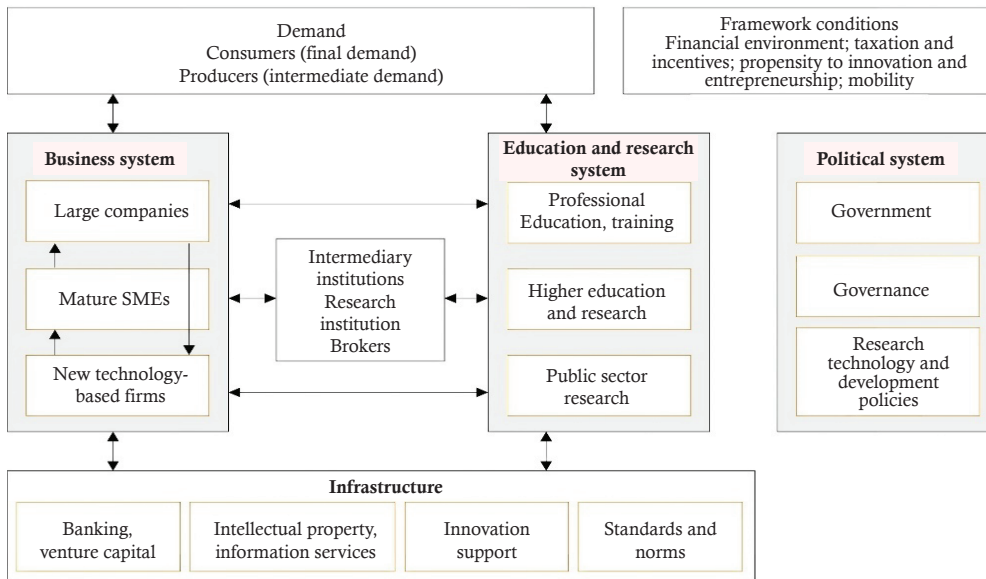
Innovation is the fundamental driving force behind social and economic development. Particularly, green technological innovation is indispensable to reduce carbon emissions and drive efficiency in natural resource utilization. In principle, the National Innovation System (NIS) is the key feature to assess the development status of the Science, Technology and Innovation (STI) ecosystem. The specific indicators must be built from the beginning for ease of monitoring and evaluation (M&E). The demand and supply sides of the STI sector are critical to be formulated with sufficient support from the policy framework and infrastructure, as seen in the figure 7.2. Particularly, the sufficiency of both quality and quantity of Science, Technology, Engineering, and Mathematics (STEM) must be ready for the business side, viz., industries or small-medium enterprises (SMEs) that demand highly skilled labor.

The endorsement of the National Policy on STI 2020-2030 by the Royal Government of Cambodia in 2019 is the first foundation of government support to promote the STI ecosystem in the country. With this continuous support from the government, Cambodia's STI Roadmap 2030 was laid out as a direct strategy for governmental ministries and relevant institutions on actions to be taken in the short and medium term until 2030. Cambodia's STI Roadmap 2030 operationalizes the National STI Policy by setting clear objectives, defining a set of key actions within a time horizon, and setting specific targets to achieve by 2030. It gives a clear role to the Ministry of Industry, Science, Technology & Innovation (MISTI) to oversee and coordinate the implementation of the National Policy on STI across ministries (MISTI, 2021). There are five pillars that are important to implement the National Policy on STI. These key pillars are to be strengthened with the following short description:

- **Governance:** Mandates of MISTI having NCSTI as a national coordinating body for guiding inter-ministerial related jobs must be enforced for the cross-cutting nature of STI activities.
- **Human Capital:** Human resources of both quality and quantity in STEM must be produced for the top priority sectors for socio-economic development.
- **Research and Development (R&D):** Investment in R&D must be made on time for knowledge creation to support the demand of the private sector.
- **Collaboration:** The working environment among government institutions, academia, and the commercial sector needs to break down silos and ensure synergy.
- **Ecosystem:** It is critical to have a favorable ecosystem with both local and foreign investments that synergize sufficient and highly skilled labor, attractive investment policies, and critical infrastructure.

Future industrial growth should rely mainly on the innovation capacity of the industries toward market demand, in which environmental aspects/criteria are

Figure 7.2 National Innovation System Framework



Source: Kuhlmann & Arnold (2001)

taken into account. At this stage, it is necessary to keep in mind that innovation is the capacity of materials/products/processes to be accessible in the market and generate profit. Cambodia has clearly defined innovation in accordance with the Oslo Manual 2018, which refers to ‘a new or improved product or process (or combination thereof) that differs significantly from the unit’s previous products or processes and that has been made available to potential users (product) or brought into use by the unit (process)’ (Organisation for Economic Co-operation and Development & Eurostat, 2018, pp. 32). Therefore, the materialization of knowledge from research in the form of market or process changes with socio-economic development is the foundation of innovation. Innovation is conducted by users, suppliers, and consumers everywhere—in government, business, and non-profit organizations, and across borders, sectors, and institutions. The manual identifies eight broad types of activities that can be undertaken to pursue innovation: 1. research and experimental development (R&D) activities; 2. engineering, design, and other creative work activities; 3. marketing and brand equity activities; 4. IP-related activities; 5. employee training activities; 6. software development and database activities; 7. activities related to the acquisition or lease of tangible assets; and 8. innovation management activities.’ It is summarized that NIS (see Figure 7.2.) serves industrialization once there are correct synergies among stakeholders, starting with high education quality, investment in R&D, sufficient policy support from the government, an entrepreneurial mindset among the private sector, and sufficient infrastructure. The future innovation ecosystem seems to cross not only the border lines of production capacity and market access, but also the environmental regulatory frameworks of trade.

3. Green Technology for Manufacturing Industries

Industrialization relies heavily on science and technology integration in the manufacturing line. Models are seen differently from country to country (Park et al., 2021). In the current context of environmental sensitivity, STI must play a major role in addressing environmental degradation, and industrial development could face challenges as a result. However, STI policies must be rigorously based on evidence of climate accountability in order to serve innovation purposes. Scholarly, the number of research articles published has been increasing, with promising trends from 1995 to 2019 (Usmani et al., 2021). This knowledge accumulation on green technologies is the foundation for transforming them into technologies that serve the manufacturing sector. This exponential growth of articles is projected to increase in the next decade due to environmental pressure. It was suggested by some studies that green technological innovation efficiently serves the utilization of natural resources, high-tech industries using green technological innovation have a correlation with efficient use of resources, and green technology transformation is a recommended new trend for resource conservation (Miao et al., 2017).

Figure 7.3 Increasing Trend of Publication on Green Technologies for Manufacturing Industries



Source: Usmani et al. (2021)

Investing in green technologies is not the direct approach to having environmental care among businesses. However, it is an indirect way of having social recognition, cost reduction, a new business opportunity, or other prerequisites to achieving economic value. Generally, firms attach less value to environmental sustainability. Thus, investment by firms in preserving the commons is something that policymakers should not underestimate. Incentivizing the ecosystem for firms to see beyond the short-term benefits of investment in green technologies is necessary. In general, change management toward green

technology investment among firms is contributed by regulation, commitment, customer demand, management concerns, and cost savings (Saunila et al., 2018). Governmental policies play a significant role in promoting the integration of green technologies to ensure a fast trade-off among firms is possible. Radical technological innovation has shown promise, as a means to reshape the current industrial landscape. Policy innovation has a strong correlation with industrial development and sustainability (Norberg-Bohm, 1999). On the other hand, the current trend reveals the need for collaboration by the private sector through different forms, viz., public-private partnership investment. Since innovation is relatively driven by the private sector, the engagement of the sector with policy support by the government sounds promising for future industrial growth and sustainability.

It is important to take note that Environmental, Social, and cooperative Governance (ESG) is a set of standards allowing measurement opportunities for investors, in which environmental characteristics are undoubtedly the major factors that address innovation. This environmental regulatory concept, particularly the ESG performance concept, positively impacts the innovation ecosystem among manufacturing industries. Furthermore, studies suggest that high ESG performance in industrial operations has a strong contribution to sustainability and innovation performance. This ESG paradigm has increased considerably in the last several years (Zhou et al., 2023). ESG can obviously be broken down into the sub-concepts of its three pillars (Clement et al., 2023).

- Environment: Impact of businesses on the natural environment and its impact on air quality, biodiversity, carbon footprint, greenhouse gas emissions, waste, and water quality
- Society: Relationships inside the firm among/between employees, suppliers, customers, and people in general. Among other things, human rights, privacy policies, and efforts to help poor communities are the main aspects taken into consideration by this pillar.
- Governance: Leadership of firms, internal controls, audits, board composition, corruption, executive compensation, and policies.

4. Conclusion

The United Nations' sustainable development concept, which is built on three pillars: the environment, society, and the economy, has struggled to achieve universal goals due to a misalignment of these elements. The tragedy of the commons, in which overpopulation trumps environmental concerns, has resulted in overexploitation of available resources to fulfill greedy in human populations. The Anthropocene's industrial revolution has ushered in a new paradigm shift in the balance between development and environmental conservation. Industry 4.0 has the potential to help manage environmental degradation in both biophysical

and social spaces, with green technologies being prioritized by high-value added manufacturing industries to address sustainable development agenda. Innovation drives social and economic development, particularly green technological innovation, which is critical for reducing carbon emissions and increasing natural resource use efficiency. Cambodia's National Policy on STI is based on five pillars: improving of governance, building human capital, investment in R&D, collaboration, and building ecosystem. These pillars are critical to the country's sustainable industrial development, which should be driven by innovation and environmental concerns. Having said that, the policy should be based on evidence of climate accountability, which includes high-quality education, innovation-driven R&D investment, quality infrastructure support, an entrepreneurial mindset in the private sector, and sufficient infrastructure. Government policies play an important role in building national innovation system by encouraging the integration of green technologies and ensuring a quick trade-off between firms. Radical technological innovation has shown promise in reshaping the current industrial landscape.

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Chapter 8: Roles of Environmental Scientists and Engineers in the Green Deal Movement

KUOK Fidero and PROMENTILLA Michael Angelo B.*

1. Introduction

The Prime Minister of Cambodia and Chair of the Association of Southeast Asian Nations (ASEAN), Samdech Hun Sen, proposed the “ASEAN Green Deal” during the 55th ASEAN Foreign Ministers’ Meeting as a crucial priority in response to the challenges faced during the post-pandemic recovery (Han, 2022). The primary objective of this comprehensive framework is to motivate ASEAN member states to invest in decarbonizing their economies, effectively addressing climate change concerns, and fostering collaboration among ASEAN member states and external partners. In a previous instance, during the global crises in 2008 concerning fuel, food, and finances, the United Nations Environment Program (UNEP) called upon governments to allocate financial resources towards supporting environmentally sustainable sectors and striving towards the reduction of carbon emissions and ecosystem degradation (UNEP, 2009). UNEP has put forth a series of measures, including mandating high-income countries such as the US, EU, and other OECD economies, along with middle- and high-income G20 economies, to allocate a minimum of one percent of their GDP over two years towards carbon emission reduction. Furthermore, developing nations should also implement domestic strategies to curtail carbon emissions while dedicating at least one percent of their GDP to enhancing access to clean water and sanitation for impoverished communities.

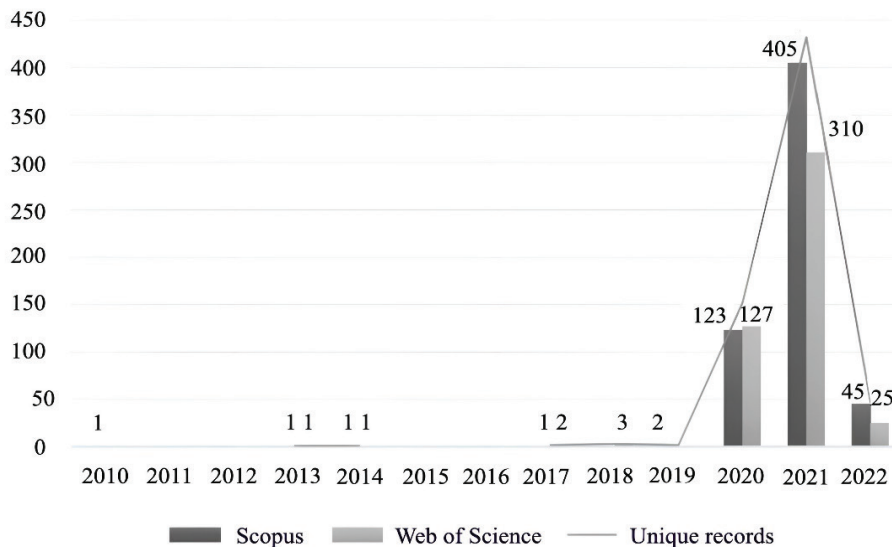
The Green Deal movement seeks to promote sustainable development, reduce carbon emissions, and address climate change. It is crucial for environmental scientists to provide robust scientific research to the public and policymakers in order to shape policies and implement actions throughout the entire process. As challenges such as climate change and emerging pandemics become increasingly complex, the role of environmental scientists as agents of change in science diplomacy is more critical than ever. They provide scientific advice to inform foreign policy objectives (Science in Diplomacy), foster international scientific collaboration (Diplomacy for Science), and utilize scientific cooperation to improve international relations (Science for Diplomacy). Environmental scientists communicate scientific facts and discoveries to the public, engaging them in discussions to achieve a sustainable future. They also play a significant role in monitoring and evaluating the effectiveness of environmental policies and regulations, ensuring that the most reliable scientific methods are employed for assessment purposes.

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2. Roles in Research and Development

Environmental scientists and engineers have a crucial role in advancing the objectives of the Green Deal movement by conducting research and development to provide scientific evidence for policymakers. Since the adoption of the Green Deal by the European Commission in 2019, various concepts related to the Green Deal have emerged as promising frameworks for addressing climate issues and achieving sustainable development goals. A recent systematic review of Green Deal research conducted by Szpilko and Ejdys (2022) revealed that researchers use the term “Green Deal” in different forms, such as “New Green Deal” and “Green New Deal.” This increased interest in Green Deal research is evident in the significant growth of scientific publications on the topic in recent years, as depicted in Figure 8.1.

Figure 8.1 Number of Publications in European Green Deal in Scopus and Web of Science Databases During 2010-2022



Source: Szpilko & Ejdys (2022)

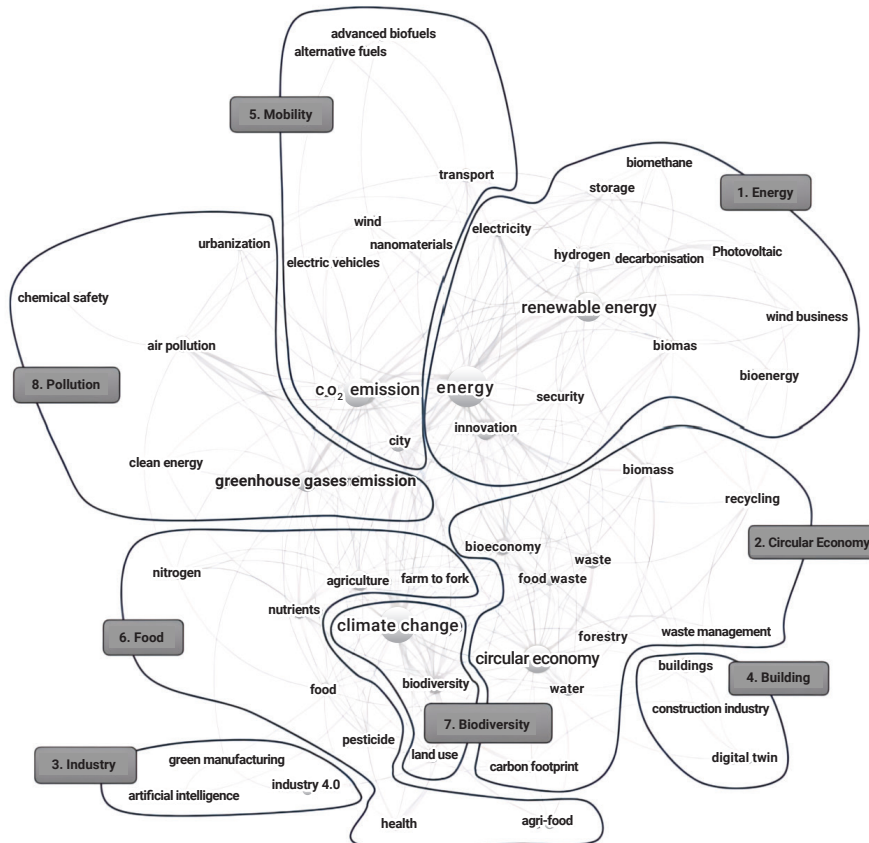
Through thematic cluster analysis of published articles by various authors, Szpilko and Ejdys (2022) identified eight research areas related to the Green Deal: energy, circular economy, industry, building, mobility, food, biodiversity, and pollution. Achieving the objective of carbon neutrality by 2050 hinges on attaining sub-targets within these research areas. As such, environmental scientists are encouraged to expand their research in key areas such as:

- Energy sector: focusing on green and renewable energy sources, energy efficiency, and security;

- Circular economy: emphasizing waste management, particularly in relation to plastic waste;
- Transportation: exploring low-carbon alternative fuels, electrification, and innovative mobility services;
- Pollution: addressing greenhouse gas emissions, chemical safety, and risk assessment;
- Sustainable food: advocating for pesticide-free crop greening and exploring alternatives while expediting the transition to sustainable, healthy, and inclusive food systems (Szpilko & Ejdys, 2022).

Figure 8.2 illustrates the thematic clusters within European Green Deal research. These research directions are crucial for scholars, governments, and the private sector to establish long-term development strategies and define actionable directions. For instance, promoting a circular economy in emerging economies like the ASEAN region necessitates improvements in economic performance and public acceptance as key drivers to encourage stakeholder engagement (Ngan et al., 2019).

Figure 8.2 Thematic Clusters on the Research in European Green Deal



Source: Szpilko & Ejdys (2022)

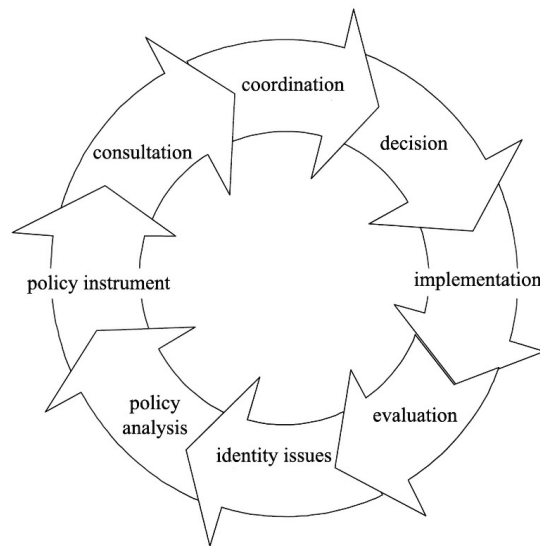
The ASEAN Green Deal presents a promising framework for addressing the urgent environmental challenges in Southeast Asia. There are several key considerations to enhance its effectiveness. Firstly, the role of the Pentahelix partnership, involving academia, industry, government, the media, and the community, should be explicitly acknowledged within the ASEAN Green Deal. It is essential to incentivize and support such partnerships in adopting sustainable and low-carbon practices and technologies. Secondly, the significance of international cooperation should be further emphasized. The ASEAN region is not immune to climate change and environmental degradation impacts, necessitating global collective action. The framework could encompass measures to foster international collaboration in climate finance, technology transfer, and capacity building. Lastly, promoting education and raising public awareness of environmental issues should be integral to the ASEAN Green Deal. Building knowledge exchange and increasing awareness among stakeholders and the general public is vital for bridging the gap between science and policy and garnering support for transitioning to a sustainable and inclusive green economy. The framework could establish a platform for knowledge sharing, including initiatives, such as sustainability science integration in schools, public information campaigns, and community engagement programs.

3. Roles in Policy Formulation and Advocacy

Policy formulation typically follows a series of stages, including problem identification, policy analysis, development of policy instruments, consultation, coordination, decision-making, implementation, and evaluation (see Figure 8.3). Civil servants are primarily responsible for carrying out most of the work involved in the policy cycle, with input and participation from all stakeholders, particularly during public consultation (Althaus et al., 2020). Environmental scientists and engineers, due to their training, are naturally inclined to identify problems that gain public attention through qualitative or quantitative research. This identification of issues often triggers the formulation of policies when problems become significant, widespread, and aligned with political priorities. In practice, policy actors strategically define policy targets and goals, which may evolve over time. Therefore, the creation of policy instruments becomes essential to achieving these goals. It is widely recognized that correctly defining the problem is the initial step, and critical thinking and language awareness is crucial for developing evidence-based policies. Undoubtedly, the role of the expert group comprising environmental scientists and engineers is indispensable for the success of the Green Deal movement's policy and advocacy. Academic research also highlights the importance of the policy mix framework, where multiple goals and complex policy instruments are co-designed with the involvement of all stakeholders (Kern et al., 2017). This dialogue between policymakers and academia indicates a growing interest in the application of policy mix, which has shown positive impacts and incremental development (Rosenow et al., 2016).

The subsequent step is policy advocacy or communication, which necessitates technical knowledge, expertise, and contextual awareness to inform society effectively. Through planned advocacy communication, policy goals can be translated into concise actions.

Figure 8.3 Policy Cycle Diagram



Source: Althaus et al. (2020)

4. Monitoring and Evaluation

Environmental scientists and engineers have a significant role to play throughout the policy cycle, not only in policy formulation but also during policy implementation. During implementation, it is crucial to monitor the progress to ensure the achievement of policy goals and to track objectively verifiable indicators. Monitoring is typically conducted annually, while evaluation is usually carried out at the mid-term or end of the policy implementation period. These monitoring and evaluation processes are generally led by specialists who assess the efficiency and effectiveness of policy implementation. It is important to define the evaluation scope at different policy cycle stages, such as input, process, output, outcome, and impact (Waller, 1996). The responsibility for monitoring and evaluation can lie within the ministry itself, the supervising ministry, agencies, or specialized think tanks with expertise in the relevant policy areas. Therefore, it is essential for the Green Deal movement to have evaluations conducted by environmental scientists and engineers who possess knowledge of both the process and objectives related to the assessment indicators. It should be noted that the selection of the evaluation body is critical, as the credibility of the evaluators will influence the quality of the conclusions and recommendations for potential policy interventions, if necessary (Gerston, 2004).

5. Challenges for the Green Deal Movement

The Cambodian government has articulated its vision of strengthening the national innovation ecosystem by approving the Science, Technology and Innovation Roadmap 2030, which aligns with the National Policy on Science, Technology, and Innovation 2020-2030. However, Cambodia currently faces challenges in research funding, with research expenditure amounting to only 0.12 percent of GDP in 2015 and 35 percent of that funding relying on support from foreign partners and international organizations (Economic and Social Commission for Asia and the Pacific, 2022). To address these challenges, it is crucial to allocate dedicated funding for research and development related to the Green Deal. Additionally, there are other obstacles to overcome, including limited financial investment and feasibility, a shortage of specialized personnel and training programs, and the need for effective partnerships (Mohammed et al., 2013).

6. Policy Recommendations

Given the important role of environmental scientists and engineers including, the researchers working in the area of sustainability science and engineering to achieve the goal of the ASEAN Green Deal, the following are the policy recommendations:

- 1) Invest in environmental science education and training: The ASEAN Green Deal requires a skilled workforce of environmental scientists and engineers. Therefore, investing in environmental science education and training programs at all levels, from primary school to university, is essential.
- 2) Support research and innovation: Investing in research and innovation in sustainability science and engineering can help develop new technologies and practices that support the goals of the ASEAN Green Deal. Governments and the private sector should increase funding for research and innovation in this transdisciplinary domain.
- 3) Create opportunities for wider stakeholder engagement: Environmental scientists and engineers should engage with a range of stakeholders, including the media, policymakers, the private sector, civil society, and the general public, to ensure that their work is relevant and effective. Governments should create opportunities for stakeholder engagement, such as public consultations and expert panels.
- 4) Promote interdisciplinary collaboration: Environmental science and engineering are interdisciplinary fields of humanities, economics, social sciences, natural sciences and other engineering. National collaborative and upskilling platforms and programs should therefore be created to promote a specialized T-profile competency with relevant skill blocks: expertise, and essential and functional skills for the Green Deal movement.

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Chapter 9: Global Health and Climate Change

GOUSHCHI Adonis and CHHEM Sirika*

1. Introduction

Climate change represents a critical challenge globally and Cambodia is no exception to its impacts. According to the World Health Organization (WHO), climate change is among the most significant threats of the 21st century, projected to result in an additional 250,000 deaths annually between 2030 and 2050 due to malnutrition, malaria, diarrhea, and heat stress (WHO, 2021). Its impacts are already being felt around the world, affecting the environment, animals, and economies. However, one of its most significant impacts is on human health. The years 2021 and 2022 witnessed a level of severity in weather events that exceeded anything previously experienced. Globally, record-high temperatures have been reported across continents such as China, Europe, Africa, and both North and South America, leading to thousands of deaths related to the heat. Meanwhile, floods ravaged Australia, Canada, China, Malaysia, Pakistan, western Europe, and Cambodia, also causing thousands of deaths as well as displacing hundreds of thousands of people. Wildfires were responsible for many injuries and deaths in the USA, Greece, Turkey, and Cambodia.

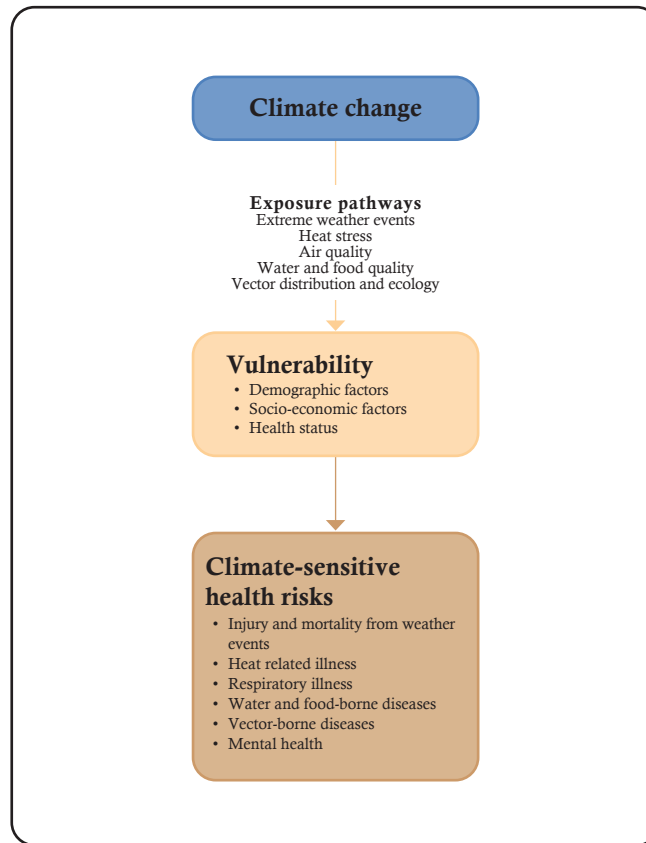
Beyond the direct effects like rising temperatures, severe weather, and increasing sea levels, climate change also indirectly influences global health in multiple ways. These include the proliferation of vector-borne diseases, the spread of food and waterborne illnesses, and the increase in air pollution levels. As the world grapples with all these effects, it is critical to both understand the link between climate change and health and develop strategies to mitigate its impact on global health. This requires a multi-sectoral approach involving a collaboration between public health officials, policymakers, climate scientists, and other stakeholders. This chapter will provide a comprehensive overview of the various ways in which climate change affects human health and the steps that can be taken to address this growing threat.

2. Health Impacts of Climate Change

Climate change is an urgent global threat that affects human health through various exposure pathways. These include direct exposure to extreme weather events, such as heatwaves and hurricanes, and indirect exposure through air pollution, reduced food and water quality, and changes in vector distribution and ecology (see Figure 9.1).

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Figure 9.1 How Does Climate Change Affect Health?



Source: Authors

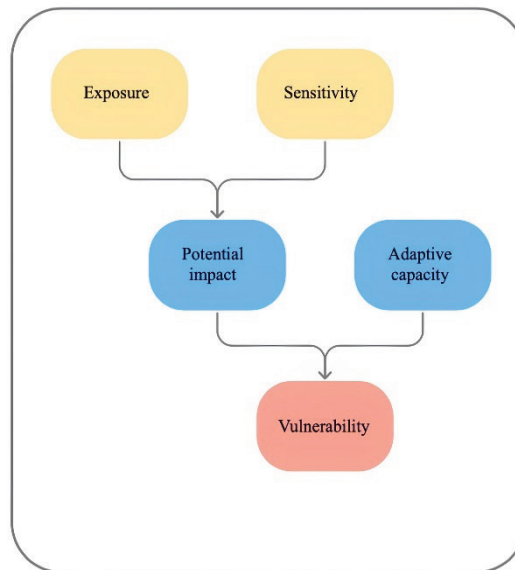
The impacts of climate change on health through exposure pathways are largely determined by vulnerability. Indeed, climate change does not affect all populations equally. Vulnerability refers to the degree to which individuals or communities are susceptible to climate-sensitive health risks (Yu et. al., 2021). It determines how severe the impacts of climate change might be. Thus, assessing vulnerability is essential for identifying regions and populations that are most at risk from the impacts of climate change, and for developing targeted strategies to mitigate these specific impacts.

Vulnerability to climate change has three aspects: exposure, sensitivity, and adaptive capacity (see Figure 9.2).

- Exposure refers to the degree of exposure to climate hazards. For example, individuals living in coastal areas are at risk from sea level rise and coastal flooding, while individuals living in arid regions may be vulnerable to drought and heatwaves. Similarly, people living in urban areas with air pollution are more susceptible to respiratory problems.

- Sensitivity refers to the likelihood of negative impacts. There are several factors that determine the sensitivity of a population:
 - Socio-economic factors: Poverty, housing, access to healthcare, and education. For example, low-income people have less access to healthcare and resources such as air conditioning during heat waves.
 - Demographic factors: The elderly, children and pregnant women are more susceptible to extreme temperatures.
 - Health status: Individuals with pre-existing health conditions, such as respiratory, cardiovascular and renal disease, are more vulnerable to air pollution and heatwaves.
- Adaptive capacity refers to the ability to cope with and adapt to those impacts. It is determined by factors, such as access to resources, health infrastructures, and social support systems.

Figure 9.2 The Concept of Vulnerability



Source: Manangan et al. (2014)

Reducing vulnerability to climate-sensitive health risks requires addressing the underlying social, economic, and environmental factors that contribute to vulnerability, and it will prepare individuals and communities to adapt to the changing climate and minimize the negative health impacts.

2.1. Heat-Related Illness

Extreme heat is the first cause of weather-related death (WHO, 2018) and is a growing concern for public health, especially in countries like Cambodia where extreme temperatures are becoming more common. By 2100, heat waves are

predicted to increase 25-fold (WHO, 2015) with population exposition increasing by several billions. One of the reasons for the increased health risks is that the human body has a limited ability to dissipate heat, leading to dehydration, heat exhaustion and heat stroke. Heat exhaustion can cause nausea, headaches, fever and muscle cramps, while heat stroke can lead to brain damage, organ failure, and death. Additionally, extreme temperatures can worsen pre-existing illnesses such as cardiovascular, respiratory, cerebrovascular, and renal ones.

Another important factor is the phenomenon of urban heat islands (UHIs). Urban areas are often several degrees warmer than their surrounding rural areas due to building density, a lack of vegetation, and increased human activity. This can exacerbate the effects of heatwaves and increase the risk of heat-related illnesses. Cambodian citizens are particularly vulnerable due to the country's hot and humid climate, coupled with high levels of poverty and limited access to cooling resources.

To reduce heat-related illnesses and deaths, key actions include: increasing access to cooling resources (such as public cooling centers and air-conditioned places), implementing heat warning systems for early alerts, reducing the urban heat island effect through green spaces and cool roofs/pavements, and conducting public awareness campaigns on recognizing symptoms, staying hydrated, and keeping cool during extreme heat.

2.2. Air Pollution and Allergens

Climate change is projected to lead to increased concentrations of harmful atmospheric pollutants (ground-level ozone and particulate matter) with significant human health impacts: increased rates of respiratory and cardiovascular diseases and premature death. A reduction in short-lived climate pollutants (SLCPs) in Cambodia could prevent almost 5000 premature deaths yearly from outdoor air pollution starting from 2030 (WHO, 2015). Ground-level ozone is a component of smog and has a detrimental impact on the respiratory system. The rise in ground-level ozone concentration has been associated with an increase in respiratory diseases such as pneumonia, chronic obstructive pulmonary disease (COPD) and asthma (Centers for Disease Control and Prevention, 2021). This leads to increased hospital admissions and emergency room visits and, therefore, increases death rates. Additionally, fine particulate matter concentrations (PM_{2.5}: particulate matter < 2.5 μm) will increase due to the smoke emitted by wildfire episodes. These small particles cause breathing difficulties for people with respiratory diseases but can also increase the risk of cardiovascular events (Ebi & McGregor, 2008).

Another significant concern is household air pollution, primarily caused by the combustion of solid fuels such as biomass and coal for cooking. In Cambodia, more than 95 percent of the population primarily uses solid fuels for cooking in rural areas, while in urban areas, 50 percent of the population does so. Women and

children are at a greater risk for disease from household air pollution, especially with rising temperatures and approximately 61 percent of an estimated 2,400 child deaths due to acute lower respiratory infections in Cambodia is attributable to household air pollution.

One other impact of climate change on the air is the increase in pollen release which worsens not only the allergic reactions in asthma and other types of allergic diseases but also the risk of allergic sensitization in exposed healthy individuals, including children. These factors make the air less breathable leading to an increase in premature deaths and hospital admissions for respiratory and cardiovascular problems. To mitigate those effects, multiple interventions can be implemented such as air quality education for patients with respiratory diseases, promotion of renewable energy and reducing greenhouse gas emissions, and broadcasting notifications for high pollution levels.

2.3. Food and Water-Borne Diseases

Climate change has a significant impact on the availability and quality of water and, thus, on the occurrence and transmission of water-borne diseases. Heavy rainfall and floods can lead to agricultural fields and sewage runoff, contaminating water sources and affecting both its consumption, hygiene, and agriculture. Inland river flooding is a potential risk in Cambodia, where the number of days of heavy rainfall is projected to increase by about 10 days between 1990 and 2100 and affect over 30,000 people (WHO, 2015). The effects of climate change on water security are especially pronounced in developing countries, where limited access to clean water and inadequate sanitation systems can exacerbate the spread of waterborne diseases such as cholera and salmonella (Akil et al., 2014).

Moreover, the rise in water temperature can promote the growth and survival of harmful algae, such as cyanobacteria, in both sea and fresh water. This algae releases harmful toxins into the water and can also contaminate fish and shellfish that consume the algae (Wells et al., 2015). Drinking water or ingesting fish or shellfish contaminated by these toxins can cause a range of symptoms, from mild stomach pain and skin irritation to more severe reactions affecting the lungs, liver, and brain. For instance, in the case of paralytic shellfish poisoning, a type of harmful algal bloom can produce a neurotoxin that affects the nervous system of those who consume contaminated shellfish, leading to symptoms such as tingling, numbness, and paralysis.

To reduce the risk of food and waterborne diseases due to climate change, several strategies can be implemented. Governments must develop new infrastructures to provide access to clean water and proper sanitation facilities. Good hygiene practices, such as frequent handwashing with soap, using clean water for drinking and cooking, and safe disposal of waste, are highly effective in preventing the spread of microorganisms in water. Improving disease surveillance systems and implementing early warning systems can also help detect and

respond to outbreaks of waterborne diseases promptly. Finally, building flood-resistant infrastructure may significantly enhance community resilience to the impacts of food and water-borne diseases.

2.4. Vector-Borne Diseases

Mosquitoes and ticks are carriers of various diseases that can be transmitted to humans, making them significant public health concerns worldwide. For instance, malaria, which is transmitted by female *Anopheles* mosquitoes, is a major cause of morbidity and mortality in tropical countries. It is estimated that there were 247 million cases of malaria globally in 2022, resulting in 619,000 deaths (WHO, 2022), and numbers in Cambodia are expected to increase to 22 million by 2070 (WHO, 2015). Dengue, another mosquito-borne disease, is endemic in over 100 countries and is a leading cause of illness in tropical regions (WHO, 2002). It is the second most worrying infectious threat in Cambodia with similarly increasing numbers by 2070 (WHO, 2015). Many other diseases are transmitted by the mosquito including West-Nile, Yellow Fever and Chikungunya.

As the global temperature rises, mosquitoes are expected to expand their habitats to more temperate and higher elevated regions of the world. This will increase the risk of mosquito-borne diseases spreading to new areas. One of the most effective ways to limit vector borne diseases is through mosquito control programs that eliminate breeding sites and kill adult mosquitoes. Vaccination campaigns play a crucial role in combating mosquito-borne diseases like yellow fever. Additionally, adopting preventive measures such as wearing long-sleeved clothing and using repellent is essential.

2.5. Mental Health

Extreme weather not only has a physical impact on people but can also affect mental health. This situation can result in various psychological disorders, such as anxiety, depression, and post-traumatic stress disorder (PTSD) (Clayton, 2021). These events can also lead to social isolation, job loss, economic instability, and displacement, all of which can contribute to poor mental health outcomes. Fortunately, there are ways to address these problems. Governments, healthcare providers, and community-based organizations can work together to provide mental health support and resources to those affected by climate-related events, including counseling services and group therapy. We can therefore help reduce the psychological impacts of natural disasters and protect the victims' mental health.

3. Climate Change and Health: Mitigating Impacts on a Global Scale

Addressing the health impacts of climate change presents multiple challenges for public health agencies for several reasons. First of all, climate change is a relatively new problem, and public health professionals may not have the necessary skills and expertise to effectively manage its complexity. Also, the scientific understanding of this issue's underlying mechanisms is still incomplete,

making it difficult to develop effective counterstrategies. Moreover, the necessary prevention and adaptation measures go beyond the remit of the health sector, requiring engagement with other sectors, such as urban planning, transportation, water, energy, agriculture and education.

The mistake, however, would be to focus solely on human health and forget about environmental and animal health, which are intimately linked together. Various infectious diseases can be transmitted through either environmental sources such as water or animals, as seen with SARS, Ebola, and the West Nile virus. The One Health approach recognizes this interdependence (Chhem et al., 2020), emphasizing the need for a collaborative and holistic approach to address climate-related health impacts. By taking this approach, better strategies can be developed for human health. For example, a One Health approach could involve improving disease surveillance systems that monitor both animal and human health, and promoting sustainable agriculture and land use practices that protect the environment and reduce the risk of zoonotic diseases.

Cambodia has already started tackling the health impacts of climate change by implementing the Climate Change Strategic Plan 2014–2023 (CCCSP) and working on several important adaptations, but much work is still to be done with regards to disease surveillance and response, climate resilience, financing and costing mechanisms, as well as the analysis of the health benefits from this mitigation.

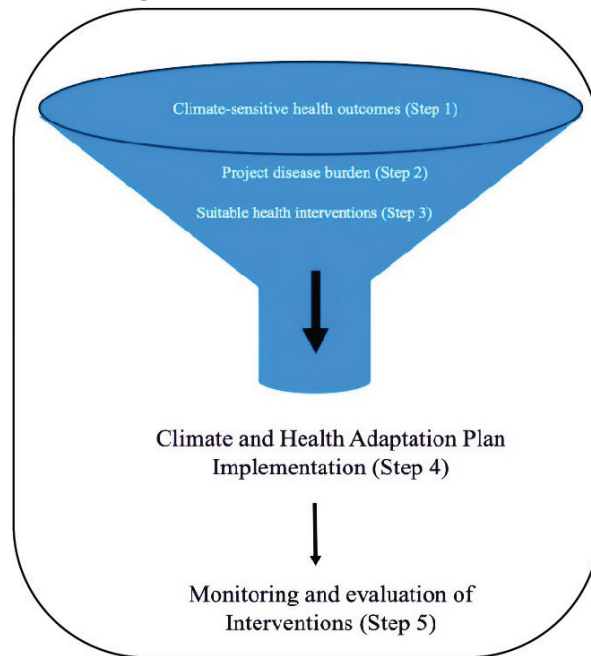
3.1. The BRACE Framework

The BRACE Framework, developed in 2010 by the Center for Disease Control and Prevention (CDC), is a comprehensive tool that allows health officials to identify key health outcomes that are sensitive to climate change and create effective strategies to address them. It is regularly updated to reflect the latest scientific evidence and best practices. The BRACE Framework is divided into five sequential steps (Marinucci et al., 2014) (see Figure 9.3):

Step 1 of the BRACE Framework involves two important tasks:

- The first task consists of identifying climate-sensitive health outcomes. To that end, public health agencies need to access climate data sources to identify trends in weather patterns, such as temperature changes, precipitation, and extreme weather. Based on these historical data and climate mathematical models, and with the collaboration of climatologists, climate change can be projected 30 or more years into the future to assess the effects of climate change on health. By the end of this task, public health agencies will be able to report a list of health outcomes that are most likely to be affected by climate change.
- The second task involves identifying vulnerable populations and regions. Various methods will be used to identify vulnerable populations, such as the analysis of demographic and health status data, and social vulnerability indices.

Figure 9.3 BRACE Framework



Source: Marinucci et al. (2014)

The climate change projections and vulnerability assessment generated in **Step 1** will provide a comprehensive understanding of the potential impacts of climate change on public health and the populations that are most vulnerable. This information can be used by policymakers and stakeholders to develop strategies and interventions to mitigate these impacts and protect public health.

Step 2 the goal is to estimate the additional disease burden associated with climate change. It can be done by linking the projected exposure pathway (extreme weather events, heat waves) to the health outcomes. For example, if an increase in extreme heat is expected to lead to an increase in heat-related illnesses, analysis of historical data and future trend projections are used to calculate the numbers incurred by this exposure pathway.

Step 3 concentrates on selecting the most appropriate measures to tackle the health issues identified earlier. To do so, the evidence-based public health (EBPH) approach can be used, allowing for an assessment of the effectiveness of various clinical and public health interventions. This involves a systematic literature review to identify relevant interventions and evaluate their efficacy. It is also important to evaluate whether interventions are well-matched to the political, cultural, and logistical landscape. Additionally, a cost-effectiveness analysis can help agencies prioritize interventions based on their cost and benefits.

Step 4 involves formulating a climate and health adaptation strategy that amalgamates the insights gathered in preceding phases. This strategy should be comprehensive, intersectoral, with a primary focus on public health outcomes. It should also harmonize with broader climate change initiatives across different sectors to promote collaboration among diverse disciplines and interest groups. Furthermore, the strategy should delineate methods for stakeholders to integrate adaptations into their existing programs and specify how activities will be assessed. Lastly, it should undergo periodic updates and be made publicly accessible.

Step 5 consists of monitoring and evaluating the impacts of interventions on public health. By collecting and analyzing data on their outcomes, we can learn from previous processes and improve future interventions. This can include identifying the most effective interventions, determining which populations benefit most from the interventions, and identifying any unintended consequences or negative impacts of the interventions. The data collected through monitoring and evaluation can also help inform policy decisions and resource allocation.

The framework has been successfully implemented in several countries given its numerous advantages. Indeed, it is an adaptive and iterative approach, allowing interventions to be refined and adjusted based on ongoing monitoring and evaluation. It is also evidence-based and grounded in scientific research and data and emphasizes health equity and collaboration between stakeholders. There are some potential disadvantages, namely its complexity and need for abundant resources, which could limit its feasibility in some contexts. Additionally, its focus on long-term outcomes may not provide immediate solutions to urgent public health problems. In addition to the use of the BRACE Framework, health officials must take into account several considerations to reduce the effects of climate change on health.

Firstly, there is an urgent need for education and training for public health and healthcare professionals on climate change. This involves integrating climate change-related topics into the existing curricula of these professionals. Various training initiatives already exist, such as the Climate and Health Literacy Consortium in the US and the Climate and Health Alliance in Australia. By incorporating climate change into professional training and education, healthcare professionals will be better equipped to identify and manage climate-related health issues.

Secondly, increasing the resilience of health systems is crucial and several measures can be undertaken. Hospital designs should include climate-resilient aspects with sustainable and resilient building materials, energy-efficient systems, and advanced water management systems. Early warning systems for extreme weather events would improve emergency response planning, enhance medical supply availability, and increase healthcare system capacity during heat waves or disease outbreaks.

Lastly, it is important to consider the varying vulnerabilities of different demographic and socio-economic groups. WHO emphasizes the need for health equity in addressing climate-related health threats. However, practical implementation of equitable adaptation solutions is rare. Stronger engagement of those working on climate change threats with social care providers can help to ensure that vulnerable populations, such as the elderly or disabled, have access to the support they need to adapt to climate-related challenges.

3.2. Connecting Global Health and Climate Change through Diplomacy

Climate change is a global phenomenon that is expected to affect every country in the world. To effectively address its health impacts, it is essential for all countries to take collective action. Global health diplomacy represents the convergence of health and international policy, enabling nations to collectively tackle global health issues through collaborative negotiations and partnerships. It involves a wide range of actors from various sectors, including government officials, diplomats, public health professionals, researchers, NGOs, civil society groups, and international organizations. These individuals and organizations work together to promote international cooperation and coordination on global health issues, such as disease outbreaks, natural disasters, and the health impacts of climate change.

In recent years, global health diplomacy has played a critical role in addressing epidemics such as COVID-19, Ebola, and Zika, by facilitating the sharing of information and best practices among countries, helping to coordinate their interventions and also by promoting access to medicines and vaccines. One concrete example of global health diplomacy in action is the Westerdam incident during the COVID-19 pandemic (Chhem & Chhem, 2020). The Westerdam cruise ship was denied entry to several ports due to fears that its passengers and crew may have been infected with COVID-19. Eventually, the ship was allowed to dock in Cambodia, where its passengers and crew received humanitarian assistance. Cambodia's decision to allow the Westerdam to dock and provide assistance to its occupants can be seen as an example of health diplomacy. By taking in the ship and its passengers, Cambodia demonstrated its commitment to promoting health and protecting vulnerable populations in the face of a global health crisis. This decision required diplomacy and negotiation with other countries, as well as coordination with public health officials to ensure that appropriate measures were taken to prevent the spread of disease.

Similarly, global health diplomacy can play a critical role in addressing the impacts of climate change on health by promoting cooperation and collaboration between countries. One example is the Paris Agreement on Climate Change signed in 2015. Its goal is to maintain the increase in global temperatures below 2°C relative to pre-industrial levels and to endeavor to limit the rise to no more than 1.5°C. The WHO has also been active in addressing climate change and

health through global health diplomacy. The WHO has developed a range of strategies and tools to help countries adapt to the health impacts of climate change, including the Climate and Health Country Profiles.

4. Conclusion

To mitigate the health impacts of climate change, urgent and coordinated action across sectors and countries is necessary. This requires a multifaceted approach that includes reducing greenhouse gas emissions, implementing energy-efficient technologies, and promoting sustainable transportation. Adapting healthcare systems is crucial, with strengthened disease surveillance, early warning systems, and climate-resilient infrastructure. Education campaigns empower individuals by raising awareness about the risks of food-borne, water-borne and vector-borne diseases, while also providing information on preventive measures to reduce these risks and protect public health. Priority must therefore be placed on research, policy development, and collaboration among different stakeholders, all the while keeping in mind the importance of health equity throughout this whole process, as different needs will require different solutions. By working together, we can effectively mitigate the detrimental health effects of climate change and pave the way for resilient communities, ensuring a better future for all.

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Chapter 10: University for Sustainability

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1. Introduction

The way that the world is doing business has changed. The evolution from Web 2.0 to Web 3.0, the increasing domination of Generation Y and Z (tech-savvy generation) in the labor market worldwide, the rise of Asian giants like China and India as the main engines of economic growth around the world, and the increasing global concern over sustainability issues are among the primary drivers of these changes (Birkinshaw & Ridderstråle, 2017). These have, without a doubt, also altered how the labor market operates and how potential employees are recruited in Cambodia. In the past, when industries mainly demanded low-skilled labor, which was also abundant due to the baby boom in the 1980s and early 1990s, the Cambodian labor market was characterized as a “buyer market” in which employers could easily hire potential workers. After all, most employees then did not need to possess very high competencies to work in a firm that predominantly required manual labor.

However, this has started to change as we speak. Since the early 2000s, the total fertility rate of Cambodia has continued to decline significantly from 4.0 in 2000 to 2.4 in 2020 (United Nations Department of Economic and Social Affairs, 2022). The number of young Cambodian workers who enter the labor market annually decreased from 300,000 in the early 2010s (Ministry of Labour & International Labor Organization, 2014) to 270,000 in 2017¹, while the demands for better skills from first-time job seekers are also intensifying, corresponding with higher wages and the changing characteristics of the Cambodian economy which has begun to compete in terms of workers’ productivity instead of low input costs such as salary. Such a change in the nature of competition is true even in the case of the garment and footwear industry, which is considered a low-skilled sector. As a result, young Cambodians must be equipped with the high skills necessary to succeed and thrive in an uncertain and unpredictable future while the country is on its way towards becoming a more developed country. It is in this sense that the way the youth is educated must also be revisited and reconsidered. The Cambodian education system cannot be the same as it was thirty years ago if the country is to advance further and grab a larger part of the global market for its products.

Nevertheless, discussing how the Cambodian educational system should evolve would be too broad to discuss in one paper. After all, different educational

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1 https://www.ilo.org/asia/media-centre/news/WCMS_573139/lang-en/index.htm (last visit on May 3, 2023).

levels and target groups (primary, secondary, and tertiary education) would need different reforming approaches, and there is no one-size-fits-all method that is usually very neutral and hence ineffective when it comes to policy impact. That said, we would narrow down the scope of discussion to only education at the university level. What, then, are the roles of universities in this regard? According to the objectives of this paper, we argue that the roles of Cambodian universities must go beyond their traditional teaching-only mission to encompass social entrepreneurship education, research and innovation, and contributions toward inclusive and sustainable development. This shift is reflected in the concept of University 4.0, which emphasizes the importance of higher education institutions in shaping the society of the future where technology, social innovation, and sustainability converge (Gorina & Polyakova, 2021; Rosak-Szyrocka et al., 2022).

The findings and discussion in this paper derive from an in-depth interview with Prof. Jeffrey Sachs, an economist leading the Center for Sustainable Development at Columbia University and the United Nations' Sustainable Development Solutions Network (SDSN), and roundtable panel discussions on University and Digital Sustainability during the 14th International Conference on Software, Knowledge, Information Management and Applications (SKIMA) hosted at the Cambodia University of Technology and Science (CamTech) on December 2 – 4, 2022. The panel discussions were public and were participated by academics in high-level management positions at seven universities across five countries. The discussion was recorded in video format and later transcribed into text for coding and thematic data analysis.

This research study echoes the concern of Prime Minister Hun Sen, who raised an important question during his opening remarks to audiences at the Cambodia Outlook Conference 2022, which consisted of leaders from the government, private sector, civil society, universities, and think-tanks, asking whether Cambodia will be able to achieve its Sustainable Development Goals by 2030, especially after the major disruptions caused by the Covid-19 Pandemic and the war in Ukraine in recent years. In addition, it seeks to contribute to filling in the gap of knowledge of a nascent initiative known as the ASEAN Green Deal, which aims to re-orient regional development towards ASEAN's sustainable, inclusive, and resilient green future while continuing to prioritize digital transformation. In short, this research is meant to identify the role of universities in promoting sustainability. Moreover, it intends to put sustainability on the radar of higher education discourse in Cambodia, as currently, only a few Cambodian higher education institutions incorporate sustainability into their mission, vision, or academic programs.

Regarding academic significance, we develop and extend Fissi et al.'s (2021) framework for a sustainable university to include international collaboration. In this regard, we would argue that sustainability issues cannot be solved by any university working in silos. More importantly, it is not a particular country's

issue because whatever the rest of the world does, it will likely generate positive or negative externalities. Take climate change as an example. If industrialized countries continue to pollute the air by emitting more carbon dioxide (CO₂), the ASEAN region will also be negatively affected. The region cannot solve the issue alone without bringing the emitters to the negotiation table. Even in academia, sustainability demands an interdisciplinary understanding. No single discipline will ever be able to deal with the problems alone due to their complexity which goes beyond science and academic research. The Fukushima Daiichi incidence in 2011 and the complicated challenges that ensued are good examples of the importance of cross-country and multi-disciplinary collaboration (Shigemura & Chhem, 2016).

2. Understanding Sustainability and Sustainable University

In 1987, the United Nations Brundtland Commission defined sustainability as “meeting the needs of the present without compromising the ability of future generations to meet their own needs.” (United Nations, 1987). However, such a loose definition with the increasing concern over climate issues is less applicable to meet the development needs of today’s agenda. It is in this regard that new ideas or concepts of sustainable development must be put in place, giving birth to the ambitious aspirations for a greener, healthier, more prosperous, equal, and peaceful world known as the Sustainable Development Goals (SDGs) in 2012. Particularly, sustainability, in this sense and in accordance with Prof. Jeffrey Sachs, the champion of the SDGs, revolves around four main basic ideas as follows:

- All societies are striving for material decency and dignity, economic prosperity, and meeting the economic needs of their people;
- All societies are or should be striving for social inclusion and social justice, and that welfare should be shared more evenly among everyone, regardless of race and gender;
- Societies need to collaborate to confront the multiple and increasingly serious environmental crises such as human-induced climate change, massive degradation of vital ecosystems, and mega pollution of air, water, and soils;
- Finally, all societies should strictly adhere to the principles of cooperation, partnership, and a multipolar world to promote peace.

2.1. From Teaching to Digital University

Now that we know how sustainability is defined, we are in an easy position to understand what it means for a university, whose role should no longer be limited to being a mere spectator for global macro-level challenges. In medieval times, a university was an institution where only lectures and seminars were given

to elitist groups of students with a focus on theology and politics. Their tasks were mostly cultivating the next generation of leaders and producing competent members of the government, and these graduates would then be the agents of gods endeavoring to rule the masses and bring prosperity (Scott, 2006).

This was mainly the case until the research function was incorporated in the 19th century to expand the scope of work of the university. Wissema (2009) calls it the University 2.0, with Humboldt University allegedly being the first of such entity in Europe. Research was then another essential source of socioeconomic development and innovation worldwide. However, academic research has often been criticized by the general public as lacking practicality and of little use in daily application. It thus does not contribute back to the community directly. In fact, research, for a considerably long time, was mainly done to boost faculty members' publications and the university's prestige. Under the publish-or-perish principle, the world of academia has also been plagued by research fraud, including plagiarism, data fabrication, data falsification, and other questionable research practices done intentionally by academics (Herndon, 2016) or unintentionally by those who pretend to know it all influenced by Dunning-Kruger effects (Dunning, 2011), causing mass confusion. As a result, only a small number of research publications find their way into good hands.

The Third-Generation University (University 3.0 or entrepreneurial university), with the University of Cambridge as the best example, is seen as the cradle of economic and entrepreneurial activities, translating know-how to commercial outcomes, supporting business activities within and outside of campus, leading innovation, and competing in the market for research contracts from the industries (Wissema, 2009). It may also assume the role of an incubator and accelerator for startups. The emergence of technology-based enterprises incubated by universities worldwide bears witness to this new overhanging role of the university, but such economically competitive characteristics of industries can be an invitation to the structural exploitation of natural resources (Kuch, 2020).

There is also the concept of University 4.0 (digital university), which tends to further extend the university's traditional duties (Strielkowski & Wang, 2020; Wissema, 2009). Such a university is the latest model of higher learning institutions, established to reflect the current teaching and learning paradigm known as the Education 4.0 (Cottafava et al., 2019). It, in turn, responds to the Industry 4.0 framework, the growing number of Generation Z entering the labor market and the digital revolution in education. One example of the new teaching methods for digital universities is the use of Massive Open Online Courses (MOOC), online learning platforms, and technological devices such as Virtual Reality (VR) sets, all of which have rendered the traditional pedagogy and learning approaches obsolete.

2.2. University for Sustainability

It is worth noting that the evolution of higher learning institutions is conceptualized based on the university's activities, but by the nature of the competitive market economy, these activities may not encourage social and economic sustainability. On the contrary, it may reinforce economic inequality (Blanden & Machin, 2004) and eventually lead to the tragedy of the commons (Grinell & Rabin, 2013; Hardin, 1968). As a result, one needs to rethink and restructure the university's functions to include the sustainability dimension in addition to the existing tasks. That is to say, a university should incorporate SDG principles into its core activities and promote such fundamentals in its daily operations. Formally, a sustainable university can be defined as *a higher education institution that is managed in respect of green economic, social, and environmental practices and that fulfills its function of teaching, research, and community engagement in ways to favor the transition of society toward sustainable lifestyles* (Velazquez et al., 2006, p. 812). Sustainable universities can also be used interchangeably with green universities or campuses, the notion of which encourages university leaders, faculty members, and students to meet social and environmental responsibilities (Dagiliūtė et al., 2018; Yuan et al., 2013). Indeed, there is still a debate on the concept of the terms “sustainable university” and “green university” or if they differ in some senses (Santa et al., 2019; Velazquez et al., 2006). Regardless of that, it is not the same as a university for digital sustainability².

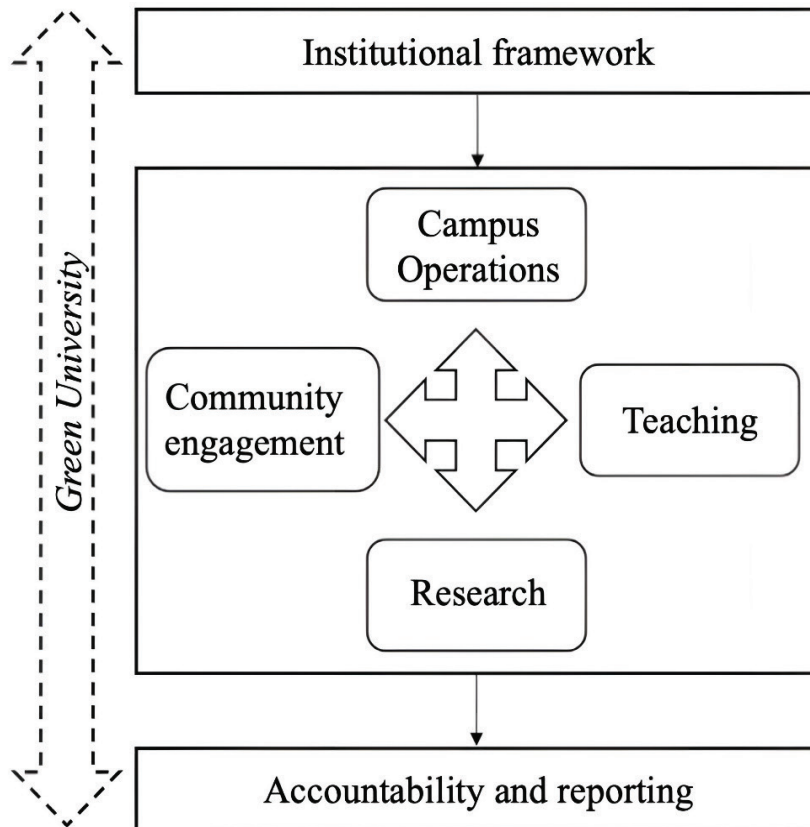
Under the definition above, a sustainable university has a complex and closely interlinked system characterized by four components: campus operations, teaching, research, and community engagement (Yuan et al., 2013). The first component comprises how the university should maintain its campus environment, waste, and use of its resources efficiently and economically. Leal Filho et al. (2019) mentioned that while teaching and research have always been the core parts of a university since the late 19th century, assuming it is, of course, not practicing the medieval-European system as a teaching-only university which is still prevalent in Cambodia (Khieng et al., 2017). Last but not least, sustainable universities must seek to engage and collaborate with the local community, government, and industries to increase their awareness of sustainability issues which the world is facing. Specific outreach activities include training, seminars, partnership building, and generating new business ideas for startups.

Fissi et al. (2021), however, argue that more than these four dimensions are needed. Many Higher Education Institutions (HEIs) have failed to become sustainable universities because they did not explicitly indicate their mission and vision when it comes to sustainability. Simply put, they need a proper institutional framework and a clear policy from the top leadership. In addition,

2 Readers are referred to Bradley (2007) for further information on such a term, which encompasses a variety of technical issues and features, such as digital preservation, that have contributed to the longevity of digital technologies and tools in the last two decades.

the stakeholders were not very aware of the university's activities and goals with respect to sustainability and hence have not shown great support. As a result, Fissi et al. (2021) propose another sustainable university framework (as shown in Figure 10.1.). With that said, the new concept includes two more executive dimensions on top of that suggested by Yuan et al. (2013), namely institutional framework and accountability. The latter involves the idea of “corporatization” of higher education institutions which refers to the practice of borrowing and incorporating the concepts and tools such as accountability and management system from the business world but tailored to the specific needs in order to effectively run the university. Such an idea is justified because universities are increasingly required to monitor and evaluate their performance and transparently document their sustainable use of financial resources, especially the taxpayer's money, in the case of public institutions.

Figure 10.1 The Dimensions of a Sustainable University



Source: Fissi et al. (2021).

It should be indicated that while we do agree with Yuan et al.'s (2013) and Fissi et al.'s (2021) frameworks on most aspects, we respectfully assert that the framework lacks an “international collaboration” dimension. The world is again

at a time of fraught relations between major global powers, including the United States, China, and Russia. While international politics is beyond the scope of discussion in this paper, it suffices to say that many international conflicts negatively impact academia (e.g., research funding is being redirected towards financing the war and humanitarian responses in Ukraine) and hinder scholarly collaboration.

3. Findings and Discussion

This section presents the findings from the panel discussion during the SKIMA conference held in December 2022. When asked how to integrate sustainability concepts and practices into the mindset of everyone on campus, ranging from the security guard at the gates to the university president, a common theme was identified by most participants, namely that a curriculum with an added focus on sustainability issues must be developed. It was also suggested that even a teaching university should seek every opportunity to promote the students' awareness of environmental and social issues, and that, in turn, will help them influence their workplace practices and perhaps also national policies in the future. It can be done by integrating sustainability into the curriculum vertically (students should take specific sustainability-related courses throughout their educational journey at the university) and horizontally (by including sustainability principles in other regular courses). Alternatively, universities can generate sustainability-responsible citizens through their undergraduate and graduate programs, particularly those who major in sustainability or sustainable development. It is worth noting that it is not simply a course in a wider curriculum but a degree program.

“The reason is that if it’s just a single course, students will learn a single course and forget it. Sustainability needs to be embedded across all courses, across all programs. Many curricula are also very jam-packed. It’s very difficult to add additional courses. Most of my students are already studying for 10 to 12 hours a day. You can just add an extra course, and they study for 14 hours a day.”

Case 6, President of a Cambodian university.

Regardless of that, the university should also be a focal point that actively promotes the concept of green-based entrepreneurship and innovation among students. This concept refers to the design and implementation of innovative methods that aim to solve societal problems and eventually improve individual and community welfare. In a broader sense, a university for sustainability should pay more attention to fostering ideas and incubating student startups whose business activities are related to the Sustainable Development Goals or aim to address the SDG problems to assist the host country in achieving them. Despite their young age, youth and students should play a part in helping to develop solutions, wherever and whenever possible. In fact, many ideas belonging to, and deriving from, the youth and students are creative and innovative, allowing the country to cope with and approach societal problems in a different way, as

their minds are not confined to an intelligence-trapping atmosphere such as the workplace culture and structure (Walter, 2012). They are also likely to understand their peers' needs more than people from another generation.

"We wish to provide a holistic education so that those whom we engage, especially our students, when they graduate, they have skills, especially transferable skills, ..., but more than skills are that they become responsible citizens. "Responsible" means responsible consumer, responsible employee, or responsible employer if they start their own startup... If enterprises just focus on the bottom line, as I already mentioned earlier, and the economy is just focused on growth growth growth without limits, an issue that we've been warned 30 years ago in 1972, when hundreds of scholars launched the book on the limits of growth. If we don't learn the lesson of the limits of growth and we continue to focus on this narrow definition of development, then it's not sustainable."

Case 7, President of a Cambodian university.

Nevertheless, entrepreneurship should not be an approach to enriching oneself at the expense of others directly or indirectly. It never is, and it should not be the fundamentally inherent nature of capitalism. It is here that the university itself, as an educational institution seeking to promote a good personality, ought to stand firm against such a notion and work to develop and shape a student's character; otherwise, having an educated person is even more destructive to the society than having many uneducated individuals.

"Now about the course [sustainability], as I mentioned that it can be added as a topic in every course, but I believe there must be a course at the first-day level, entry-level, that would teach ethics, leadership, and entrepreneurship."

Case 3, Professor at a university in Australia.

"Should sustainability courses be compulsory for all academic programs? Of course, we should in the foundation year so that all students have this mindset while they study with a very good positive attitude in terms of sustainability."

Case 5, Vice President of a Cambodian university.

Establishing partnership and collaboration across universities is another common message raised by the top leadership of the universities that participated in the panel discussion. Overseas cooperation here also means being a hub and acting as an intermediary to create networking opportunities, foster communication, and facilitate partnership and collaboration among and between local and foreign higher learning institutions and other stakeholders in order to promote integrated and holistic approaches to provide solutions to sustainable development challenges.

"So, I think we need to push this somehow and either through our academic research in such a way that politicians and those who are in power will believe that this is a real crisis and that we should act now. Otherwise, we will be in difficult situations in the coming years. Also, I think university education systems until now had some problems because, as you have talked about, working in silos. For

example, in my time when I studied in universities. I'm a technical guy; I studied technical issues and technical issues all the time, not knowing that my engineering solution would have an impact on the social aspect, political aspect, and all this. So this is critical, like [name] mentioned, that there are the political system, social system, and engineering system – that need to work together. Then we can solve the problem of economic prosperity, social inclusion, environmental crisis, and corporations. So, I think we need to somehow cooperate or put pressure somehow through academic institutions or change our attitude and also the attitude of the upcoming generations through education.”

Case 4, University professor and director of a research center in Scotland.

It should be highlighted that tackling the technical complexity of the problems would generally need interdisciplinary understanding and participation from scholars of different fields. For example, the climate crisis cannot be solved solely by climatologists. They can only tell what will happen if we emit more greenhouse gases, but they are not energy engineers. They will be able to advise on which type of (clean) energy sources we should utilize to reduce pollution or how we should communicate this with the general public in a way that would change their environmentally-unfriendly behavior and earn their trust and support and not cause panic. International collaboration among universities would also bring about the internationalization of higher education, and it is almost certain that through such efforts, the university would prepare the students for a globalized workforce and improve their cultural tolerance and ability to work and interact in a multi-ethnic environment (de Wit et al., 2015), which in turn would encourage more collaboration across societies in the future.

“And the second one, to be quick, is about the uncertainty of science, of course, because it's another issue. And so we know that with the media, the social networks, and things like that, there's more and more information everywhere. So we need to push also our students to be able to have critical thinking and to have also a holistic approach in their problem analysis, so that means that I agree totally with Professor [Name] about the transdisciplinary approach, which is very very very important. Of course, I'm a computer scientist, and I think most of us are from the STEM domain, but we need to work hand-in-hand with human science, sociology, philosophy, medicine, etc., to really have a global approach, a holistic approach of the sustainability to training our students.”

Case 2, Professor at a French university.

A university also needs to revisit its missions and stop considering itself as just a place, in the eyes of the commoners, where men with glasses spend long hours reading books, talking unintelligibly with one another, doing esoteric and bizarre research, and publishing incomprehensible texts. However, we have, by no means, suggested that the university needs to forfeit its roles, including research and entrepreneurship development. In fact, they need to expand their scope of work in each of these roles. There should be a much more active pursuit of applied research at the university level that responds to policy and industrial

demands. Simply put, research should seek to make an impact on society and contribute in terms of policy suggestions to solve sustainability issues, regardless of whether or not it would be publishable in a top journal.

“So the third point is quite similar to what [case 6] mentioned, many of our faculty members were trained in an era that was very different from the students that we are educating. So we need to continue to retrain and rethink, especially after Covid, what is the role of the university again. Is it purely to provide skills through teaching and no research, no relevance to industry and society? So, we need to train faculty members every year [...]. Third is an echo of Professor [Name] and some of our panelists in terms of building partnerships. So we can do so much alone, but we can go so far going together, and in particular, at the moment, we are proud but at the same time humbled to be a member of UN Sustainable Development Solutions Network, a member of the university rectors and vice-rectors around the world, connecting resources and expertise, but we are also partnering with UN agencies such as UNESCAP and International Labor Organization to research and to develop a program in social innovation and social entrepreneurship because these are the explicit actions that may address the issue Professor Sachs has mentioned – a complete change of mindset in terms of how we run the economy.”

Case 7, President of a Cambodian university.

Last but not least, as Prof. Jeffrey Sachs suggested, one needs to strictly adhere to exercising diplomacy to promote international collaboration to solve global issues. A combined effort to strengthen political liaison is, without question, the first thing one needs to consider in a world of conflict; otherwise, it will obstruct other competition due to the lack of trust, resulting in mutual losses. For instance, climate and environmental issues would require international cooperation across continents because many sustainable challenges one country faces, are likely shared by its neighboring states, such as Cambodia with Lao PDR, Vietnam, and Thailand. Economically speaking, it would be equivalent to a public good which is both non-rivalrous and non-excludable. That is to say, if Country A manages to clean the air, people from neighboring Country B will also benefit, and that Country A cannot exclude Country B from breathing in clean air. However, Country A would not want to do this alone and would prefer teamwork, but if Country B refuses to partner in this situation, Country A will find absolutely no incentives to clean the air, not even for themselves.

4. Challenges

Promoting sustainability even internally within the university campus has always been challenging (Ávila et al., 2017; Dahle & Neumayer, 2001), let alone promoting international collaboration across countries to deal with these issues which have not received a strong political will, especially from the developed world, which is the main source of greenhouse gas emission. On some occasions, these major emitters also announced their withdrawal from the 2015 Paris Agreement on

Climate Change Mitigation. In fact, there are also very few incentives for any higher education institution to do so in the first place. Universities are judged and given ranks based mostly on the number of their lecturers, students, research impacts and activities, including grants received and publications. They receive little to no rewards for being a civic and green university, at least in the context of the ASEAN region. Therefore, it comes with little surprise that even though the ASEAN universities do collaborate, to a certain degree, they do so in order to compete for funding, students and faculty members (Chan, 2004) and not to solve sustainability challenges, which have also been given low levels of attention (Clark et al., 2018).

In Cambodia in particular, only six institutions, namely the Cambodia University of Technology and Science (CamTech), University of Puthisastra (UP), University of Cambodia (UC), National University of Management (NUM), Royal University of Phnom Penh (RUPP), and Asian Vision Institute (AVI), are part of the Sustainable Development Solutions Network (SDSN) championing sustainable development resolutions as of April 2023. They have yet to be recognized for their efforts in taking initiatives and scaffolding sustainability, not even locally. Consequently, the university ranking system should give more priority or higher weight to the sustainability aspect and related activities which generate impacts on society on top of the mainstream academia. There must be an incentive system there to stimulate certain pursuits and discourage harmful behavior, or universities are likely to resort to detrimental habits just to put on a front, the same way that a country can increase GDP by chopping down trees and extracting mineral resources uncontrollably. It apparently looks grand from the outside, but in reality, it is very unhealthy on the inside.

Another challenge when it comes to the implementation of the model is that both lecturers and students typically have a very strong academic interest, believing that their priorities should be to embed the students' attributes in as many technical skills as possible (Pham & Soltani, 2021), disregarding other important (social) issues, including sustainable development unless their field of study is highly relevant with them. For that reason, it is usually difficult to convince students to recognize the value of contemporary general knowledge, which may help improve their critical and holistic thinking and chances of landing a job right after graduation. The students, however, often only realize the importance of taking these subjects later when they enter the real world of the labor market and come into contact with employers (Pham & Soltani, 2021). Such insufficient and poor preparation of employability skills for students eventually causes much frustration to both the students themselves and their potential employers (National Employment Agency, 2018). It is worth noting that some research has also indicated insufficient government policies to encourage the implementation of an educational system for sustainability as a part of the problem (Brandli et al., 2015; Filho et al., 2015). In this regard, the university needs to revise its

current curriculum and introduce a revolutionary interdisciplinary approach to education. Certain courses relevant to sustainable development should be made compulsory for all university students, regardless of their major. Doing so will help students grasp today's complex phenomenon and global issues and scaffold sustainability.

Finally, despite the fact that some HEIs have taken some initiatives to support the SDGs, such as establishing a center for social innovation to promote social enterprise and entrepreneurship and a school for a sustainable built environment to improve urban planning and regulations, and conducting policy research studies to provide a better and more comprehensive understanding of the current issues and solutions, such initiatives often face major resource challenges since the environmental and sustainability field of research is not an area of priority (Velazquez et al., 2005). In Brazil, Brandli et al. (2015) have also documented a substantial lack of financial aid for sustainability projects for universities that are being run mainly based on the goodwill of their faculty members and staff and the work of volunteers. Similarly, respondents from 27 Canadian universities assert that the greatest barrier to sustainable university programs is the lack of funding (Elliott & Wright, 2013).

5. Conclusion

Because of their impact on the economy, society, and environment, the role of the institutes for higher learning in scaffolding a more sustainable society has recently been a research topic that has been increasingly studied by academics (Marques et al., 2019). A few models for sustainable universities have been proposed, but these models are limited in the sense that they need to include an important component of international collaboration and interdisciplinary approaches to dealing with sustainability issues. In the age of globalization and economic connectivity, the sustainable development challenge is not a single or a few countries' problem. It is a cross-border topic of concern and discussion. For example, if we continue to cut down trees in the United States and Canada, the temperature in Europe will also rise, while some cities along the coast in Asia are also likely to submerge completely. Addressing the issue in one country in Asia alone would not be enough. Likewise, in order for the ASEAN Green Deal Initiative to have major impacts, involvement from ASEAN universities is a must, and they have to collaborate to implement the tasks. It should be highlighted that the ASEAN Green Deal is an instrument to address increasing concerns over the discussion on climate change, environmental degradation, carbon neutrality, circular economy, and green technologies in addition to economic competitiveness and education and TVET reform which have always been ASEAN's priorities.

This backdrop gives rise to a framework for a university for sustainability. The model, which comprises four components, including campus operation,

teaching, research, and community engagement, is extended to place significant importance on international cooperation and concerted efforts while extending the conventional roles of a university directly and indirectly beyond just producing the next generation of economically driven executives and skilled workers. Instead, it aims to generate socially responsible and environmentally friendly civic pioneers, leaders, and innovators with transdisciplinary and interdisciplinary competencies to effectively cope with both local and global issues and ensure sustainability and survival in the Anthropocene era, where human activities have led to much damage to the environment and the whole earth. Note that the model recognizes that sustainability and economic growth can go hand in hand, contrary to what was suggested by the Limits to Growth report in 1972 (Meadows et al., 1972). It is worth noting that the report has also received much criticism regarding methods and assumptions. We have also discussed how to incentivize universities to contribute more to sustainability by rearranging how they are ranked and recognized for their efforts. The key constraint remains lack of financing to implement such visions. University's revenue comes from several sources, namely research, the funding for which is, at the moment, very limited to accomplish the tasks at hand and make a serious effect. Therefore, there should be more global and national funding for projects aimed at providing solutions to SDGs. It should be highlighted that research grants should be provided to a consortium of universities from several countries to eliminate the practice of universities working in silos and artificial arrangements in academia.

For Cambodia, as the country embarks on emerging technologies and digital transformation, accelerated industrialization, and rapid socioeconomic development, the university for sustainability model should be put into practice, at least in some universities. A very close collaboration between the government, the academia, and the industries, is also required. This is to (1) address business challenges, (2) encourage and promote industrial R&D and innovation, (3) mitigate the negative impacts of industrial development on the environment, and (4) promote responsible and green development for a more sustainable world. Such complex and multilayered challenges cannot be undertaken without an enabling and supportive environment, including human resources and funding for universities to realize its vision, missions, and objectives, especially those of the ASEAN Green Deal, which was initiated by Cambodia in 2022 the rotating chairman of ASEAN. What comes next would be the execution of the model, which requires the operationalization of all the components, especially campus operations, and the teaching and learning aspects. It involves the design of a new curriculum, restructuring the university's institutional functions, and changing faculty, staff, and students' behavior and perception toward green campuses. The latter conventionally takes quite a lot of time to achieve, for there are always people who resist changes in their way of living.

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Chapter 11: Sustainable Tourism: How to Make It Work?

SENG Touch and ROS Sayumphu*

1. Introduction

People have been debating various aspects of development since the beginning of time. It is fascinating to observe how people have utilized their environment and surroundings to develop the community. Even though the forms of development used in the 21st century have been modernized and a significant emphasis has been placed on environmental protection, it may be difficult to conceive of human development without environmental dependence. In any case, the overexploitation of natural resources has sparked widespread environmental degradation concerns. As a result, people worldwide are looking for ways to reverse, stop, or lessen environmental degradation, which could eventually lead to the extinction of the human race. A multilateralism initiative has gradually shaped the concept of sustainable development through the numerous initiatives undertaken. Sustainable development applies to several industries, including tourism. There has been a shift from “mass tourism” to “sustainable tourism,” which has moderated the tourism industry’s operations. Green deals in Europe can serve as a model for ASEAN to use in unifying the collaborative approach among member states, especially in securing green finance, providing clear roadmaps, rules, and regulations. Even coordinating the green innovation and STI in the region as a means to sustainable tourism can be supported by making a conversion to renewable energy, determining a circular economy, or enhancing green infrastructure (Ciot, 2021). In light of this, Cambodia, despite being a small nation, views its tourism industry as “Green Gold.” Cambodia has adopted various tourism development master plans, strategies, and roadmaps which clearly outline the plan to strengthen sustainable tourism through a variety of measures. This chapter will begin by examining the concept of sustainable tourism, followed by a discussion of sustainable tourism’s challenges and potential solutions to those challenges. Later in the chapter, Cambodia’s experience with sustainable tourism will be discussed.

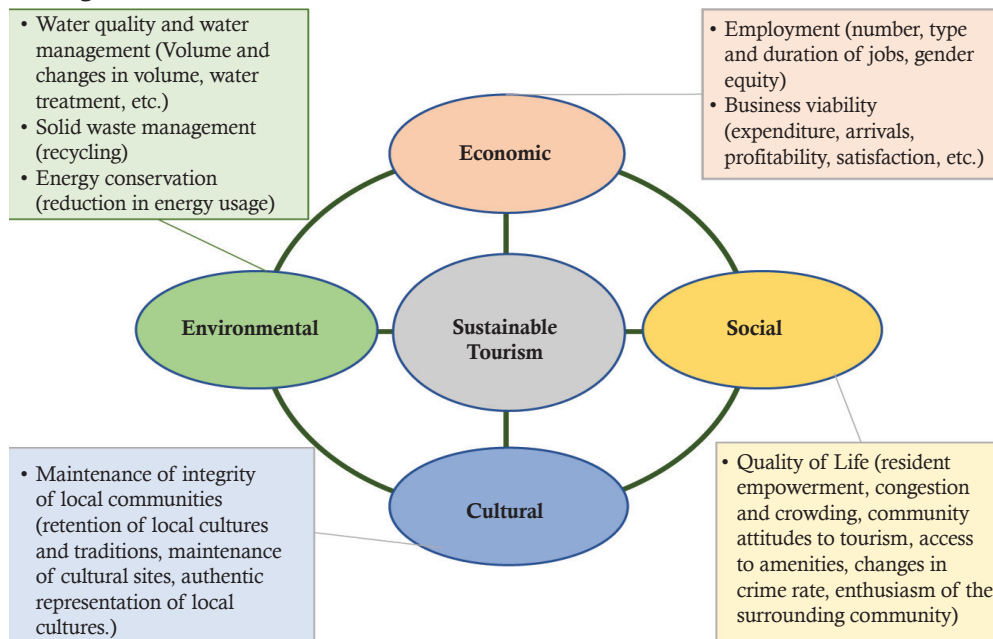
2. Definition and Indicator of Sustainable Tourism

Sustainable tourism includes activities, management, and development that safeguard natural, economic, and social integrity and conserve cultural resources. Thus, implementing sustainable tourism principles requires ensuring viable and long-term economic operations, making optimal use of environmental resources, and respecting the socio-cultural authenticity of host communities (Niedziółka,

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2012). Economically, sustainable tourism ensures viable and long-term economic development that can provide socio-economic benefits to all engaged residents, privates, and the government through stable employment, income-earning, social service provision, and poverty alleviation (Ady et al., 2022; Niedziółka, 2012; Palacios-Florencio et al., 2021). Sustainable development can also reduce environmental damage (Tahiri et al., 2022). In addition, tourism revenue can be used to protect and conserve the environment, making it an essential part of sustainable development (Baloch et al., 2023; Shien et al., 2022). It also unites the government, commercial sector, NGOs, and communities to preserve and monitor the ecological environment, natural heritage, and biodiversity (Baloch et al., 2023; Niedziółka, 2012). Culturally and socially, sustainable tourism may boost social and cultural engagement, especially between visitors and locals. Tourists also help residents and the country market their image, reputation, and culture (Muaini et al., 2021). This is a win-win situation for all parties involved. It conserves the built and living cultural heritage and traditional values and contributes to a more favorable perception of the nation in the long run (Niedziółka, 2012; Palacios-Florencio et al., 2021). Thus, measuring sustainable tourism requires considering the dimensions, indicators and measures in Figure 11.1.

Figure 11.1 Dimension, Indicator and Measures for Sustainable Tourism



Source: Niedziółka (2012) and Wibisono et al. (2021)

3. Problems of Sustainable Tourism

Balancing economic, social, cultural, and environmental perspectives is the most challenging obstacle to achieving effective long-term management and sustainable

tourism development strategies (Ene & Baraitaru, 2010; Palacios-Florencio et al., 2021). The critical success of this effort heavily relies on meaningful participation from all stakeholders. While the government can set rules, regulations, and policies, it needs private actors like hotels, transportation services, and locals who benefit from tourism to implement and lead in consumer satisfaction and environmental protection (Amoako et al., 2022). If a stakeholder only focuses on short-term gains or financial gain while ignoring the environment and social impacts, then the environment will deteriorate. The sustainable tourism goals, particularly the intended long-term sustainability of incomes, the environment, and social-cultural norms for future generations, will not be met (Janusz & Bajdor, 2013; Khan et al., 2021). In Poland, hotels will only switch to clean energy or water-saving equipment if they believe it will lower operating costs (Niedziółka, 2012). In order to ensure the efficacy and productivity required to provide long-term benefits to future generations, the complexities of sustainable tourism must be taken seriously and addressed collaboratively by a broad range of relevant stakeholders and actors.

In addition, as soon as the location for sustainable tourism becomes widely known, hordes of tourists will begin to visit, which may lead to new issues such as water pollution, air pollution, and noise pollution (Roblek et al., 2021). Without proper management, this will undermine sustainable tourism and aggravate the situation. Local traditions have steadily eroded or even disappeared since residents emphasize financial gain and fit their work to international visitors' interests (Poort et al., 2021).

The revenue generated by tourism activities has been accumulated by foreign firms and individuals but not redistributed to locals, which also defeats the purpose of sustainable tourism (Garrigós-Simón et al., 2015). This will worsen the economic, social, and environmental situation, which may result in a decline in the social well-being of the locals and may also contribute to the destruction of the environment or natural resources for financial gain (Andries et al., 2021).

4. Science, Technology and Innovation in Sustainable Tourism

Science, technology, and innovation (STI) play a crucial role in bridging the gap between the economic, social-cultural, and environmental spheres, as well as in enhancing the overall efficacy and effectiveness of the policy (Sharif et al., 2023). Various actors, including the state, the private sector, and locals, can utilize the currently available technological or digital resources to enhance monitoring, generate income, and promote products or destinations (Anderson et al., 2021). The government may utilize digital formats or scientific technologies to enhance a destination's environmental health, electronically monitor and take action against any environmental intrusion, and digitally facilitate international tourists (Anderson et al., 2021; Sulistyo et al., 2022). Private hospitality and food companies can use technology to save money and the environment. They may

limit clean water use, generate electricity with solar panels, or stay ahead with online marketing (Anderson et al., 2021).

Over the last decades, it has been remarkably observed that tourism development has significantly benefited from the development of innovative technologies added to the market (Gössling, 2021). These technologies have significantly improved the efficiency and effectiveness of business operations by sharing information between different departments in the hotel industry (Koo et al., 2015). In fact, the transformation of technology in tourism can be seen in different periods from the 1970s to the present, and the tourism industry has been transformed accordingly (Table 11.1).

Table 11.1 Development of Technologies Applied to Tourism

Time	Key Technology
1970-1980	CRS, GDS
1980-1990	Personal computer and Emails
1990-2000	Internet, Website, Software (e.g., financial and office software), Email, Mobile phones, eCash, Email, Intranet, Internet banking World Wide Web and video conferences
2000-2010	Smartphones, social media, Web 2.0, Text messages, database and computer software, online booking systems, CRS, web/mobile apps, and online video streaming instant messaging
From 2010 to present	AR/VR, web/mobile apps, cloud computing and online data storage, virtual assistants, wearable technologies, social media, Google analytics, Blogs and review websites, collaborative online environments, Web 2.0, Chatbots and instant support, digital platforms

Source: Gössling (2021) and Xiang & Fesenmaier (2017)

5. Cambodia's Sustainable Tourism and Policy Supporting

Cambodia's tourist attractions are impressive despite their size. Tourism drives economic development. International visitors climbed gradually from 4.2 million in 2013 to 6.6 million in 2019. By 2022, approximately 2.2 million international tourists visited this small nation due to Covid-19 (Ministry of Tourism, 2023). The Royal Government of Cambodia (RGC) has put efforts into promoting sustainable tourism by implementing policies and legal instruments, including Cambodia's 2010 National Tourism Policy, Tourism Development Strategic Plan 2012-2020, Roadmap for Recovery of Cambodia Tourism During and Post Covid-19 (2021-2025), Tourism Development Master Plan Siem Reap 2021-2035, and Tourism Development Master Plan Siem Reap 2021-2035. These legal instruments mainly emphasize the necessity for sustainable tourism and STI action plans to attract tourists, smart tourism, smart destinations, and digitalizing tourist firms (Ministry of Tourism, 2021). Recently, RGC has just launched the TourismTech Roadmap, one of the primary legislative tools that propel Cambodia as the

top tourist destination in the region by integrating modern technologies with sustainability and inclusion and offering strategic advice to diversify and add value (Ministry of Industry, Science, Technology & Innovation, 2023). It improves goods and services and generates new economic and social possibilities. These legal instruments seek to make Cambodia's tourism industry a "quality, safe, and sustainable tourist destination." New technologies and creative techniques will help Cambodia's tourist sector adapt to the fourth industrial revolution (Ministry of Tourism, 2021). Furthermore, services and the digital economy are scientific and technical goals; thus, tourism and technology will be long-term priorities. Cambodia also developed STI and established sustainable tourism efforts, including "Clean City, Clean Resort, Good Service and Good Hospitality (Ministry of Tourism, 2021, p. 19)." Despite this optimism, additional resources should be allocated to ensure that public tourism infrastructure is both adequate and fully functional and that collaboration between private and public entities is strengthened (Economic and Social Commission for Asia and the Pacific, 2021).

6. Conclusion and Recommendations

It is essential to emphasize that sustainable tourism cannot be approached uniformly on any level. This strategy will not yield the desired outcomes. If it is not effectively resolved, the concept will continue to be fluid and will amass an increasing number of potential dangers. Various public and private entities and individuals must collaborate to achieve sustainable development. Everyone needs to be on the same page and make progress toward the shared long-term objectives of generating income, protecting and conserving the environment, and preserving social and cultural well-being. Proper education, effective management, and the judicious application of science and technology will undoubtedly contribute to achieving an effective and efficient sustainable tourism industry and reducing undesirable potential risks.

Policies and legal instruments play an integral role in supporting technology use in Cambodia's tourism industry. They should therefore provide clarity and transparency about the rules and regulations surrounding the use of technology in the tourism industry, encourage innovation and competition among tourism operators, enhance technology integration in tourism services to remove any hurdles or barriers when using technology, motivate tourism operators to adopt certain standards that can enhance the quality of their services, provide assurance to potential investors and invest in related tourism businesses in Cambodia. In addition, Cambodian tourism operators require technology adaptation and adoption in response to the future global tourism trends and tourism technology transformation since these technologies will play a crucial role in sustainable tourism in Cambodia.

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Chapter 12: Green Built Environment

*NATH Kananda**

1. Introduction

In the past decades, the planning, design, and construction communities have been embroiled in a vigorous discussion of sustainability and green buildings. The link between green building and climate change is, in a way, inseparable. Many environmental crises greatly influence us to alter the design and practice of the built environment. Pollutions caused by human activities from, for instance, the burning of fossil fuels or the rise of global warming, show that we are now in emergency mode and need to begin to consider reducing our use of energy.

Sustainable architecture is about the assurance of components that will strive, structures with prolonged existence and renewable consumptions of both electricity and water. A green built environment is more than just about sustaining the use of energy. In the early 2000s, it was about doing what was right. From the mid-2010s, considerations were given to client demand, market demand, lower operation costs, and branding/public relations 2010s (Charles, 2016, p. 2). These trends are what mark the rapid movement of green buildings, but how green are they really? To explain what a green built environment is and the strategy to tackle the issue, I will raise some key features that need to be considered.

Recent studies from the U.S. Energy Information Administration showed that the process of constructing buildings is responsible for almost half of total energy usage and carbon emissions (Ching & Shapiro, 2014). This is in relation to the production of materials and the processes of construction, cooling, and lighting of buildings. Thus, architects and designers need to design buildings that incorporate effective utilities, natural lighting, as well as air ventilation.

Before diving into a more thorough explanation, we should take a step back and ask what a green building is. Do architects and designers need to follow a set of rules and regulations to meet the requirements of a sustainable building standard? Or does the building need to have zero or close to zero energy consumption to be considered a green building? The factors that urge people to choose green buildings over a non-green buildings include improving their quality of life. At the same time, some consider reducing expenses to keep up with building utilities. In contrast, some choose to conserve the importance of other historic buildings and to live somewhere that is reasonably priced.

Despite categorizing all the stated goals, there have always been ongoing discussions (Rinkesh, 2020) about the purposes and main priorities of green building. In general, green building projects support either one or more of the

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goals sensibly. However, in most cases, there will always be parts that contradict each other and, in the end, show what is significant to the quality of life that people seek.

2. Primary Essence

The first and foremost thing to consider when we start designing sustainable buildings is their loads, which refers to how buildings shelter their dwellers from various kinds of factors, both externally and internally. Extreme temperatures are the main reasons for human heat and cool buildings. Other external factors include wind, rain, and the sun's ultraviolet effects. Surrounding factors such as animals, insects, and pollution from human activities are also considered loads (International Energy Agency and United Nations Environment Program, 2019). The basic practice is to shield and embed our buildings with components from those loads. In their book, Ching & Shapiro (2014) mentioned that the most commonly used method to cut down on the effects of extreme temperatures is the installation of thermal insulation between wall cavities or right under the roofs. Screenings such as windows, either of glass or mesh net, are selective types that designers use to purposely let in chosen elements and leave out others.

This practice is only fundamental to what green building means. Kriss (2014) noted that "green building is a holistic concept that starts with the understanding that the built environment can have profound effects, both positive and negative, on the natural environment, as well as the people who inhabit buildings every day". The design of the components mentioned earlier completes most of the goals to improve human health and comfort. However, if not tackled properly, it would lead to further negative effects on the environment. For instance, all those thermal insulations need to be manufactured, using raw materials, energy, and carbon emissions and produce waste, which in the end only ends up making the project less green than it should be. Thus, hereinafter, this chapter will focus on the design process itself, material design, other utilities, and waste management systems in architectural green built environment.

3. Integrated Design

Alternatively, incorporated design is a new approach to building sustainably from the foremost stage to finishing the construction. In order to do this, many involved participants need to work collaboratively with each other to confirm that every component contributes to green design. In the later stages, architects, owners, and other consultants will need to address the true aims and concept of the project. The team needs to clarify as much detail as possible on some of the components of the building, such as the number of residents and how much of the space is needed to have wall insulation or other temperature control methods. With early discussions and confirmations on these steps, the expenses on many other utilities and materials can be significantly reduced.

Contrary to the earlier obstacles, if an assessment can be made prior to these assumptions on the operation costs of a sustainable building during its long uses, we can confirm that the utility expense of a sustainable building is generally less than that of a normal building. For example, Asim et al. (2022) noted that a carefully calculated thermal temperature of a room can significantly reduce the cost of an air-conditioner. Lowering the ceiling of a building may reduce the cost of materials for the construction, and with more natural lighting, it would also help reduce the cost of energy needed during the day.

4. How Can We Build a More Efficient Structure?

The primary goal of starting a project is to choose a site location. Leadership in Energy and Environmental Design (2020) indicates in their rating system that it is vital to consider the impact and effect when constructing buildings on the perimeter of sensitive sites, such as conservative historic sites or other fragile forests and greenery. Another aspect is to think about the linkage between the site and its nearby community to reduce the use of energy for transportation. Usually, when initializing the process of a project, owners start conceptualizing how the interior space works and how it would look aesthetically, both externally and internally. In sustainable design, instead of working on the interior and exterior of the building, architects and designers would instead visualize how the whole building interconnects with the surrounding community, as well as the availability of public transportation, shopping centers or grocery stores, schools, or public parks.

In order to decrease the long-term risks to an individual building, we seek to solve the issue of all relevant energy consumers at the site location. Besides the tenant living in a building, there are more people who consume energy and release carbon emissions by delivering services. So, how exactly can we build a more efficient structure? Besides the site selection and its community, we should redirect our focal point to the design of the building shape. Reducing and reforming the shape of a building through the use of floor heights, wall openings, and open space can have significant influences on the cost of construction materials and the overall energy consumption of the building. Based on this, buildings in mountains and colder temperature areas can be built with much lower ceilings as there will not be too much heat from the roof into the rooms.

5. How Can We Manage Materials?

Building materials have an adverse effect on the environment because of their energy consumption and emissions, the depletion of limited resources, and the unintentional build-up of waste in landfills. An article, *Nature Communications*, by Zhong et al. (2021), shows that the mining and harvesting of raw materials, the processing and production of finished goods, the transportation of resources, the usage of hazardous compounds, and the production of construction debris

are the activities responsible for these effects. We may significantly lessen these effects by choosing the quality and quantity of building materials wisely when designing the building. From a quantitative perspective, using less material is also a greener option. Typically, our production processes begin with energy-intensive mining, crushing, smelting, refining, and shaping. Next, additional stages of treatments, shielding coatings, and adhesives are frequently added. Unfavorably, this intense process is different from the manufacturing that occurs in nature. It seems like a worthwhile objective to strive to mimic nature's efficiency in our manufacturing processes given our current problems with resource depletion, a growing population, and climate change.

In the context of architecture, as shown in the book, *Biomimicry in Architecture* by Pawlyn (2016), studies show that where these issues may be more clearly distinguished, and where conventional production methods and drawing packages both support this distinction, the idea that structure and materials are inseparable in nature requires a fundamentally new way of thinking. Structure and materials are grouped together in nature via hierarchy. Visualizing various bridge designs could be a useful method to start understanding hierarchy. Two sturdy steel beams placed on piers at either end might be used to crossing a minor distance. This would be a non-hierarchical monolithic approach. A pair of steel trusses would be a more effective approach to cross the distance. That denotes a single hierarchy level. Imagine going a step further and turning each compression element into a little box truss and each tension member into a cable made of stranded steel. That would indicate a hierarchy with two tiers. The structure becomes more effective in terms of the quantity of material needed to accomplish a particular goal as the hierarchy levels rise.

The site selection stage, which is not concerned with the design of the building itself but rather with its placement and infrastructure, offers a significant potential for material savings. One can utilize the infrastructural elements that are already in place by placing structures in already established locations. Since they are shared, infrastructure like roads and municipal water systems is essentially reused, avoiding the potential material consumption of building new infrastructure. Opportunities for material reduction may be found when careful structural design is done rather than employing rules of thumb or conventional wisdom. For instance, as mentioned in the book, *Building Structures Illustrated* (Ching et al., 2013), depending on the load distribution, the amount of beams and columns needed can be reduced.

Avoiding finishes where they are not required is another strategy for cutting down on material consumption. An example of this method, which is also known as "structure as finish" (Canadian Wood Council, n.d.) enables structural components to fulfill both the functions of structure and finish. If the exposed structure has a low lighting reflectance, trade-offs with the demand for additional illumination should be carefully considered. Another strategy for conserving

resources is to leave visible parts like ducting or plumbing. Gypsum board and related furring for chases, soffits, and plenums, as well as finishes, are used less frequently as a result.

6. How Will We Manage Energy and Water?

6.1. Energy

One of our biggest problems is energy, which is caused by a combination of poor strategic planning and the more serious reality of climate change. Over the coming decades, how will we decarbonize our economies? What does this mean for the layout of cities and buildings? Let's start with the most divisive tenet: where should our energy originate from? The energy flows in nature reveal that all of the energy needed by living things is now generated by the sun. Could we make the same transition from a global economy based on fossil fuels to one based on solar energy instead of nuclear energy? A statistic from the International Energy Agency (2010), reports that every year we receive energy from the sun that is around 10,000 times greater than what we now consume. This abundant source of energy has kept life on earth going for billions of years and has enough energy to last an eternity (MacKay, 2009). Future energy sources should be available other than nuclear-powered plants. For instance, we could meet all of our energy requirements by erecting concentrated solar power plants across around 5 percent of the world's deserts.

A solar economy is one that relies only on renewable energy sources like concentrated solar power and direct solar energy, particularly photovoltaics, wind, waves, biomass, and infrared radiation from the sun. Tidal and geothermal energy are complementary natural resources.

As a first step in any logical strategy to reduce greenhouse gas emissions, substantial improvements in efficiency are necessary, and advancements in the built environment provide some of the largest prospects. Even if there are still many obstacles to overcome, the solar economy is now conceivable on a practical level, and renewable energy technologies are developing and becoming significantly more affordable. As a result, tackling our energy concerns might launch the most inventive period in human history if done correctly.

6.2 Water

As climate change has become undeniably real, the subject of water is becoming more and more divisive on both an environmental and political level. Reducing water use, collecting rainfall, and reusing water are three important techniques that can assist us in managing our water resources effectively. The most efficient way to reduce water usage is to first look at water loads. This is possible because of newly developed fixtures and appliances that consume less water while yet producing the same results.

It is possible to utilize rainwater in buildings instead of city water or well water by collecting it. A conveyance system for moving rainwater to storage is part of a rainwater harvesting system. This conveyance system often involves the building roof, a cistern or storage tank, a filtration system, and perhaps a disinfection system, a backup system to supply water during dry spells, space for overflow, and a delivery system to get the water to the water loads. For the collected water to not freeze in cold areas, the storage must be inside or underground. Various materials, such as steel, concrete, wood, fiberglass, or plastic, are used to construct tanks (Texas A&M AgriLife Extension, 2013).

Similar to how we utilize fossil fuels, many of our typical methods for dealing with water suffer from a technical laziness that stems from the same belief in an endless supply that defined our view of resources at the beginning of the Industrial Revolution. For our soils to recover their fertility and to ensure long-term food security, especially in those regions that may see less rainfall as a result of climate change, we must rethink our water treatment systems.

7. Is Zero-Waste Systems Achievable?

The concept of waste as a whole is riddled with inconsistencies. In an article by Bergstrom (2020), she explained the management and key issues of waste in construction field work. Though unattractive, it has a great deal of potential for creating closed-loop systems that are highly desirable. Designers tend to overlook it, but projects that investigate it show incredible innovation. It uses the term “waste,” which may mean both useless material and anything that indicates the possibility of something. We should try to reuse materials wherever feasible in order to reduce the embodied energy needed to gather and process new materials as well as the depletion of raw material sources. It is recommended to use recycled-content materials. During production, pre-consumer recycled materials are removed from the waste stream. Materials made from post-consumer recycled products are derived from garbage produced by end consumers. The most popular building material with recycled aggregate is crushed concrete that has had the reinforcing and other embedded elements removed (United States Green Building Council, 2009).

Is waste water actually wasted? Water is a crucial vector in the worldwide cycle of all nutrients, so taking a cool, strategic look at this process might have a big impact on how we handle “waste” water. Water may now be separated into grey and black water from human usage. Grey water is that which accumulates from sinks and showers and may be used to maintain the landscaping and gardens. Black water, commonly referred to as sewage water, may be processed and utilized for agricultural and landscape upkeep.

8. Green Beauty, Green Built Environment

The first consideration in architectural design has always been aesthetics. This is the gold standard for green building design, and, in some respects, it could even be more crucial. Why must buildings be visually appealing? Calm is evoked by beauty. Pride is fueled by beauty. A sense of order is brought about by beauty. Our relationship to nature can be facilitated by beauty. The immense potential for achieving harmony inside, among, and with the rest of the world is expressed in beauty. We allow the poets to continue defending beauty. We operate on the basis that aesthetics matter.

Beauty is frequently a matter of perspective. When comparing the traditional and contemporary definitions of beauty, we might also consider the aesthetic value of a building's performance. Perhaps a structure with low energy consumption is beautiful. In the winter, a building without ice dams hanging from its roof looks stunning. A peaceful structure is lovely. These are all traits of high-performance green structures. Buildings should have beauty that goes beyond the surface. New elements introduced by green design, such as solar panels, must be elegantly incorporated into structures. These parts are a thing of beauty for many of us, but not everyone may feel the same way. As designers, it is our responsibility to make sure that these elements are combined in a way that is both visually pleasing and functional.

A new design, namely a green architecture, is necessary in light of the numerous negative effects of climate change and other environmental challenges. Building energy consumption has been highlighted as a significant contributor to greenhouse gas emissions as well as a significant potential to minimize these emissions. Those in the design and construction industry have to decide whether to take responsibility for the climate change effects of buildings or to drive the change required to lessen those effects.

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Chapter 13: A Farm-To-Fork Strategy: An Approach Towards Sustainable Food System

SRUN Marnick, SOTH Sereyboth, and CHHEM Kieth Rethy*

1. Introduction

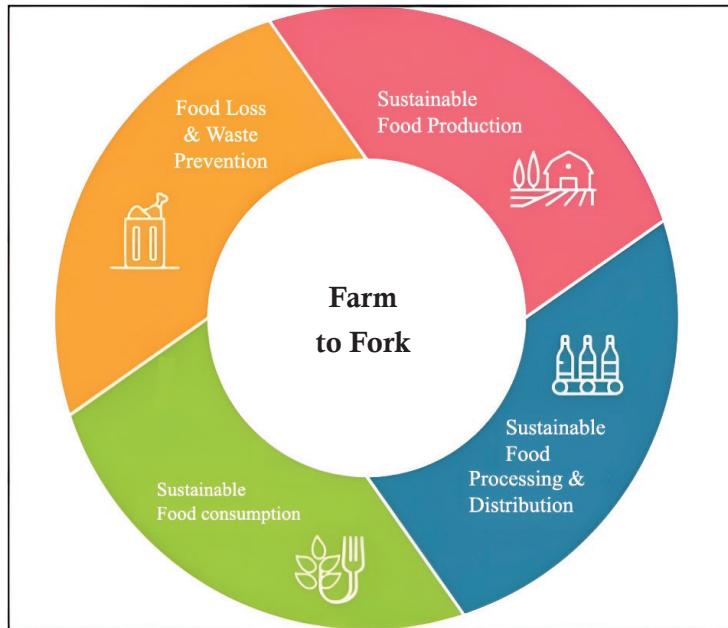
The interrelationship between climate change and food systems is a complicated issue that negatively affects the world. On the one hand, rising temperatures, droughts, and heat waves caused a decline in crop yields and livestock, and decreased fisheries and agroforestry, which is susceptible to our food systems (Food and Agriculture Organization of the United States et al., 2017). However, food systems have also contributed to climate change. Anthropogenic activities in each stage of the food supply chain, commencing with production, manufacturing, distribution, consumption, and disposal, have caused an increase in greenhouse gas emissions (GHGs).

Food systems contribute significantly to global GHGs, with industrialized countries accounting for 27 percent and developing countries for 73 percent of these emissions (Crippa et al., 2021). The main greenhouse gas emissions in the agricultural sector are methane, nitrous oxide, and carbon dioxide. Methane is generated by activities such as ruminant digestion, rice farming, and managing manure. Nitrous oxide is released from natural processes in agriculture and manure management. Carbon dioxide emissions result from food processing and transportation, which rely on the use of fossil fuels (Sjörs et al., 2016). The environmental and health impacts of these emissions highlight the importance of incorporating food policies to achieve the Sustainable Development Goals (SDGs).

In order to establish a balanced food system, the Farm-to-Fork (F2F) strategy should be considered. The Farm-to-Fork strategy is a significant element in the EU Green Deal, which examines the issues of sustainable food systems from production to consumption. It acknowledges the strong connections among healthy individuals, societies, and a healthy planet (Mowlds, 2020). The Farm-to-Fork Strategy is specially designed to meet six main goals: 1). sustainable production; 2). food security; 3). sustainable food processing; 4). promoting sustainable consumption; 5). minimizing food waste and loss and 6). reducing food fraud along the supply chain. To achieve these goals, the food supply chain from production to processing and distribution to consumption must be taken into account. Figure 13.1 illustrates the farm-to-fork strategy in more detail.

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Figure 13.1 Farm to Fork Strategy – For a Fair, Healthy, and Environmentally-Friendly Food System



Source: European Union (2021)

2. Adopting the Farm-to-Fork Strategy in the Cambodian Context

Cambodia is a country that is highly vulnerable to climate change in Southeast Asia. With 79 percent of its population residing in rural areas and depending on agriculture for both livelihood and food security, climate change has a significant effect on the country. The country undergoes annual droughts and floods, and it faced severe droughts that ruined approximately 41,000 hectares of rice fields in 2015. Similarly, floods in 2011 caused damage to about 171,000 hectares of rice fields (Mishra et al., 2018). These events are predicted to increase in frequency and intensity in the coming years. If this persists, Cambodia's agricultural productivity, the foundation of its food security, will be impacted, eventually leading to a disruption in the country's entire food system. Therefore, building a resilient food system that can adapt to climate change is vital for a country like Cambodia.

The Farm-to-Fork Strategy is in alignment with ASEAN's and Cambodia's sustainable development goals and food security objectives. It serves as a significant mechanism for realizing Cambodia's vision of attaining high-middle income status by 2030 and becoming a high-income country by 2050, while also

contributing to the achievement of its SDGs. As previously mentioned, the farm-to-fork process encompasses all aspects of food preparation, from production to consumption. The following section will provide a brief overview of the six components: 1) Sustainable agriculture; 2) Ensuring food security for all; 3) Sustainable food processing; 4) Sustainable consumption; 5) Reducing food waste; and 6) Diminishing food fraud along the supply chain.

3. Building a Sustainable Agricultural Production

The majority of agricultural practices still rely on rainfall, and most farmers cultivate only a single crop annually. Traditional techniques and seeds are still widely employed, which makes Cambodia's rice yields low and impedes the improvement of local livelihoods (Ros et al., 2011). Furthermore, the supply of other agricultural products, particularly fresh produce, remains limited and seasonal, with approximately 70 percent being imported (United States Agency for International Development, 2015; Sim et al., 2021). As such, it is essential for all relevant stakeholders, particularly the public sector, to consider modernizing the agricultural sector through agricultural technology (Agri-Tech) and utilizing bio-products.

Agri-Tech is transforming the way farmers operate around the world, including in Cambodia. Automated systems and sensors can help farmers monitor crop yields, optimize production output, and manage soil health with minimal manual labor and human error. For instance, with Agri-Tech, tasks like controlling irrigation levels and managing soil health can be accomplished with the click of a button, minimizing manual labor and human error as well as reducing labor costs (Ahmad & Sharma, 2023). Agri-Tech also has applications for tracking livestock health and behavior, enabling farmers to make informed decisions about animal welfare. By reducing farmers' workloads and promoting sustainable practices, Agri-Tech has the potential to revolutionize the agricultural industry.

Bioproducts such as biopesticides, biofertilizers, and biostimulants are important for sustainable agricultural production. These products are derived from plants, animals, and microbial sources and are safer and more environmentally friendly than traditional fertilizers that contain potentially toxic chemicals (Upadhyay et al., 2020). Bio-products can help maximize crop yields and replenish soil health while providing essential nutrients to crops. Incorporating Agri-Tech and bio-products into farming practices can help farmers produce healthier and better-yielding crops with less risk of contamination or harm to the environment, leading to more sustainable practices in the agricultural industry (Spanner & Napolitano, 2015).

Incorporating Agri-Tech and bioproducts into farming practices has the potential to transform the agricultural industry, in line with the Agriculture 4.0 roadmap and the agricultural sector's plan for 2030 (Ministry of Industry,

Science, Technology & Innovation, 2022), aimed at promoting a sustainable and efficient agricultural industry.

4. Ensuring Food Security

Food security is achieved when all individuals have consistent access to enough safe and nutritious food that meets their dietary requirements and preferences, allowing them to maintain good health and an active lifestyle. However, Cambodia, with its diverse food systems and rich culinary history, faces several challenges in achieving food security.

One of the major obstacles is the country's vulnerability to climate change and natural disasters, which frequently disrupt agricultural production and threaten food availability. Rice, as the staple food in Cambodia, plays a vital role in providing calories and protein to the population. However, droughts and climate anomalies pose significant risks to rice farmers, leading to production losses and impacting overall food production. Cambodia also grapples with persistent inequalities, despite progress in poverty reduction. A significant portion of the population remains unable to afford a healthy diet, exacerbating food insecurity. Unfortunately, the ongoing COVID-19 pandemic has further heightened these challenges. The global food supply chain disruptions have impacted both food availability and affordability, affecting the overall food security situation in Cambodia.

In this context, it becomes even more crucial to prioritize the promotion of agriculture and the adoption of healthy dietary practices. By strengthening Cambodia's food systems, the country can enhance its resilience to external shocks and ensure food security during and after the pandemic. This entails investing in climate-resilient agricultural practices, providing support to rice farmers, and addressing inequalities in access to nutritious food. Moreover, efforts should be made to improve the efficiency and stability of the food supply chain, ensuring that safe and nutritious food reaches all segments of the population.

5. Encouraging Sustainable Food Processing

Food processing, an integral part of the farm-to-fork process, includes a range of activities that typically use abundant materials and energy. The majority of food processors in Cambodia uses traditional processing methods that rely on non-renewable energy, contributing significantly to carbon emissions and exacerbating the negative environment. To address this challenge and transition towards a more sustainable food system, the adoption of the Food Processing 4.0 concept is increasingly being seen as a viable solution.

The Food Processing 4.0 concept emphasizes the application of Industry 4.0 technologies to process food in this digital age, with the aim of enhancing food quality and safety, decreasing food processing expenses and time, conserving

energy and resources, as well as reducing food loss and waste (Hassoun et al., 2023). This includes innovative technologies such as robotics, smart sensors, artificial intelligence, and the Internet of Things (IoT) that help the food industry optimize its processes, improve efficiency and reduce its environmental footprint.

One such approach is the use of smart sensors and IoT devices throughout their production line to monitor and analyze data in real-time, such as temperature, pressure, and pH levels. With this information, they are able to fine-tune their production processes to reduce energy usage, minimize waste, and improve the quality of their products. Integrating the Time-Temperature Indicator (TTI) in food packaging, for instance, can monitor the temperature and time that a product has been exposed to. This TTI utilizes color changes that are not reversible, which allows consumers to quickly check if the product is safe and prevent unnecessary food waste due to premature discarding. Additionally, they can promote optimal storage and transportation conditions, which can reduce the likelihood of food spoilage and waste (Ahmed et al., 2018; Wang et al., 2015).

Integrating Food Processing 4.0 into Cambodia's food industry can support the government's Industrial Development Plan (IDP) and the National Strategy for Food Security and Nutrition (NSFSN). It promotes industrial growth, diversification, and sustainability while facilitating access to technology, credit, and markets for small and medium-sized enterprises (SMEs).

6. Promoting Sustainable Consumption

Our current diet and eating patterns contribute significantly to our carbon footprint, making the consumption stage the most impactful in the food system. Unfortunately, unsustainable food choices often involve unhealthy diets characterized by high consumption of processed, calorie-dense foods and animal products, further exacerbating the carbon footprint. By transitioning to a sustainable and healthier diet, we can mitigate these impacts and pave the way for a more sustainable future, benefiting both our well-being and the planet.

Information dissemination is vital to promote sustainable food consumption. Educating consumers about the environmental impacts, local and seasonal food options, and benefits of plant-based diets empowers them to make informed decisions (Ammann et al., 2023). This knowledge enables individuals to understand the broader consequences of their food choices and encourages engagement in sustainable practices. Additionally, the farm-to-table movement should be widely promoted as part of sustainable food consumption efforts. It prioritizes locally sourced and seasonal ingredients and reduces the carbon footprint associated with transportation. Restaurants in this movement highlight flavors of locally sourced produce, meats, and dairy, supporting nearby farms and minimizing GHGs. It encourages organic and sustainable practices, such as avoiding synthetic pesticides and fertilizers, minimizing water usage, promoting

biodiversity, contributing to soil health, water conservation, and ecosystem preservation.

A market-based approach promotes sustainable food consumption through certifications and labels. Organic and fair trade certifications ensure environmental practices and fair labor. Carbon footprint and water usage labels aid consumers in selecting low-impact products. These market signals incentivize producers to access niche markets. Collaboration between producers, retailers, and policymakers is crucial. Retailers promote sustainability through placement and partnerships, while policymakers support adoption through regulations and incentives.

In conclusion, market-based approaches, through certifications and labeling schemes, empower consumers to make sustainable food choices. Collaboration between producers, retailers, and policymakers is crucial to creating a supportive market environment. By combining information dissemination and market-based strategies, we can build a sustainable food system that benefits human health and the environment in the Anthropocene era.

6.1. Minimizing Food Waste and Loss

Food waste is a significant contributor to global GHGs, accounting for 8-10 percent of the total amount (United Nations Environment Programme, 2021). In Phnom Penh, research has shown that households, shops, and markets are responsible for 70 percent of food waste, with fruits, vegetables, sugarcane, coconut shells, and maize comprising 76 percent of this waste (Chinda & Thay, 2020; Sang-Arun et al., 2011). These findings underscore the importance of addressing food waste in Cambodia and highlight the need for stakeholders, particularly consumers, to play an active role in reducing waste.

To tackle this issue, Cambodia needs to develop regulations and enhance public understanding. Experts recommend implementing incentives for recycling in restaurants and strengthening food bank capabilities. Following Europe's example, extending product shelf life can also be beneficial. Raising awareness is vital to changing consumer behavior and reducing emissions from waste disposal. The government should take part in implementing regulations and establishing food banks, while youth groups can lead awareness campaigns. Ensuring coordination is crucial for the efficient distribution of food from retail shops to those in need.

Moreover, adopting a circular economy model would greatly benefit Cambodia's waste management system. This approach focuses on minimizing waste and maximizing resource value through practices like reuse, recycling, and recovery. Implementing circular practices such as composting organic waste or repurposing food scraps for animal feed or energy generation can move Cambodia closer to a zero-waste system.

6.2 Reducing Food Fraud Along the Supply Chain

Food fraud poses a significant obstacle to achieving a sustainable food system, as it leads to consumer misinformation and undermines both food safety and fair market practices. In Cambodia, the Consumer Protection, Competition and Fraud Repression Directorate-General (CCF) is currently tasked with inspecting food products available on the market. Consumer protection is a pressing topic in Cambodia and the ASEAN region, and existing frameworks governing this area include the United Nations Guidelines for Consumer Protection, the ASEAN Strategic Action Plan for Consumer Protection (2016-2025), and the government's Rectangular Strategy - Phase IV.

7. Cambodia's Progress Towards a Sustainable Food System: An Overview of the Current Situation

Cambodia has started a great move so far in building some policy frameworks and roadmaps to support a resilient, sustainable, and efficient food system, as follows:

- Cambodia has adopted Agriculture 4.0, which involves the use of smart greenhouses, drone technologies, smart irrigation, and a digital marketing platforms. Recently, the AgriTech Roadmap was launched by the National Council of Science, Technology & Innovation and aims to fully adopt Agriculture 4.0 technologies by 2030.
- The Ministry of Agriculture, Forestry, and Fisheries has partnered with Agri-Sambath Khmer to build 10,000 smart greenhouses by 2030, aiming to produce 400,000 tons of vegetables annually.
- Digital marketing platforms like Chamkar, Tonlesap, CamAgriMarket, and Angkor Salad Apps connect farmers with customers.
- Oxfam has implemented blockchain technology to link small-scale rice producers with buyers in the Netherlands.
- The International Fund for Agricultural Development supports digital technology start-ups through the SAAMBAT project, accelerating agricultural value chains in Cambodia.
- Cambodia's Food Systems for Sustainable Development 2030: Cambodia's vision for 2030 focuses on expanding access to health services, sustainable agriculture, and empowering youth and vulnerable groups. The four priorities are healthy diets, empowerment, resilient livelihoods, and inclusive governance. This vision will drive innovation and infrastructure development, leading to a sustainable and inclusive food system.
- Cambodia's National Strategic Development Plan (NSDP) (2019-2023): The plan will encourage sustainability and resource efficiency by implementing the concept of sustainable consumption and production.

- Cambodian Sustainable Development Goals (CSDGs) Framework (2016-2030): The framework aims to align national development priorities with the global SDGs, guiding the country towards sustainable development that leaves no one behind.
- Food Safety Law, endorsed by King Norodom Sihamoni, 18th May 2022: The Food Safety Law consists of 11 chapters and 43 articles, providing the necessary legal framework and mechanisms to regulate and ensure the safety, quality, and sanitation of the food production chains and businesses.

8. Recommendations

To enhance the adoption of these policy frameworks and effectively implement the Farm-to-Fork Strategy, the Cambodian government should focus on:

Investing in human resources:

- Enhance stakeholders' understanding and expertise in the farm-to-fork approach, sustainability considerations, and their advantages.
- Focus on education, awareness, and skill development.
- Engage and support the younger generation in STEM subjects.

Allocating resources for research, innovation, technology, and investments:

- Speed up the transition to sustainable, healthy, and inclusive food systems.
- Direct financial support towards the agri-food sector and the green and digital transformation of farms.
- Improve technology access and transfer to enhance productivity.
- Increase investment in R&D (currently, one percent of GDP is allocated for R&D by 2030).

Enhancing communication channels:

- Ensure effective communication between national policies, local authorities and farmers.
- Establish feedback mechanisms to facilitate coordination and collaboration between different stakeholders.
- Avoid a lack of coordination that can limit the effectiveness of the strategy.

Strengthening policy frameworks and enforcement:

- Implement strict regulatory frameworks to support the farm-to-fork strategy.
- Ensure strict enforcement of the recently endorsed Food Safety Law (2019) to safeguard consumers and maximize the strategy's potential.

Fostering collaboration and avoiding working in silos:

- Encourage the government, farmers, and processors to collaborate and share information.
- Address the mismatch between market demand information and crop quality data.
- Minimize food waste and seize economic opportunities by promoting cooperation among stakeholders.

Improving infrastructure:

- Invest in adequate transportation networks, storage facilities, and processing plants.
- Strengthen infrastructure to support the farm-to-fork strategy effectively.

Ensuring access to clean water and reliable electricity:

- Support food processors' operations and livelihoods by ensuring access to clean water and reliable electricity or energy sources.

9. Conclusion

The Farm-to-Fork Strategy is a sustainable and eco-friendly food system that covers all aspects of food production, processing, distribution, consumption, and waste reduction. In Cambodia, where agriculture is a significant sector, establishing a sustainable food system is essential for future generations. Although Cambodia is still in the early stages of implementing sustainable measures in its food system, it is crucial to continue investing efforts to overcome the challenges imposed by climate change. By embracing the Farm-to-Fork Strategy and implementing comprehensive measures throughout the entire food supply chain, Cambodia can effectively work towards its sustainable development goals, ensure food security, and minimize the adverse environmental impacts caused by its food system.

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Chapter 14: Youth and Sustainable Development in the World and ASEAN

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1. Introduction

Global peace and stability are becoming progressively endangered by both human-driven and natural causes such as climate change, resource depletion, geopolitical tensions and socioeconomic disparities, amongst others. The potential ramifications of these issues will transcend future generations if they are not given urgent and serious attention now. Given that their future is at risk, young people naturally play a critical role in sustainable development. Rich in innovative ideas and brimming with creativity, it is today's youth population that is most capable and has the greatest desire to construct a better world for themselves and generations beyond. At the international level, the United Nations (UN) has been and continues to be a crucial agent in facilitating sustainable development in the twenty-first century, ensuring that processes of sustainability are provided with direction and supported with structural means to achieve success. The Association of Southeast Asian Nations (ASEAN) has also shown a strong commitment to empower and onboard youth as a driving force for the region's long-term development. However, there continues to be much untapped potential in many areas across the globe, including ASEAN. It is important to understand the different contexts in which sustainable development can best be advanced, and how today's youth can contribute within these contexts. It is equally crucial to establish settings that empower and equip historically marginalized groups with the necessary measures to transform desires to make the world a better place into tangible and productive actions. This chapter opens with a discussion of the current state of youth and their role within the 2030 Agenda for Sustainable Development. It then identifies the areas of sustainability where youth engagement and leadership can be amplified. Finally, the chapter concludes with an assessment of the ASEAN Green Deal, highlighting the opportunities in which Southeast Asian youths can further engage as essential stakeholders in new processes of sustainable development.

2. Current State of Play of the UN SDGs and Youth

Adopted in 2015, the UN Sustainable Development Goals (SDGs) constitute the latest global call for action to address the myriad of development challenges facing countries worldwide. The SDGs seek to dissolve the divide between the developed and developing worlds by engaging the efforts of countries at all

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income levels. Building on the momentum of the Millennium Development Goals (MDGs), which primarily targeted poverty alleviation, the SDGs set out a much more comprehensive global program that focuses on sustainable development. Comprising 17 goals, 169 targets, and over 300 indicators that aim to combat social, economic, and ecological issues, the SDGs emphasize the importance of inclusivity and a united effort to ensure prosperity for all. The precedent for this broadened agenda not only comes from a heightened understanding of global interconnectivity but also from a newfound recognition that the world has entered a modern geological epoch driven by human impact on global environmental changes – the Anthropocene (Griggs et al., 2013; Sachs, 2012). At the same time, there is an increasing awareness that implementation of the 2030 Agenda necessitates the empowerment of youth participation in propelling sustainable changes. Leveraging the energies of young people can bring about innovative ideas that have the potential to revitalize often stagnant or unimpactful processes of development.

The UN system itself has demonstrated an active commitment to enhance youth engagement in the SDGs and relevant processes. In 2018, UN Secretary-General Antonio Guterres launched ‘Youth2030, the UN Youth Strategy’ to strengthen youth participation. Although Youth2030 progress reports since 2021 have revealed improvements, challenges such as normalizing meaningful and continued participation, acknowledging diversity among youth, and engaging marginalized young people continue to impede youth involvement (United Nations Youth2030, 2023). To combat these persisting obstacles, the UN established a Youth Office that aims to empower participation in development processes by gathering ideas and insights from young people to better inform UN activities. Such a level of commitment by the UN to positioning youth at the core of twenty-first century sustainable development is also evident by the fact that the SDGs themselves are anchored within a rights-based framework promulgated by several international declarations, including the Convention on the Rights of the Child (CRC). Effective since 1990, the CRC recognizes the capital and potential of young people in tackling pressing global challenges and situates them as key stakeholders in development processes (UN Commission on Human Rights, 1990).

The flexibility of the SDGs – in the freedom that it gives states to interpret the goals and implement action – provides opportune spaces for young people and youth-led initiatives to accelerate sustainable development. Historical trends suggest that young people have been generally marginalized from high-level decision-making processes (Bastien & Holmarsdottir, 2017, p. 7). However, many current national development strategies are now underscoring the importance of their participation in driving transformative change to socioeconomic systems. In 2023, the High-level Political Forum (HLPF) on Sustainable Development published a Voluntary National Reviews (VNR) Synthesis Report highlighting

that several countries, including Bahrain, Bosnia and Herzegovina, Brunei Darussalam, Cambodia, Iceland, and Lithuania, actively sought youth involvement, outlining diverse initiatives in their VNRs. Other countries also found creative ways to engage young people in national-level processes. Guyana created a Youth Advisory Council, Turkmenistan appointed Young SDG Ambassadors, and Timor-Leste solicited advisory messages from youth addressing various development issues and their expectations for implementation (High Level Political Forum on Sustainable Development, 2023). At present, several countries across the globe have placed serious attention towards empowering youth participation in decision-making processes related to the SDGs.

ASEAN member states in particular have shown sustained efforts in establishing and implementing development policies and programs that tap into the potential of young people. The ASEAN Youth Development Index (YDI) was launched in 2017 under the ASEAN Work Plan on Youth 2016-2020 to support such efforts. Based on seven domains, including education, health, participation and engagement, and more, the YDI serves as a measurement of progress in youth-focused initiatives across ASEAN. The most recent report of the YDI recorded an increase from 0.535 in 2013 to 0.606 in 2022. Cambodia showed the greatest improvement, with its YDI increasing from 0.440 to 0.568 in the same period (ASEAN, 2023). While we have witnessed improvements to youth engagement in sustainable development across all ASEAN member states, certain areas continue to hold opportunities for greater progress.

The state of the world that we see today is one that is unequivocally most conducive to successful youth engagement in sustainable development processes. Youths have a plethora of opportunities to actively engage in development processes. Social media – its proliferation, immediacy and reach – enables young advocates to raise awareness and mobilize collective action towards a more sustainable and environmentally conscious future. Educational institutions at all levels are increasingly integrating sustainability into their programs and curricula, equipping youth with the knowledge and skills necessary to address contemporary development challenges. Such learning experiences nurture both human-human and human-nature relationships that lead to a deeper sense of environmental responsibility among young people (Sterling, 2017). Social entrepreneurship provides a space for young business leaders to align economic activities with sustainable goals. Technology can be leveraged by young innovators to develop solutions and products grounded in sustainability. These opportunities collectively contribute to building a generation of informed, empowered, and proactive individuals dedicated to sustainable development. As the contemporary landscape continues to evolve, youths play a vital role in shaping a more sustainable future for themselves and the coming generations.

Although there are many opportunities for youth engagement in sustainable development, multifaceted challenges exist that continue to impede progress. A

significant hurdle lies in the lack of awareness and understanding among youth about the technical intricacies and importance of sustainable development. A study done by Arnold and Barbara Pabian (2023) revealed that while social media serves as a powerful tool to disseminate information, content on sustainability circulating across platforms is limited and often superficial or lacking depth. This results in a persisting lack of knowledge that subsequently translates to a lack of responsibility and action among many young individuals towards engaging in sustainability. Moreover, the prioritization of immediate concerns such as education, employment, and simply, the trials and tribulations of growing up, often takes precedence over long-term goals for sustainable living. In many cases, sustainable lifestyles are viewed as costly and difficult to maintain. Research by leading consulting firm Deloitte United Kingdom (2022) finds that 52 percent of consumers are hesitant to adopt greener purchasing choices due to high costs. Eco-friendly products serve as alternative consumption options that are much more environmentally and socially beneficial relative to conventional products. But these added benefits often come with heftier price tags, making them, on average, 75 to 85 percent more costly. The highest price markups can be found in fashion, beauty, and food products (Kearney, 2020). Ironically, these are the industries that resonate most with the present-day youth demographic. Socioeconomic inequalities, coupled with a lack of inclusivity, can further deter segments of the youth population from engaging in sustainability. Such a phenomenon is ever present across the world, but especially in developing countries where those inequalities are more profound. Transformative changes need to be made to socioeconomic systems to ensure that young people are supported with the financial, educational, and structural means to meaningfully contribute to sustainable development.

3. Areas of Impact: Where Youth Can Lead

Given that they comprise more than 1.2 billion people, around 16 percent of the global population, youths aged between 15 and 24 years naturally play a pivotal role in the global pursuit of sustainable development (UN, 2020). Their role is further emphasized by their inclusion among the nine Major Groups recognized by the UN as the main segments of society through which broad participation in sustainable development processes would be promoted, alongside women, indigenous peoples, non-governmental organizations, local authorities, workers and trade unions, business and industry, and the scientific and technological community (UN, 2022).

The desire for environmental action is substantial among youth, with 7 out of 10 young people expressing a desire to actively engage in the green transition (United Nations Department of Economic and Social Affairs [UNDESA], 2023b). The sheer size and enthusiasm of the youth demographic positions them as a force capable of instigating meaningful change. Their population is expected

to further grow by seven percent to approximately 1.3 billion by the conclusion of the SDGs Agenda in 2030 (UN, 2020).

Over the past few years, there have been many exemplary youth pioneers who have embarked on green initiatives. Nzambi Matee, the founder of Gjenge Makers, revolutionized waste management in Kenya by turning plastics into bricks to pave roads, thus addressing both plastic pollution and infrastructure needs (United Nations Environment Programme, 2022). Fatemah Alzelzela from Kuwait co-founded Eco Star, a non-profit recycling initiative aiming to address the country's high trash generation through using social media to educate and inspire sustainable waste management practices (United Nations Environment Programme, 2022).

Although many young people have demonstrated an active commitment to engage in green initiatives, the current level of youth participation is still insufficient to mitigate the rapidly developing negative impacts of climate change. The urgency of involving youth in sustainable development is underscored by the heightened vulnerability of children to the impacts of climate change (Eskenazi et al., 2020). The detrimental effects of climate change have and will continue to threaten the futures of youth. Its impacts can be both direct and indirect, jeopardizing not only young people's physical health and well-being, but also their future educational, social, and economic opportunities (Sanson et al., 2019). Furthermore, the Paris Agreement's goal of limiting global temperature rise to under two degrees Celsius, which is essential for averting catastrophic consequences, is now in jeopardy (Sachs, 2016). It is projected that, at the current rate, none of the SDGs will be achieved by the intended year of 2030, with less than 20 percent of the associated targets on track for fulfillment (Sachs et al., 2023). As such, it is even more imperative that youths actively participate in finding innovative solutions to combat climate change and contribute to sustainable development overall.

For these reasons, green skills are essential for young people to succeed in the green economy. These competencies comprise the knowledge, abilities, attitudes, and values required to support, foster, and live in a society that is resource-efficient and sustainable (UNDESA, 2023a). With 40 percent of employment linked to a healthy environment under threat due to climate change (UNDESA, 2023b), the green transition is expected to generate 8.4 million employment opportunities for young people by 2030 (UNDESA, 2023a). However, a concerning trend emerges as 60 percent of young individuals are anticipated to lack the skills essential to prosper in this evolving green economy (UNDESA, 2023a).

Therefore, empowerment is crucial to harness the potential of youth in driving sustainable development. Educational initiatives should focus on cultivating green skills and equipping young individuals with the knowledge and tools needed to address environmental challenges. Additionally, mentorship programs,

partnerships with businesses, and access to funding can foster the entrepreneurial spirit among youth, enabling them to initiate and scale green projects.

4. ASEAN Green Deal and Youth

During his chairmanship of ASEAN in 2022, former Cambodian prime minister Hun Sen declared the initiation of the ASEAN Green Deal (Khmer Times, 2022). It has garnered considerable attention and has been lauded as one of ASEAN's most comprehensive overarching green programs. As of yet, no formal document detailing the intended activities or itinerary of the ASEAN Green Deal has been produced (Chheang, 2023). However, it is crucial to acknowledge that the ASEAN Green Deal bears strong resemblance to the European Green Deal, a set of policy initiatives that was officially adopted by the European Union in 2020. The European Green Deal encompasses the EU's dedication to tackling environmental and global warming issues while concurrently promoting sustainable growth in the bloc via a green economy. It also serves as a mechanism for promoting sustainability and strengthening security, by specifically focusing on addressing Europe's security concerns, namely in energy (Rybski, 2023). On the contrary, although there have not been any written policies laid out, the Chairman's Statement for the 12th ASEAN-United Nations Summit held in Phnom Penh, Cambodia on 11 November 2022, provides an overview of the objective of the ASEAN Green Deal:

“Aim to gradually transform ASEAN region towards a green future, sustainability, effective use of resources, resilience, and economic competitiveness.”

Chairman's Statement of the 12th ASEAN-United Nations Summit (2022)

The excerpt above underscores that the ASEAN Green Deal is designed to bolster environmentally conscious endeavors within the ASEAN region, while simultaneously functioning as an instrument towards achieving the 2030 UN SDGs. This vision, which incorporates the ASEAN Green Deal and UN SDGs, can also be found in the ASEAN Economic and Socio-Cultural Community and the ASEAN Community Blueprint 2025 (Ting & Yongkun, 2023).

Given the potential long-term consequences of climate change and environmental degradation, it is essential to consider their impact on future generations in ASEAN. Young people have a crucial role in leading the green drive by actively contributing to the resolution of the potential intergenerational problems mentioned section 1 and 2 of this chapter. Out of the roughly 660 million individuals in ASEAN, 224.2 million were young people in 2020 – this accounts for one-third of the population. Among these youths, 53 percent fall within the age range of 15 to 25, while the remaining 47 percent are between 26 to 35 years of age (ASEAN, 2023). With this, despite being the third biggest population in the globe, and although ASEAN has enacted various policy measures, as mentioned above, the area's progress toward sustainable development objectives remains

bleak. For example, the ASEAN SDGs snapshot in 2022 showed that climate-related disasters had a greater effect on people in the region in 2020 compared to 2016. In 2020, on average, the number of individuals who were reported missing, directly affected, or whose lives were taken by climate-related catastrophes was 2,546.8 per 100,000 population. This is a significant increase compared to the rate of 554 per 100,000 population in 2016 (ASEAN, 2022). Furthermore, the outbreak of the COVID-19 pandemic has been a reason for the reversed financial backing in terms of financial assistance for ecologically friendly initiatives (Ahmed et al., 2022). Concurrently, the global energy crisis, which began in 2021 in the aftermath of the COVID-19 pandemic, has impeded the capacity of ASEAN member states to shift towards renewable energy or provide support in other areas of sustainable development such as healthcare and education (Jahanger et al., 2023). ASEAN's insufficient advancement towards the SDGs can potentially result in further consequences, one of which is the threat that climate change in 2023 may affect up to 11 percent decrease of regional economic growth and the mortality caused by climate change to increase to 70,000 in 2030 (Jahanger et al., 2023).

Although overall progress in ASEAN has been slow in terms of achieving the SDGs by the intended year of 2030, there is still indication of favorable, albeit modest, progress in the fulfillment of certain goals. These goals include: Goal 2) Eradicate Food Insecurity, Goal 3) Overall Physical and Mental Well-Being, Goal 4) Quality Education, Goal 7) Affordable and Clean Energy, Goal 8) Decent Work and Economic Growth, Goal 9) Industry, Innovation, and Infrastructure, and Goal 17) Partnerships for the Goals (ASEAN, 2022). Young people in ASEAN have played a crucial role in advancing the SDGs by actively participating in various ways. Examples of youth initiatives for green skills include the Young Forces for Agricultural Innovation (#Y4AGRI) of the Southeast Asian Regional Centre for Graduate Study and Research in Agriculture (SEARCA), the ASEAN Student Leaders Forum (ASLF) of the ASEAN University Network (AUN), and other prominent regional academic networks and initiatives (Nelles, 2023).

Creating opportunities for youth engagement, empowerment, and awareness in sustainable development has been a priority for ASEAN. However, it is also important to emphasize to all parties involved, particularly young people, that sustainable development must focus on human interests and the long history of core principles and practices of environmental sustainability in Southeast Asia (Sofjan, 2022). Youth must comprehend the significance of recognizing that there is no one-size-fits-all solution or standard of conduct for environmental sustainability. It is critical to understand the implications of capitalism, which may lead to an absence of true commitment to environmental conservation and life preservation (Nelles, 2023). By bringing measures for reducing poverty into line with the goals of the private sector, the SDGs could represent a move

towards a more firmly free-market orientation. Particularly, concentrating on global markets, global demands, and global consumption in accordance with the neoliberal agenda, rather than a real concern for the sustainable development goal, may also lead to the abandonment of unique local cultures (Cervantes, 2013; Lassou et al., 2019). Therefore, it may be seen that neoliberalism can contribute to a phenomenon known as “greenwashing,” in which multinational firms exclusively present the positive impacts of their products or operations on the environment while hiding, neglecting, or failing to disclose the repercussions to nature (Bowen & Aragon-Correa, 2014). Global community and ASEAN should exercise caution regarding the phenomenon.

5. Conclusion

The engagement of youth in sustainable development is a multifaceted and crucial aspect of global and regional agendas. The world and ASEAN stand at a crossroads, where the choices we make today will inevitably determine the future we inherit. Moving forward, it is important to understand that investing in the potential of youth for sustainable development at every level is not merely a moral imperative, but a strategic necessity. In the endeavor to build a sustainable world for all, policies and initiatives should be implemented that empower and equip young people with the knowledge, skills, resources, and platforms they need to fully realize their potential. So far, ASEAN has demonstrated an unwavering commitment to youth engagement, as demonstrated by the Youth Development Index, the forthcoming ASEAN Green Deal, and the several initiatives already in place for youth to contribute to the 2030 Agenda for Sustainable Development. Economic dynamism, structural improvements to socioeconomic systems, and environment cooperation are all positive avenues for the holistic engagement and empowerment of youth in sustainable development processes. ASEAN as a region is on the right path in facilitating and fulfilling this. However, more can always be achieved. We have yet to maximize our fullest capacity in utilizing the potential of youths across Southeast Asia in sustainability. The ASEAN Green Deal is just one of the many golden opportunities we have to further explore and create ways to enhance young people’s participation. It is only when we acknowledge and realize the role of youth as key stakeholders in development processes that we can construct a more just, equitable, and sustainable future for ourselves and generations beyond.

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CONCLUSION

ASEAN Green Ideals book aims to bring together perspectives from writers in various fields. Through these incisive contributions, the book has proven to encourage conversations, creativity, and ideas, as well as develop a broader understanding of the Green Deal's revolutionary potential for Cambodia, the region, and the globe. Throughout the book, it is important to note that there is no uniform approach to tackling climate change or maintaining sustainable development. On the other hand, it is clear that each sector, ranging from technology, food, health, energy, water, and architecture, experiences different impacts, while actors such as universities, scientists, and even youth will all contribute, albeit in different capacities and levels, to mitigating climate change and environmental issues.

As stated in the preface, the objective was not to enforce a particular formatting style on each chapter; however, a certain standard must be met. With that said, readers should be aware that, due to the issue's multifaceted nature and the need to gather multidisciplinary experts to take a holistic approach to addressing pressing issues such as climate change, environmental sustainability, and socioeconomic development, it is critical that people from various backgrounds and disciplines have the freedom to express their research and ideas creatively. The book aims to provide ideas in a new but comprehensive manner to investigate and advocate for the urgent need for a green deal.

Furthermore, the book aimed to be accessible to policy-makers, students, and anybody interested in creating a sustainable and resilient future. The 14 chapters' discussions eventually lead to an academic analysis of the Green Deal's rich and intriguing characteristics. The vast range of ideas from many viewpoints may serve as a platform for undertaking an extensive multidisciplinary study on the Green Deal with the goal of informing future policies that transcend national boundaries, stressing a more systematic and global approach.

Aside from establishing the book's goal and approach, the combined 14 chapters have produced a few conclusions worth contemplating. Climate and environmental concerns touch all nations and players, regardless of geography, and show that a green strategy and sustainable development are the way ahead for a sustainable future. However, said can be easier than done. Issues arise at all levels, from local to worldwide, owing to political issues, resource constraints, a lack of knowledge and education, or a lack of commitment. These concerns seem to arise in every chapter of this book. For example, education has been repeatedly emphasized in numerous chapters as a crucial instrument for defining the norms of future generations and raising public awareness of the possible consequences of climatic and environmental issues. On the other hand, resource-related factors such as cultivating technology, green energy, and financial resources for green development are ideal; however, there are gaps between the powerful and the rest of the world in the aforementioned factors, complicating the path to sustainable development. Despite this, the bulk

of the chapter emphasized the need for collaboration and working together rather than working in silos, since preventing climate change can only be accomplished by working in solidarity and across all domains of specialty.

Last but not least, the book demonstrated that ASEAN's Green Deal will be an essential diplomatic equivalent to the European Green Deal, emphasizing that ASEAN will need and can play a role in promoting sustainable development. These perspectives have been shared by largely Khmer writers and some international professionals, demonstrating Cambodia's expertise and vision as a gift to the world. As a framework, as a member of ASEAN and a global player, Cambodia, via this book, shares the idea that there are diverse approaches to mitigating climatic and environmental concerns and that localization of measures is critical rather than adopting a certain norm. At the same time, some chapters questioned and raised awareness of the prevailing narrative of sustainable development, while others used it as a greenwashing tactic to accomplish a specific goal rather than the benefit of the whole global population. Such concerns are crucial, particularly when participating in an intellectual discussion with the global community, to share their modest viewpoint with the world, with the hope that all voices may be heard as part of the decolonization of concepts and information.

Several potential policy suggestions that would be well-suited for the implementation of the ASEAN Green Deal have been identified based on the comprehensive data collected from all 14 chapters. Nevertheless, the community is currently in the process of transforming these ideals into concrete and implementable plans, suggesting that they have not yet reached their full potential. This book advocates for the ongoing participation of all stakeholders in order to broaden perspectives and guarantee a comprehensive approach. While also suggesting that specific aspects of the investigation below may be worth further investigation in order to substantiate this idea for ASEAN Green Deal:

- To ensure equitable participation and benefits for all member states, how can ASEAN countries effectively integrate diverse cultural and social values into a unified Green Deal framework?
- What governance structures and mechanisms promote transparent and accountable regional collaboration for ASEAN Green Deal implementation despite differing national interests and capacities?
- How can ASEAN create innovative financing models for large-scale green technology and infrastructure investments while ensuring that developed and developing member countries have access to resources?
- How can ASEAN use regional diplomacy to align the Green Deal with global sustainability initiatives such as the European Green Deal while also addressing local environmental and economic concerns?
- How can the ASEAN Green Deal be put into action to prevent greenwashing and ensure genuine, measurable sustainability efforts that benefit the environment and society for the long term?

DISCLOSURE

Responsible Use of ChatGPT in the Creation of This Book

To varying degrees, the authors of this book have responsibly and transparently utilized ChatGPT as a tool to enhance the clarity, coherence, and accessibility of the text. While ChatGPT played an instrumental role in refining language, brainstorming ideas, and ensuring stylistic consistency, the book's conceptual framework, logical structure, and substantive content remain deeply rooted in the author's extensive knowledge, experience, and rigorous research. The following points outline our approach to the responsible use of AI tools in the writing process.

1. Augmenting Communication, Not Replacing Expertise

ChatGPT has been used primarily to refine language and enhance the accessibility of the text, particularly since none of the authors are native English speakers. The tool served as an aid to ensure that our work effectively communicates the depth of our knowledge without language barriers detracting from its impact. By sharing knowledge localized to our context, this book provides a valuable opportunity for voices from a small but rapidly developing country to contribute to global discourse.

2. Acknowledging Bibliographic Sources

All references, ideas, and sources of information have been meticulously cited according to the highest academic standards. The use of ChatGPT does not compromise this principle. The authors remain fully committed to proper attribution and transparency in acknowledging all scholarly and non-scholarly influences on this work.

3. Commitment to Academic Integrity

The authors have ensured that ChatGPT was employed solely to augment clarity and precision in expressing ideas. At no point was it used to generate substantive content that could compromise academic or intellectual integrity. Plagiarism or misrepresentation of originality has been categorically avoided, with the tool integrated into the writing process responsibly.

4. Transparency in Methodology

Recognizing the importance of transparency, we openly acknowledge the use of ChatGPT in the editorial process. It has been employed as a supplementary tool, not as a primary source of ideas or arguments. This approach underscores our intent to maintain accountability and highlight the central role of human authorship in conceptualizing and structuring the book.

5. Educational and Ethical Intent

This book aims to educate, inform, and inspire readers, rather than serve as a means for personal promotion, undue recognition, or financial gain. As non-native English speakers, we have used ChatGPT to help ensure our ideas are communicated clearly and professionally, in line with our broader educational and ethical goals.

6. Ethical AI Use

As advocates of ethical technology use, we consider ChatGPT an assistive tool akin to a digital collaborator that streamlines the writing process. Its use reflects our commitment to leveraging technology to enhance human creativity and productivity while maintaining full accountability for the final content. This aligns with broader principles of responsible AI adoption, demonstrating how emerging technologies can augment human expertise without substituting it.

7. Authorship and Accountability

ChatGPT, or any generative AI, cannot be recognized as a co-author on this or any other publication. Authorship requires intellectual contributions, responsibility for the content, and accountability - uniquely human capabilities that AI does not possess. AI tools like ChatGPT cannot engage in intellectual discussions, interpret data within a research framework, or make decisions about the direction of research. While ChatGPT was valuable in assisting with drafting, editing, and brainstorming, the intellectual and conceptual framework of this book is entirely human-generated.

Ethical guidelines, such as those from the Committee on Publication Ethics (COPE), affirm that authorship criteria cannot be met by AI. In alignment with these standards, we have included this disclaimer to ensure ethical transparency while preserving the integrity of human authorship. Rather than listing ChatGPT as a co-author, we have detailed its role in this section to reflect its contribution as a tool and not as an intellectual collaborator.

This book exemplifies how generative AI can be responsibly incorporated into academic and professional work, augmenting human expertise while maintaining the highest standards of academic integrity and authorship. Through transparency and accountability, we hope to inspire others to adopt similar ethical practices in their engagement with emerging technologies.

(Back Cover)